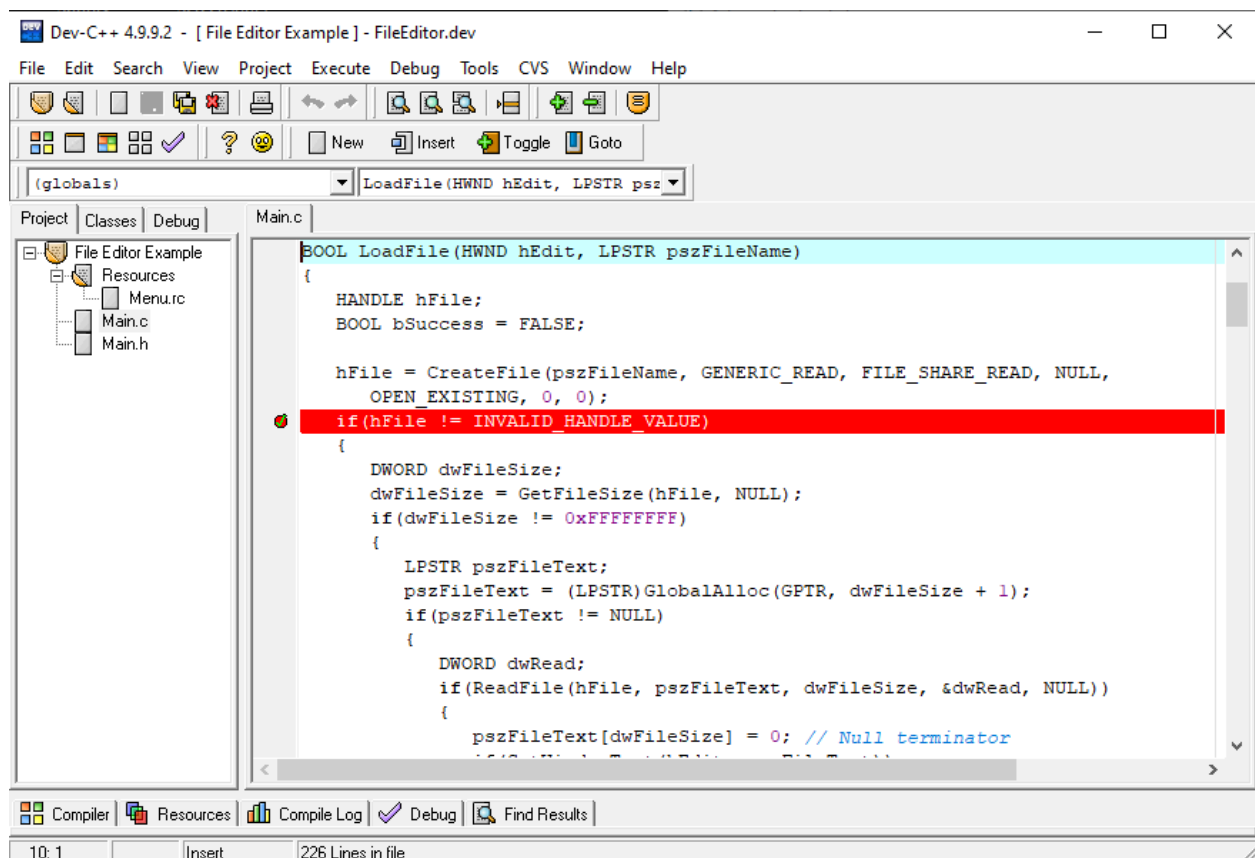


DEV C++

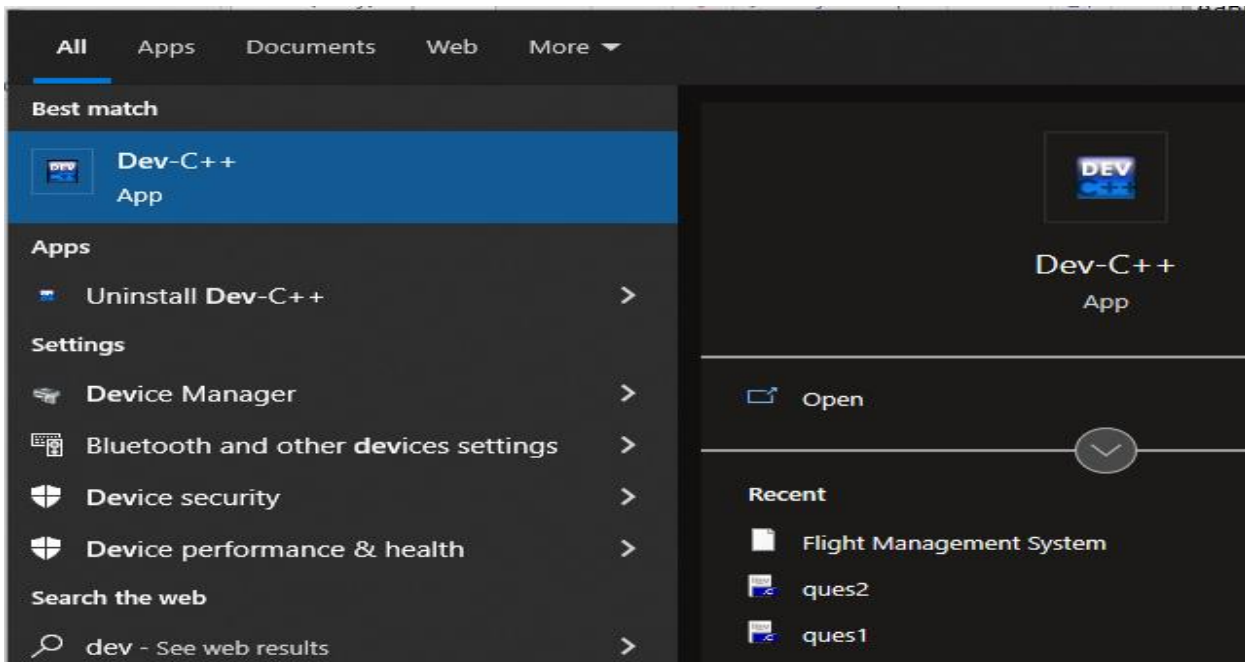
Installation and Getting Started under Windows OS

Dev-C++ is a full-featured C and C++ Integrated Development Environment (IDE) for Windows platforms. Millions of developers, students and researchers use Dev-C++ since the first version was released in 1998. It has been featured in dozens of C++ and scientific books and remains one of the favorite learning tool among universities & schools worldwide.

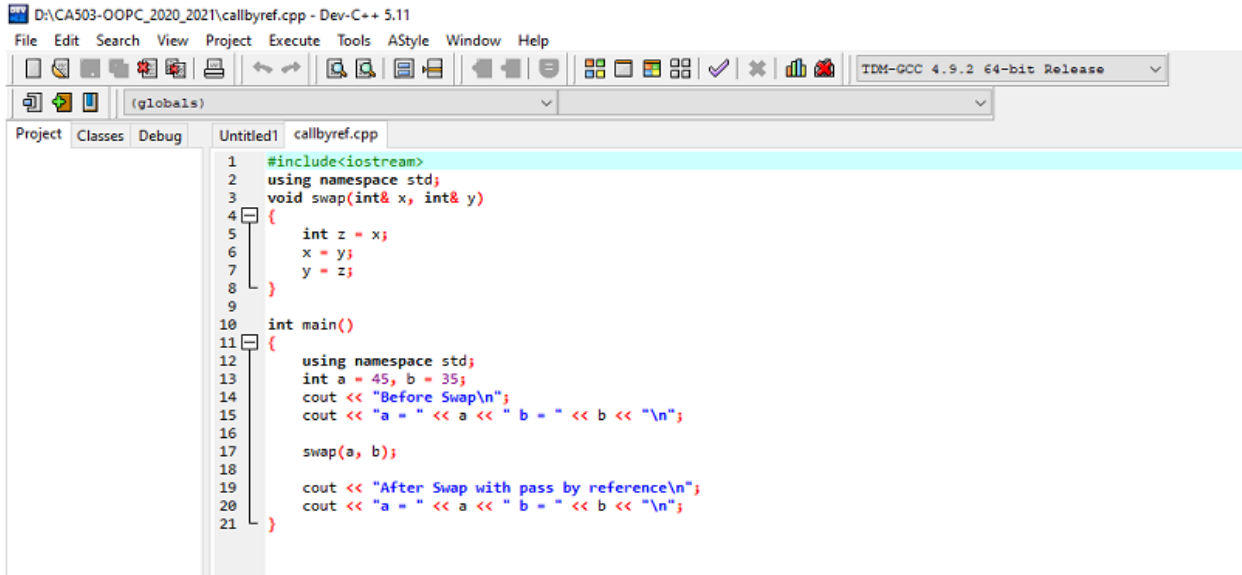
- 1 • Go to <https://www.bloodshed.net/>
- 2 • Click on [Download Original Dev-c++ for Windows](#)
- 3 • Click on [install Dev C++ for the first time](#)
- 4



Tool Based Manual || BCA || SEM II || CA129-Fundamentals of Object Oriented Programming



Select Dev-c++ from start menu,it open Dev-c++ editor



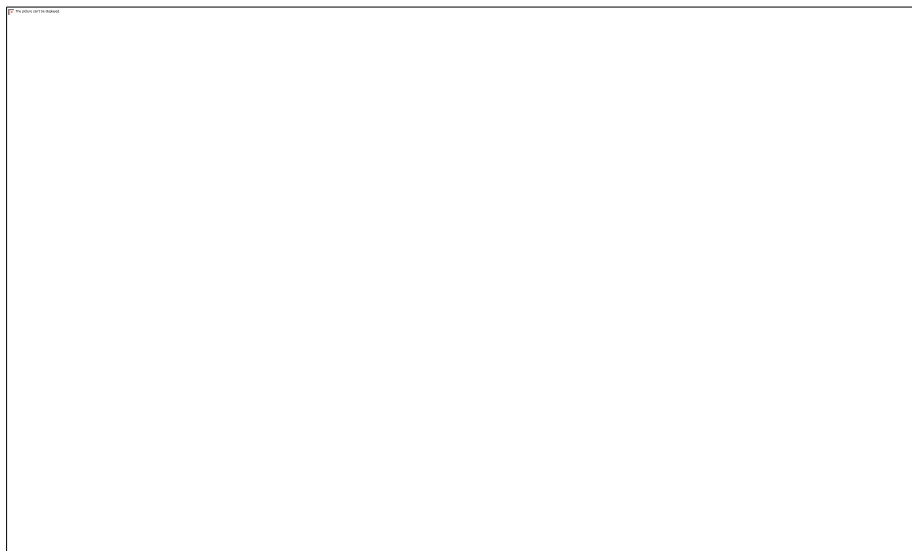
Write your code in Editor, save it and Select Execute->Compile and Execute->Run (F10)

Download

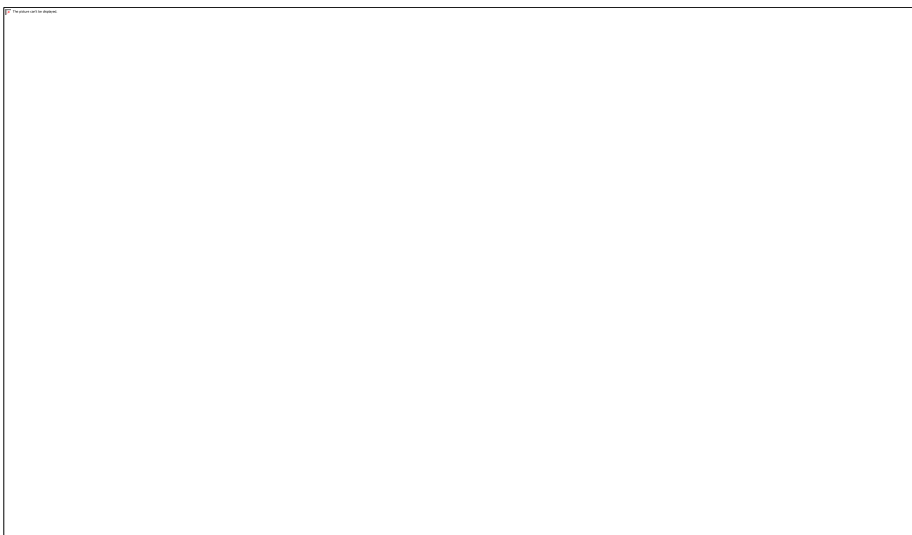
To download the latest version of Cygwin, follow this [link](#) to its official website. You can download the version suitable for your Windows installation and system's architecture. For the 32-bit version, download <https://cygwin.com/setup-x86.exe> and for the 64-bit version, download from here https://cygwin.com/setup-x86_64.exe.

Installation

- Launch the downloaded executable file from the previous step. Accept any prompt and warning that is displayed on-screen from Windows User Access Control. This will launch the setup program as shown below.

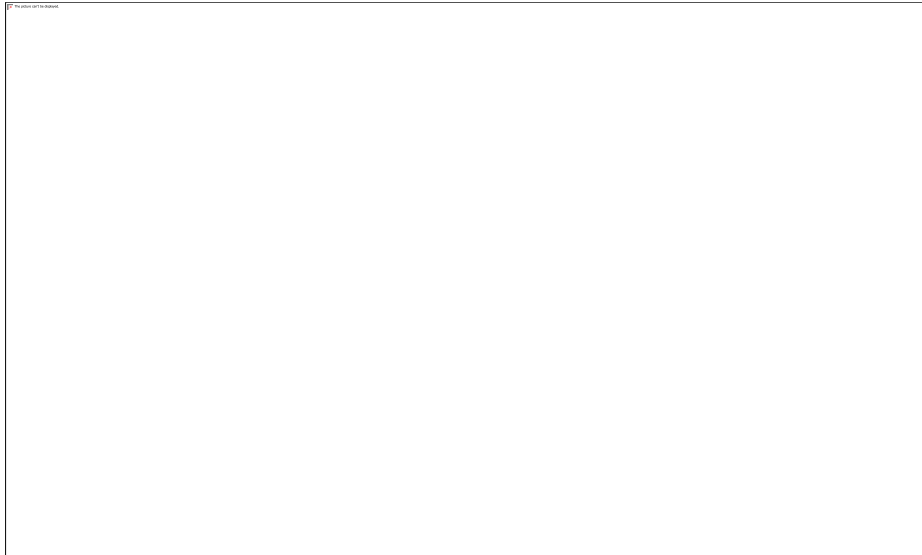


- Click "Next" to continue with the setup process.

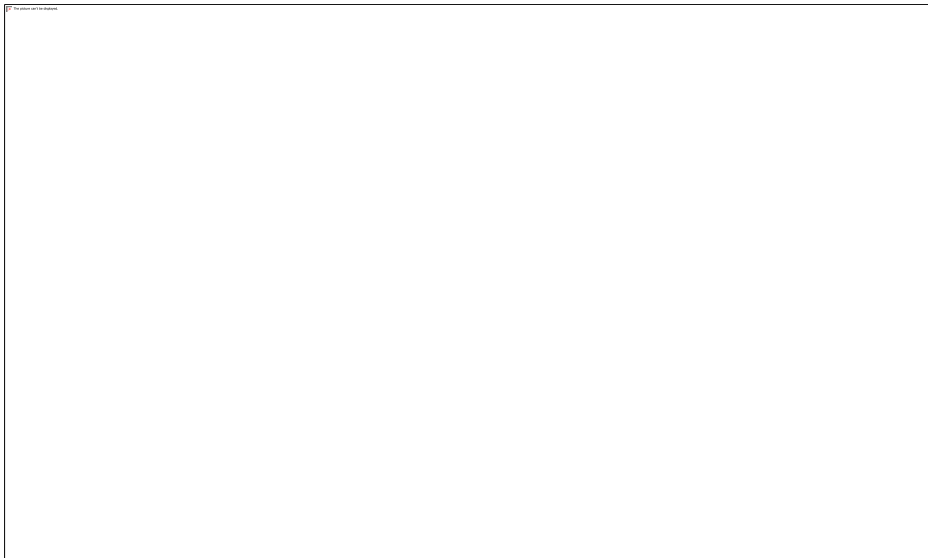


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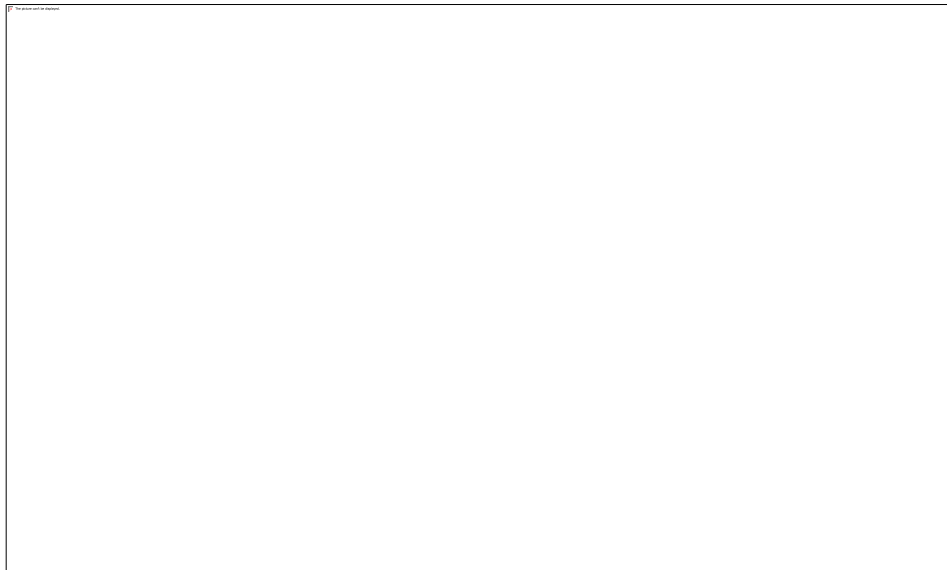
- A prompt to select a download source will be displayed. For most cases, the default option to "Install from Internet" is okay and should be kept. If you've a local downloaded repository or just want to download and install Cygwin later, choose one of the other available options. Click "Next" to continue.



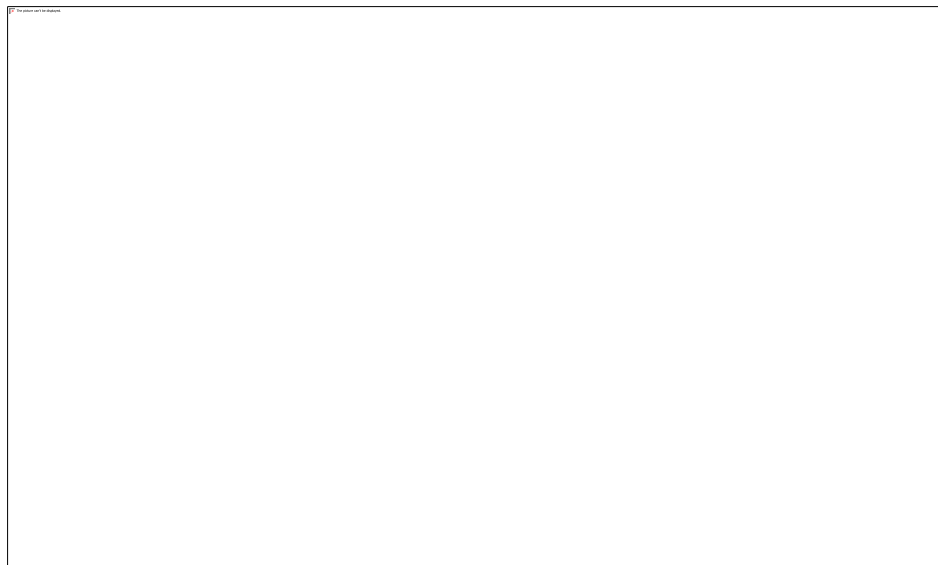
- It prompts you to specify "Root Directory" for the install as well as whether it'll be installed for "All Users" or "Just You". Again if there isn't a specific need, keep the default options and click "Next" to continue.



- To download the package files, the setup will ask you to specify a "Local Package Directory" which by default points to your system's "Download" folder. In case you want to specify another location, specify its path here and click "Next" to proceed with the setup.



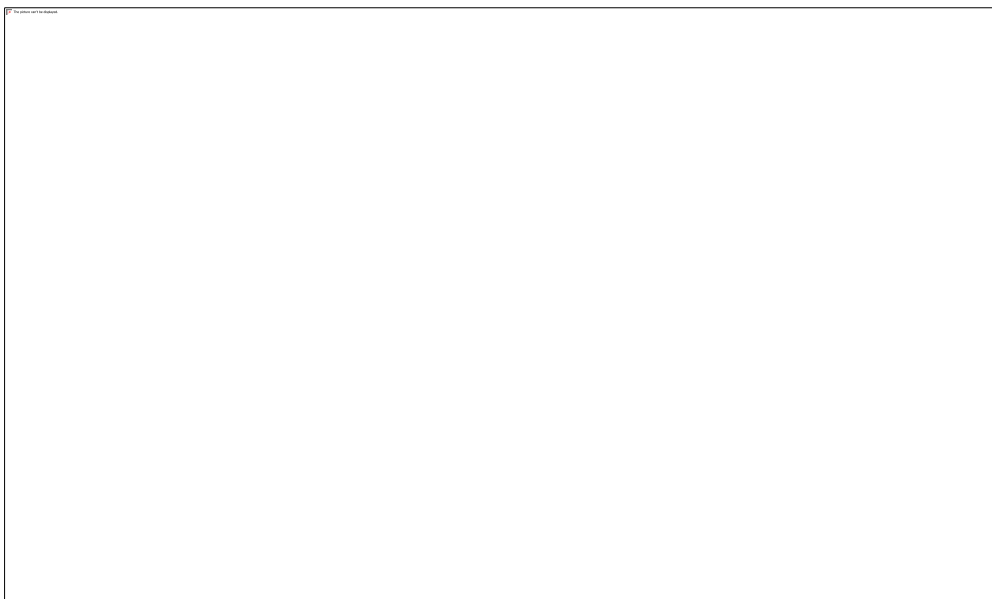
- For most cases, "Using System Proxy Settings" should work for the downloading of packages. If you don't intend to use any proxy and use a direct connection to the Internet, use the "Direct Connection" option, or you can specify a custom proxy setting as well. Click "Next" to continue.



- Next, it'll prompt you to select a mirror from where Cygwin will download its package files. If a mirror is near your location, which can provide faster speed, choose that one from the list. If you're unsure, select anyone from the list and click "Next" to proceed.



- Cygwin provides a lot of UNIX/Linux-based utilities that run on Windows. Not all packages get installed by default. Only a minimal set of important packages gets installed. In this step, you can customize your Cygwin setup by selecting/un-selecting different categories of packages. Remember, this can also be done later to add/remove packages to your Cygwin installation. If you have a specific list of important packages for your work, you can select them now and continue further by clicking "Next".



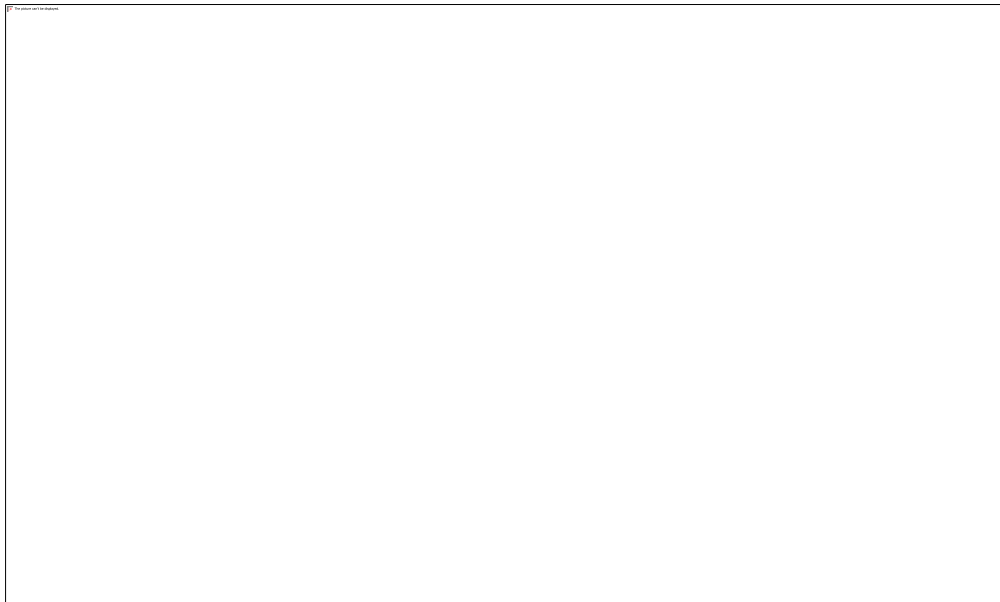
- The next screen lists the selected packages and asks you to review and confirm. Click "Next" to confirm and continue.



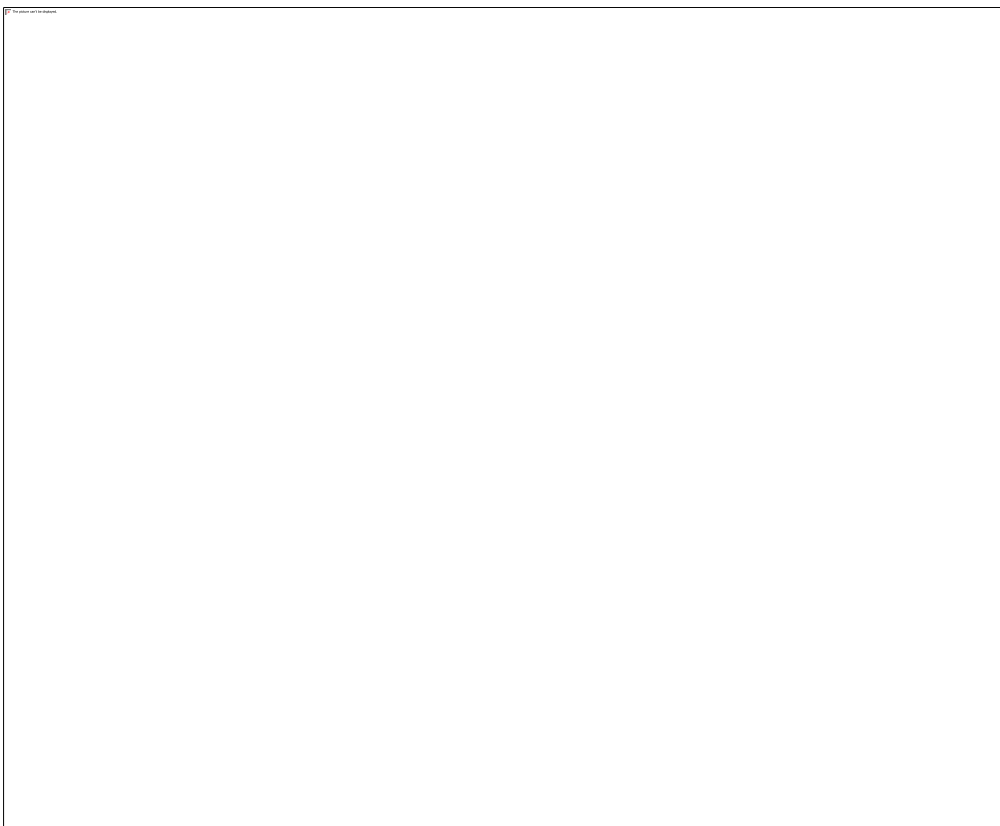
- The selected packages will begin to download, and this may take a while depending upon the speed of the selected download mirror and the number of selected packages.



- Once all packages are downloaded, the setup will continue to install the packages.



- When setup is complete, you'll be shown the option to create icons on Desktop and Start Menu for easy access. Click "Finish" when done to end setup wizard.

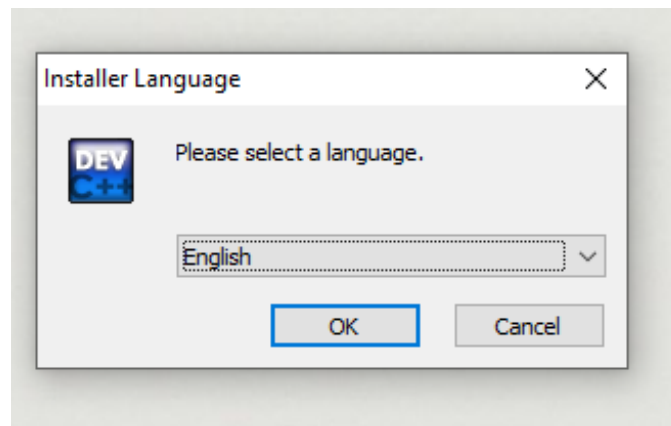


- You can now launch Cygwin by searching in Start Menu as shown above or double-clicking the Desktop icon (if created during setup). This will launch the Cygwin shell.

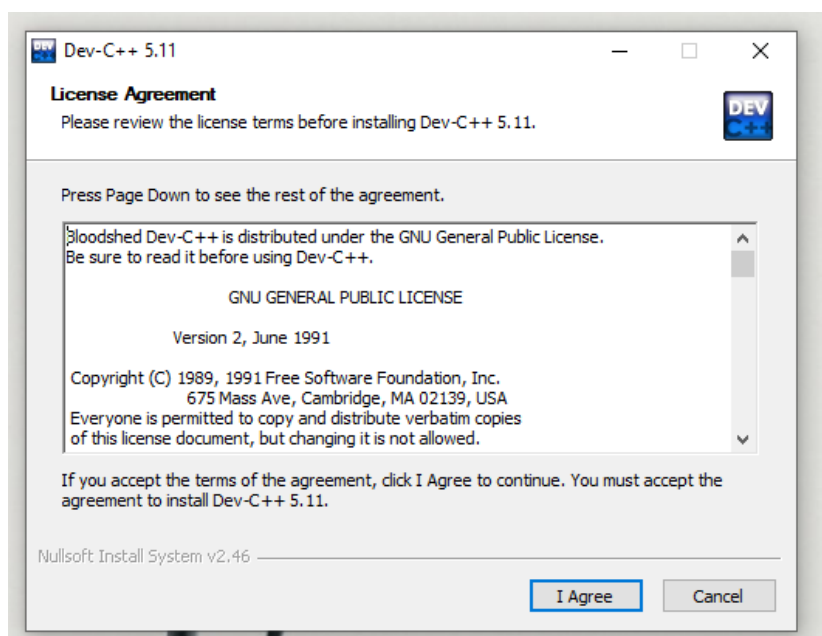
Installation and Getting Started under Windows OS

Follow the steps mentioned in following figure to install Dev c++ on Windows Operating System.

1. Go to <https://sourceforge.net/projects/orwelldvcpp/>
2. Click on Download and wait for five seconds.
3. Double click on Dev-Cpp 5.11 TDM-GCC 4.9.2 Setup file

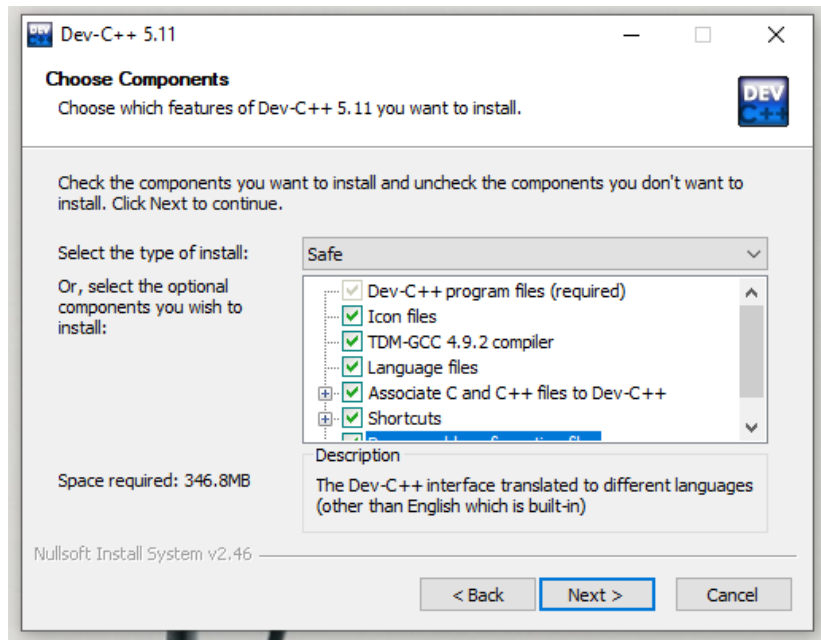


4. Click On Ok.

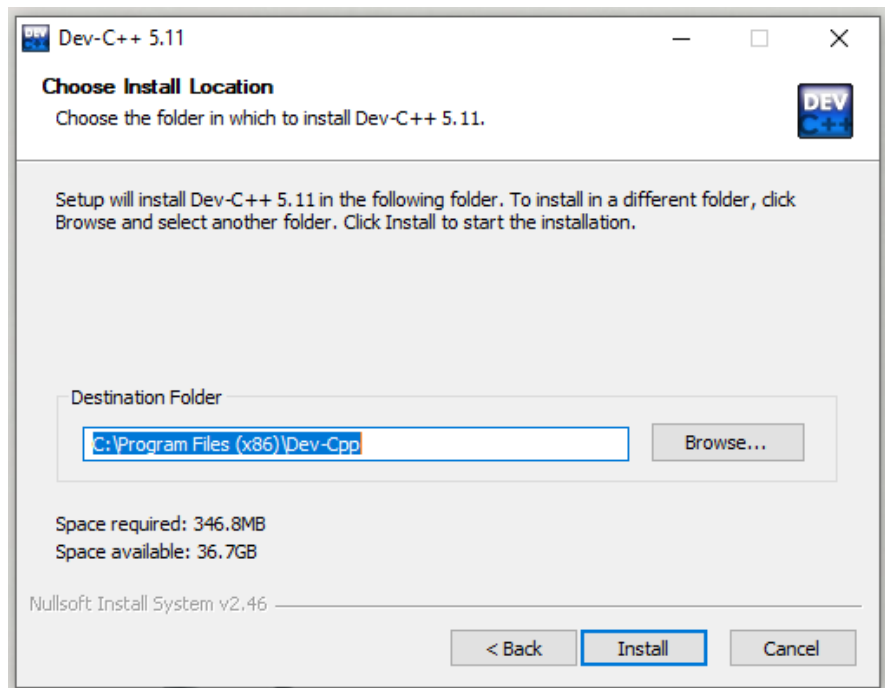


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5. Click On I Agree.

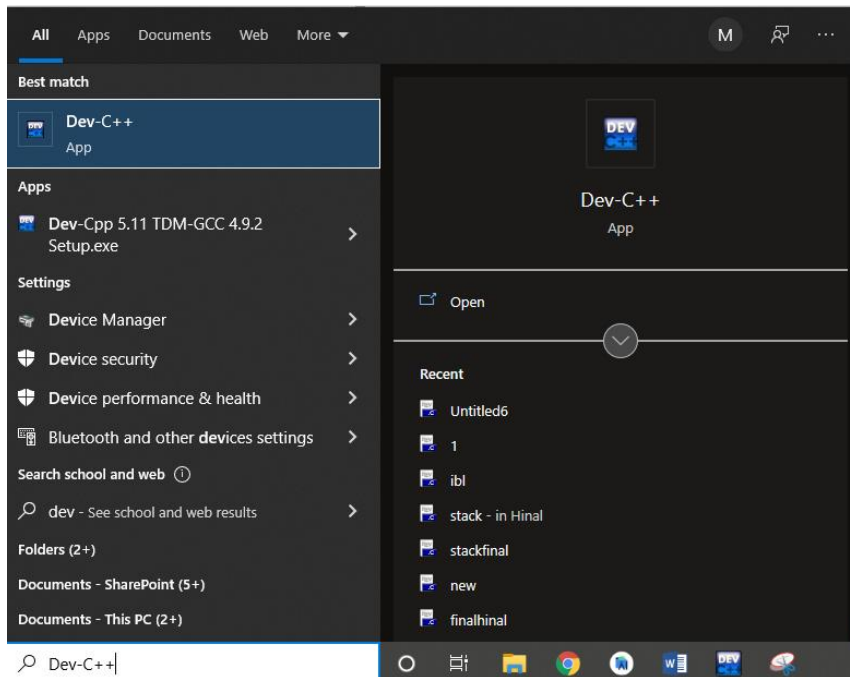


6. Click on Next.

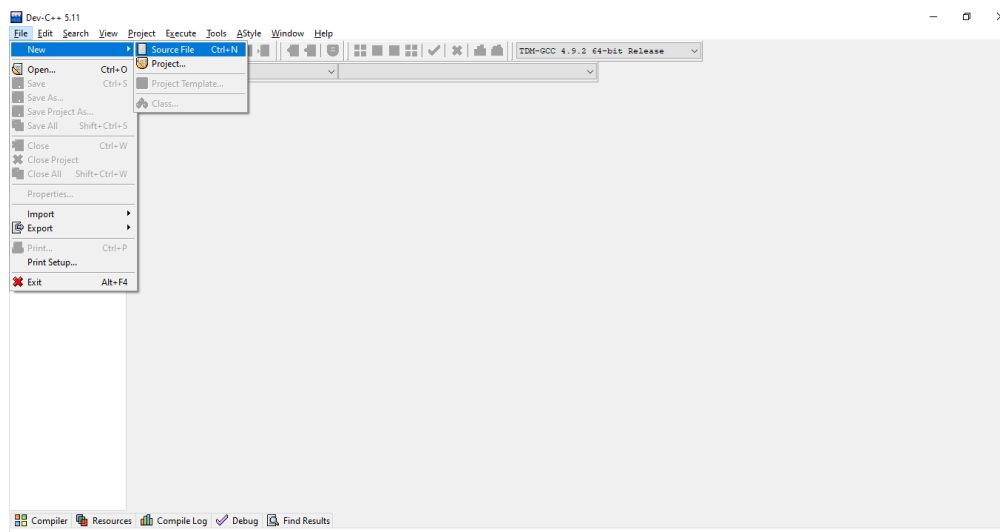


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9. Click on Next, Next Okay.

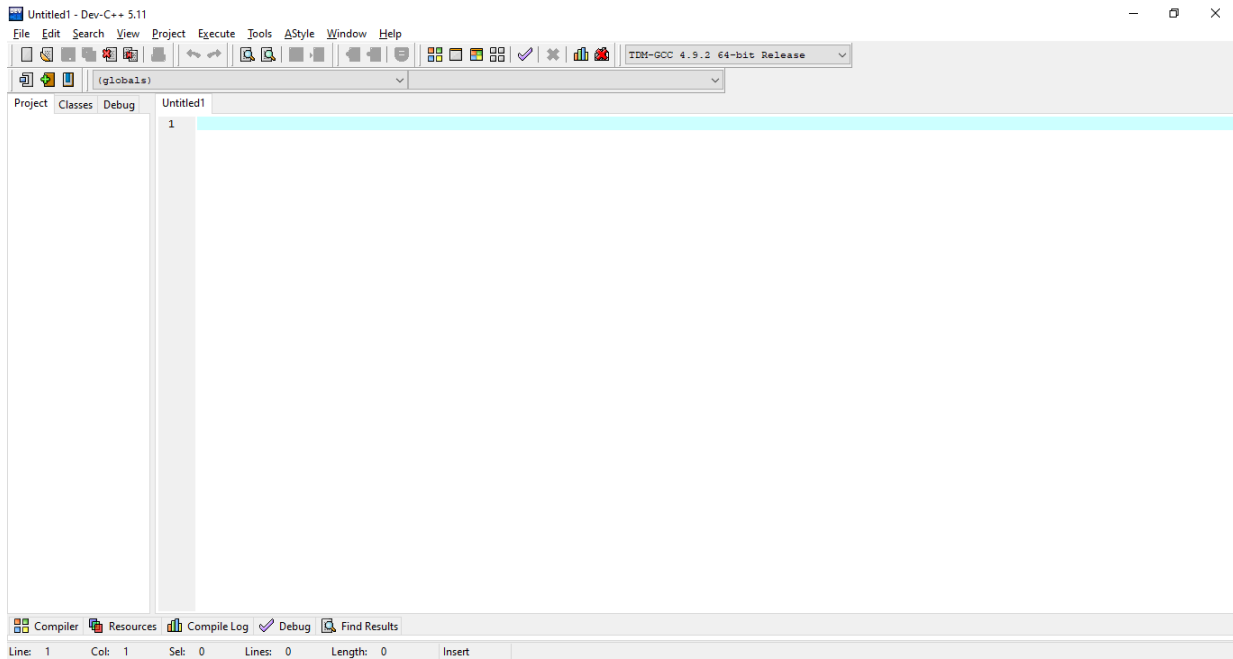


10. Open Dev C++.

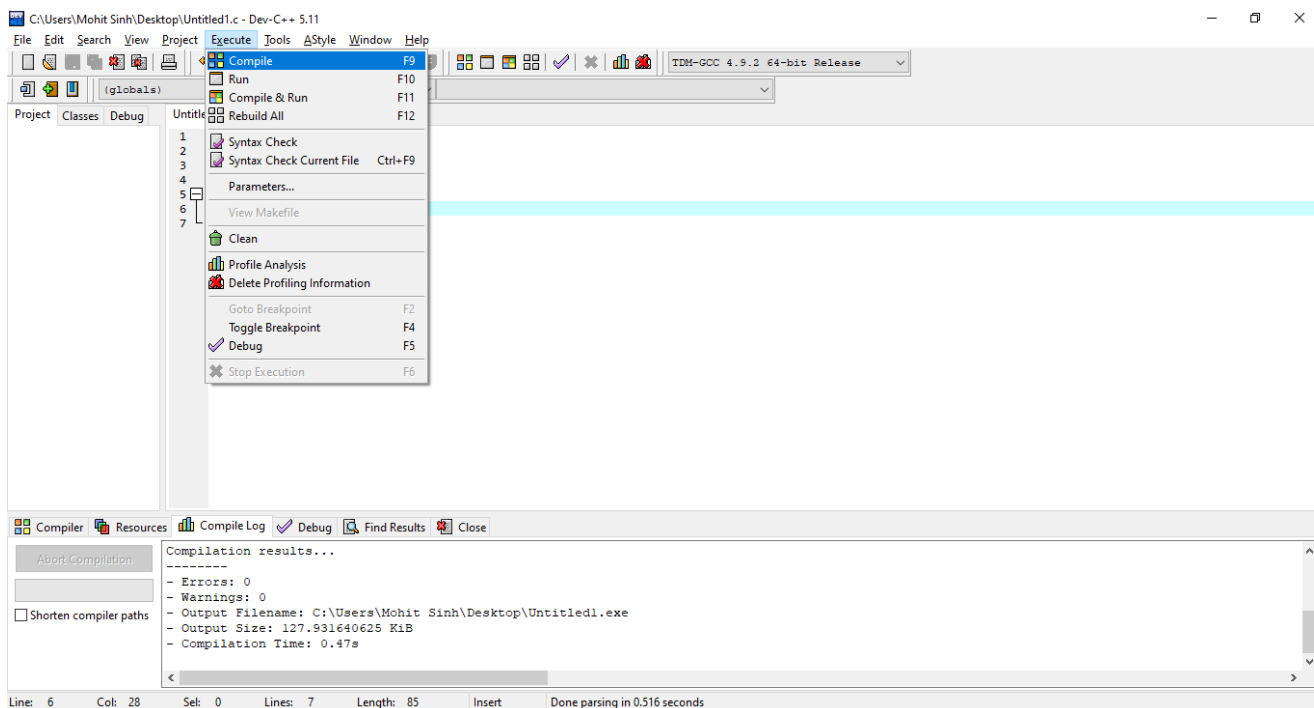


Tool Based Manual || BCA || SEM IV || CA221 : Data Structure and Algorithms

11. IDE of Dev C++.

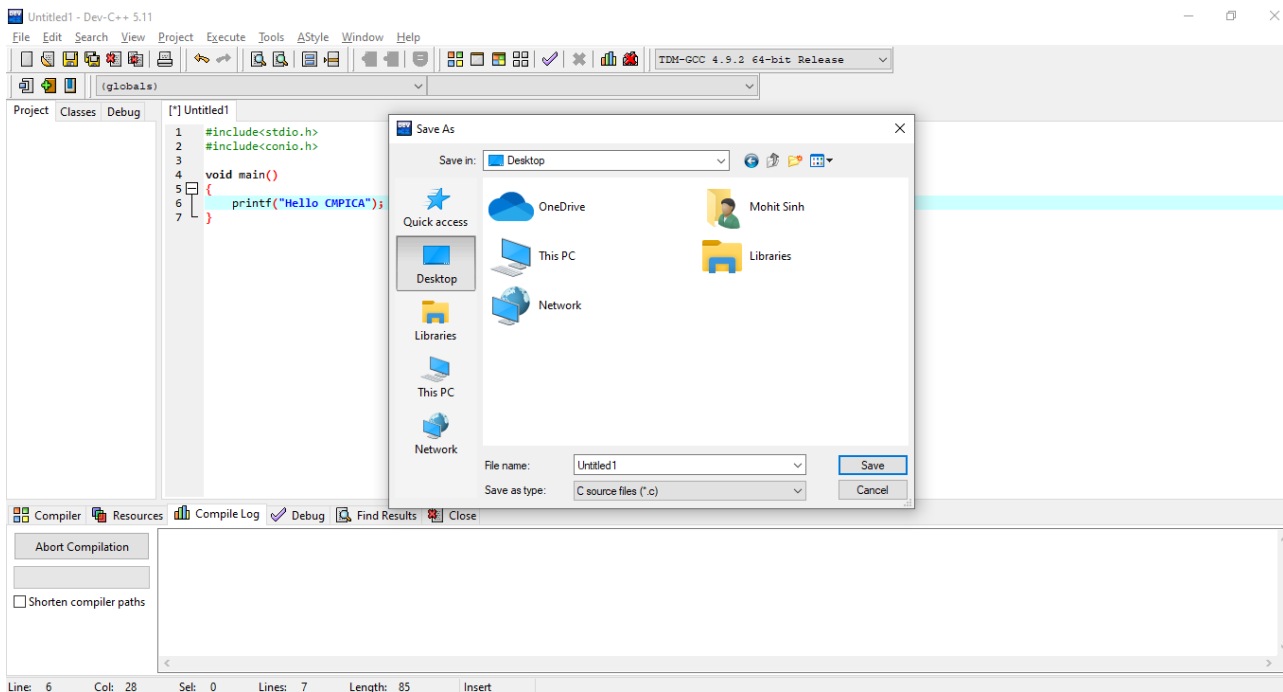


12. New file for c programming.

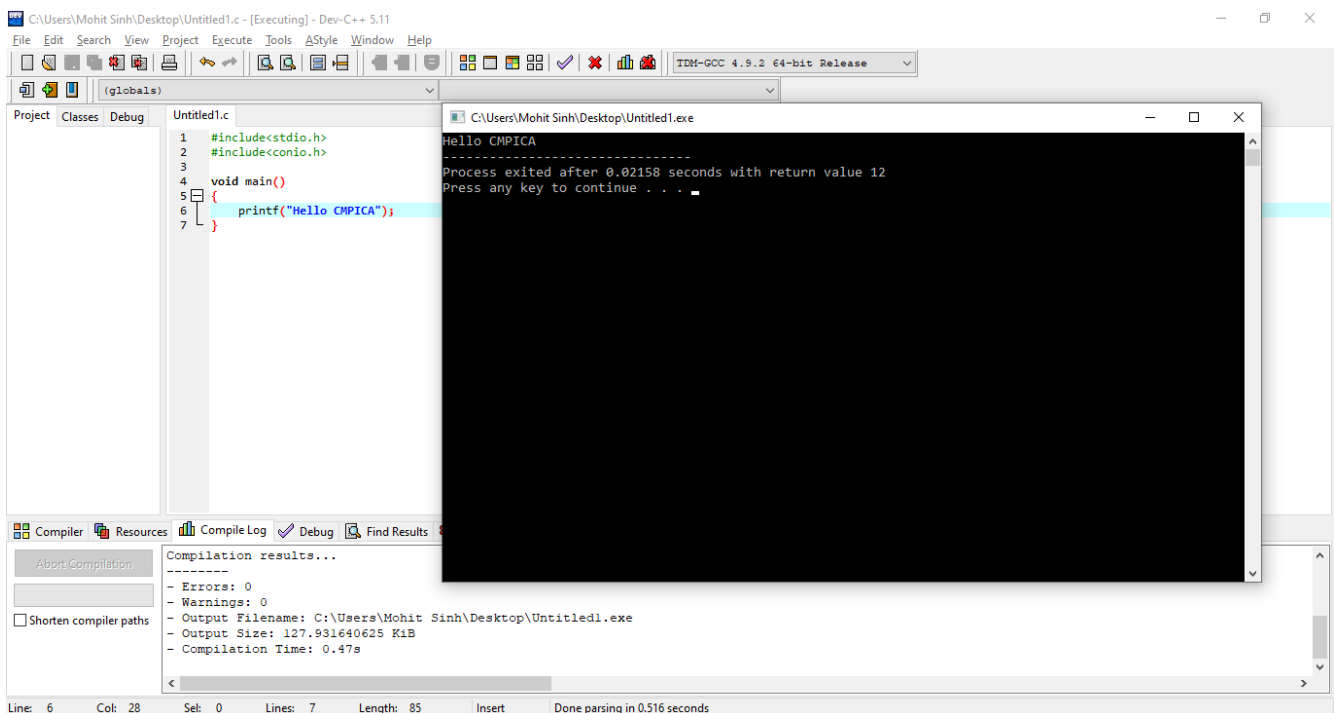


Tool Based Manual || BCA || SEM IV || CA221 : Data Structure and Algorithms

13. From the execute tab of menu bar you have to select compile option.



14. To run the program, save the file with appropriate name and extension.



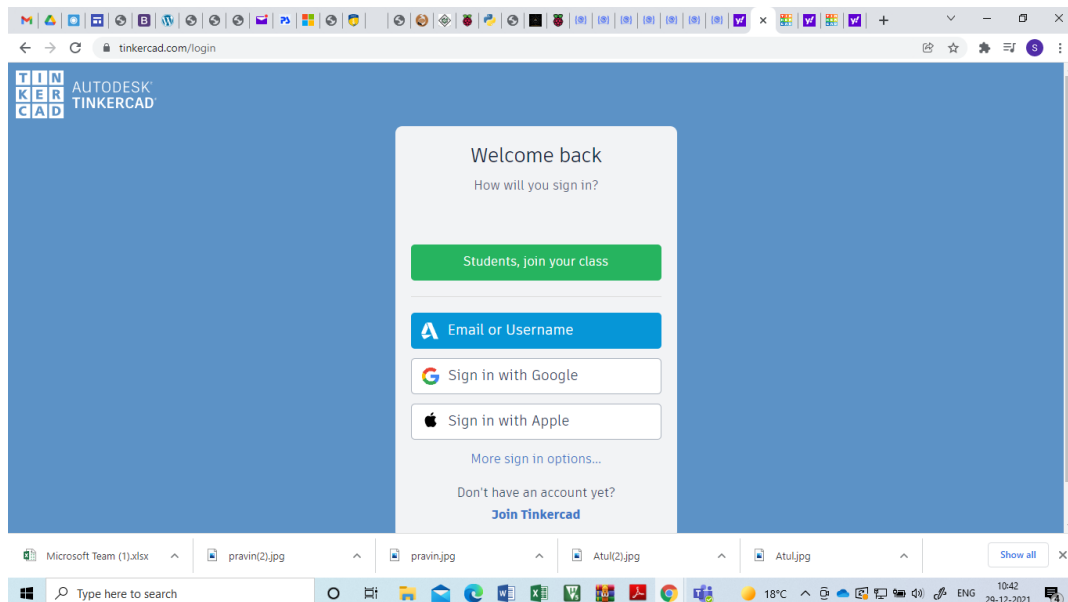
The black screen window is the console window that display the output of the program.

Tool 1:

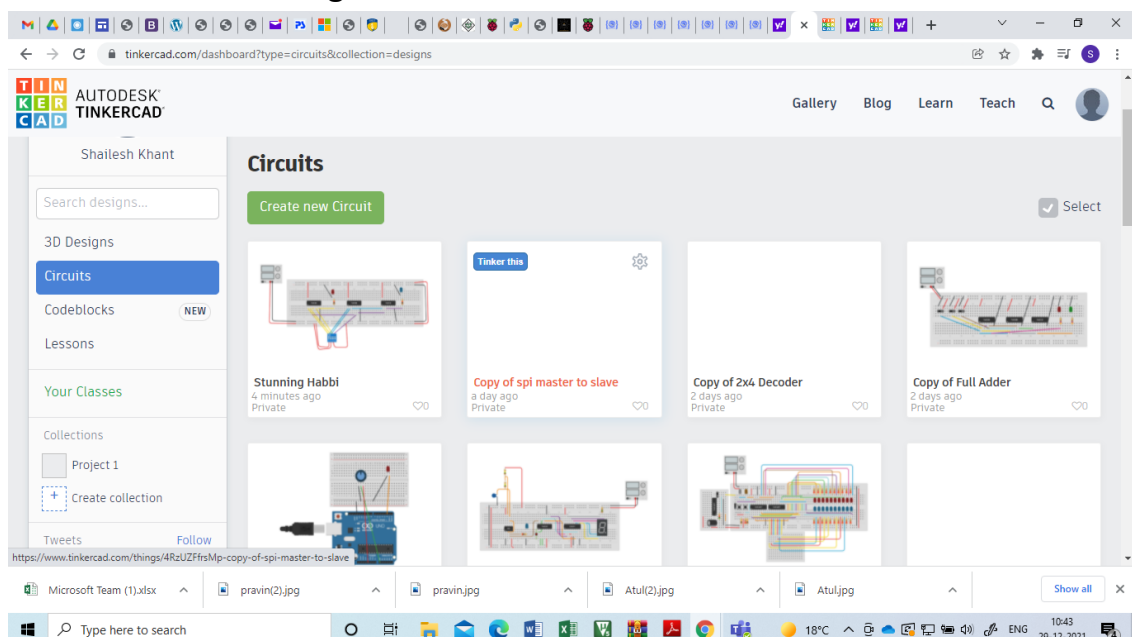
Simulation Using Tinker cad Tool.

Getting Started with Tinkercad

1. Open www.tinkercad.com
2. Login using Google or Facebook account.

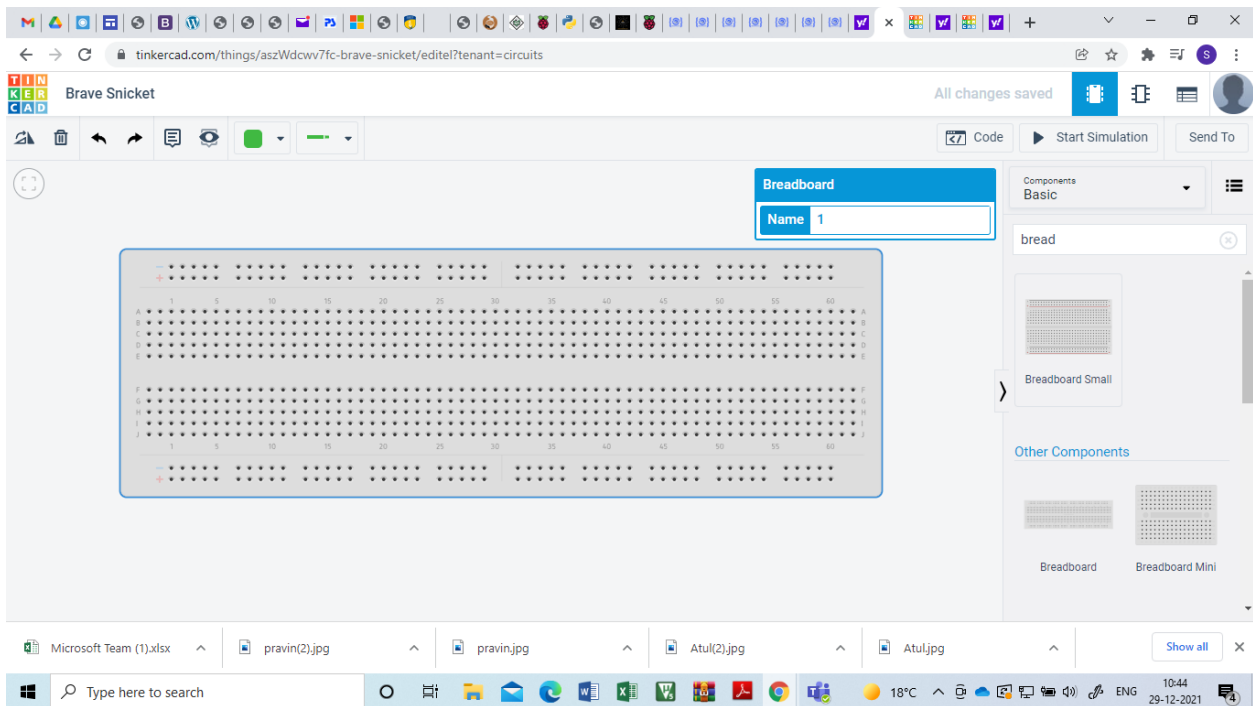


3. Click on new circuit to design new circuit based on electronics hardware.

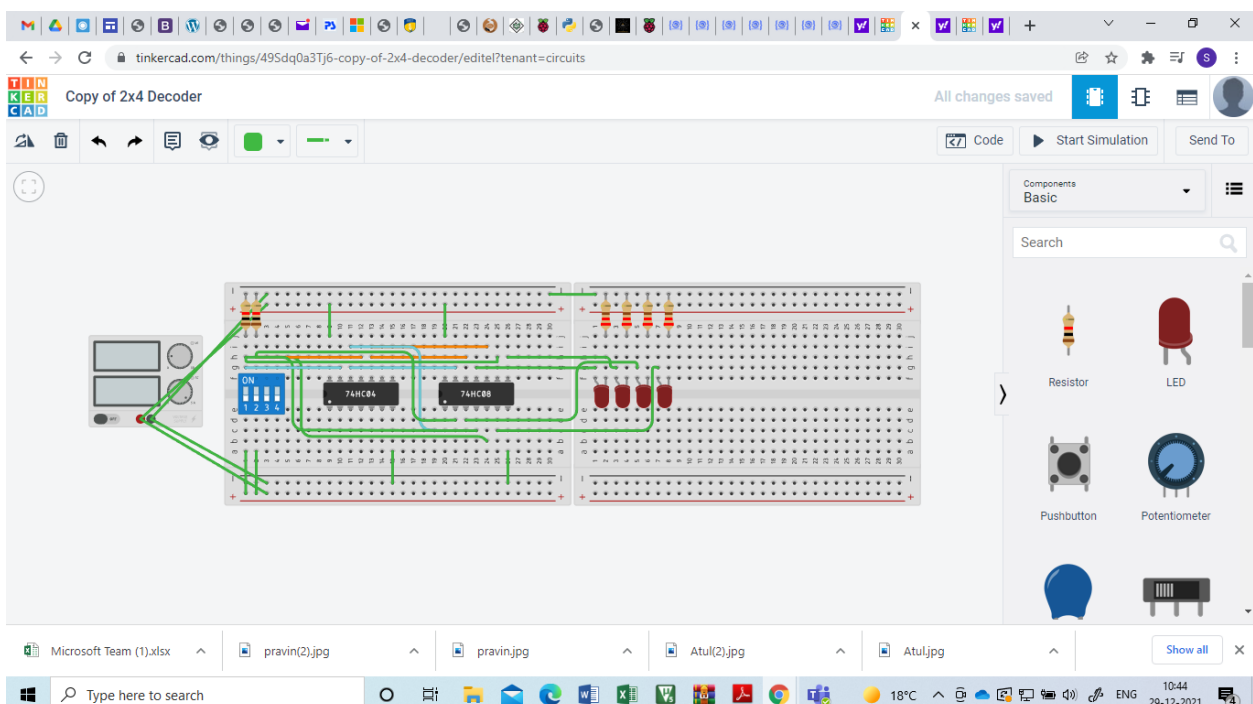


Tool Based Manual || BSCIT || SEM VI || CA615: Embedded Programming using Raspberry PI

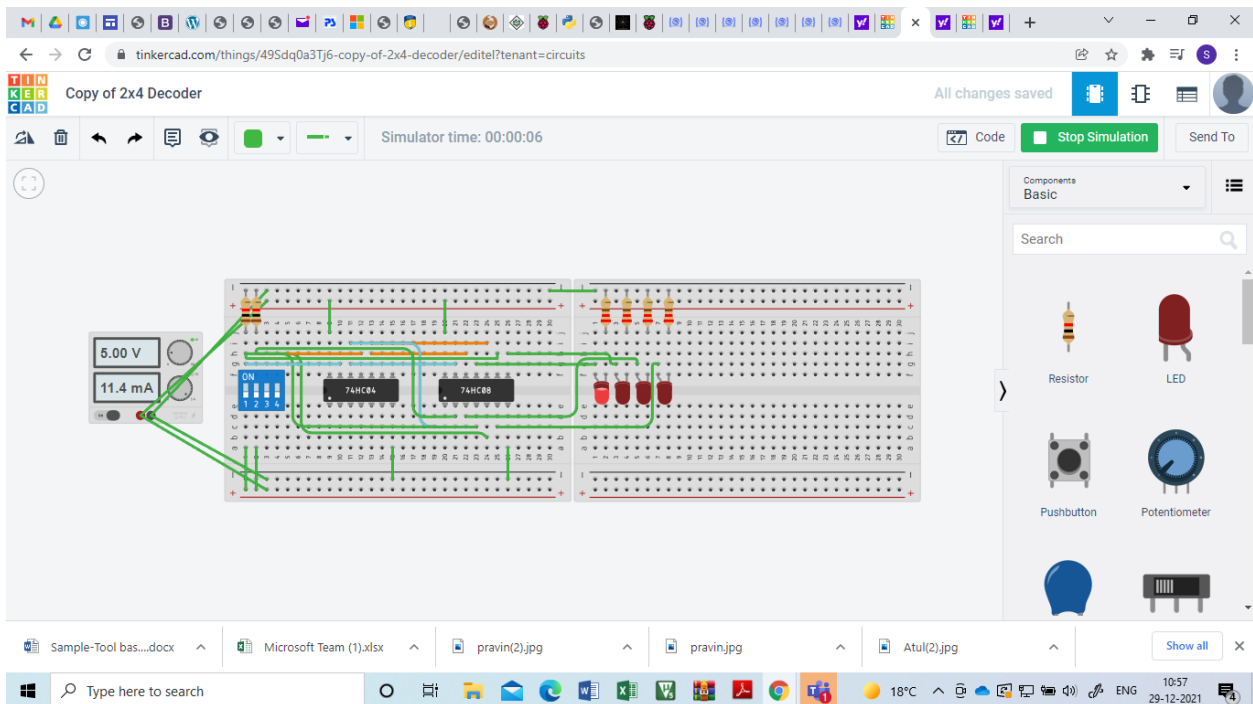
- Once new circuit is opened then find out the desired components and placed on board.



- Design whole circuit and check out proper connections.



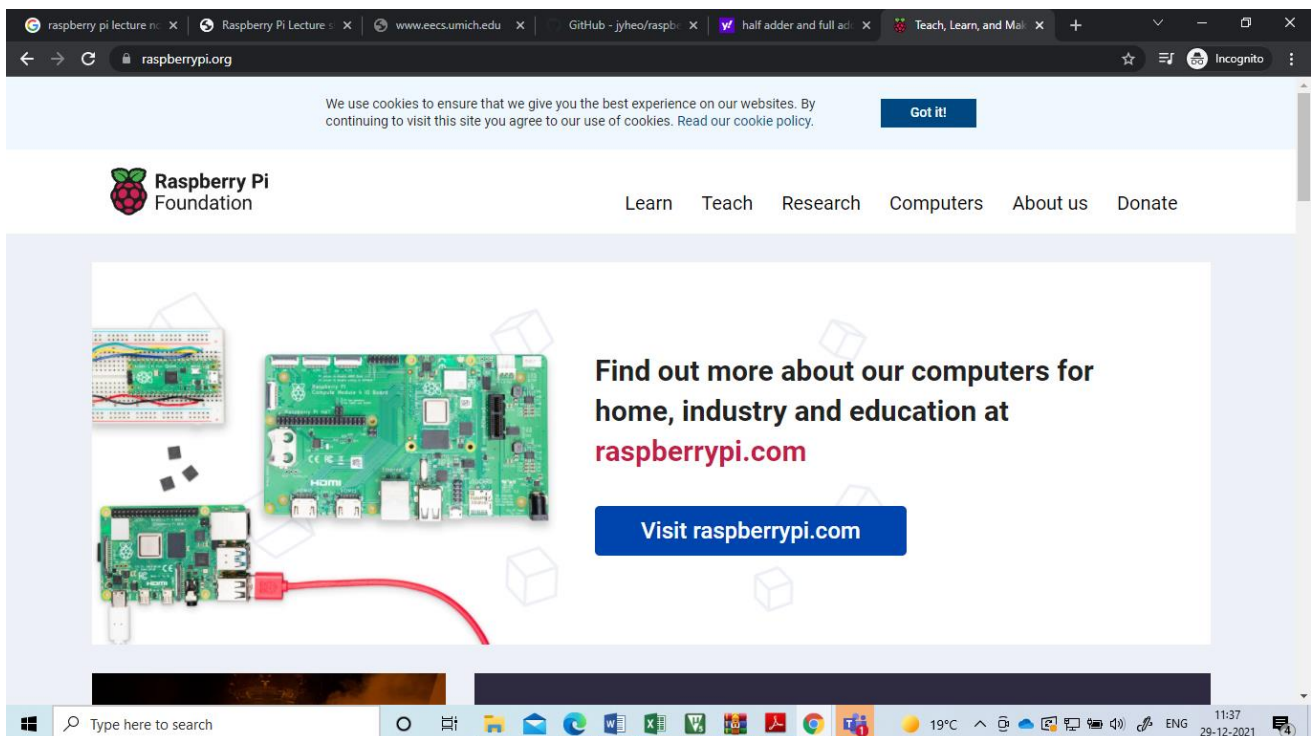
6. Click on start simulation to check the output.



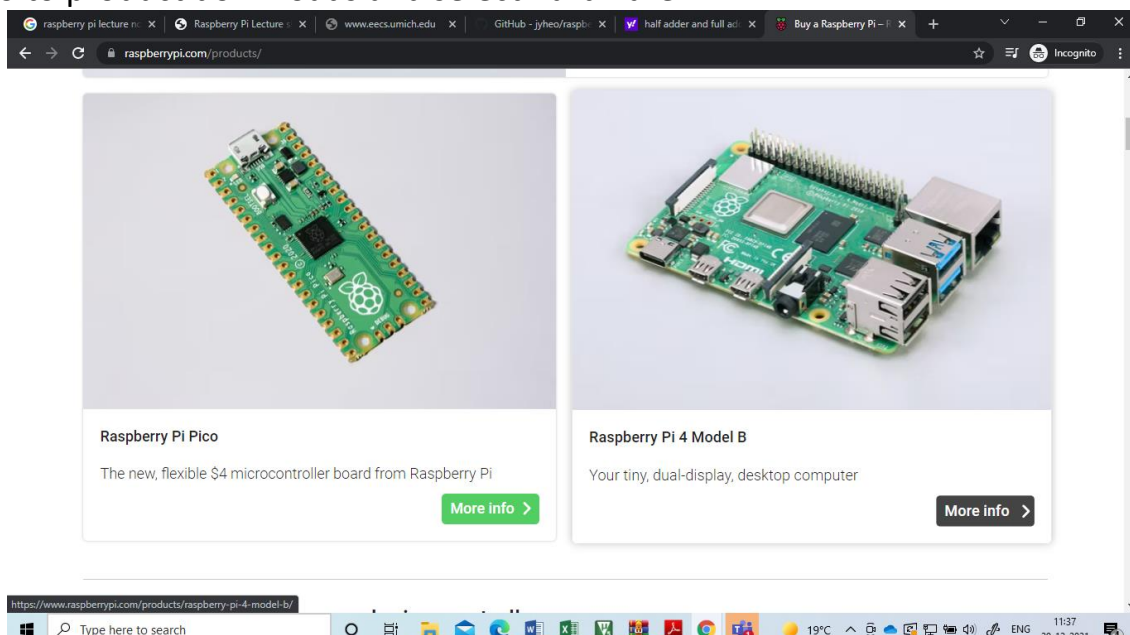
Tool 2:

Working with Raspberry PI

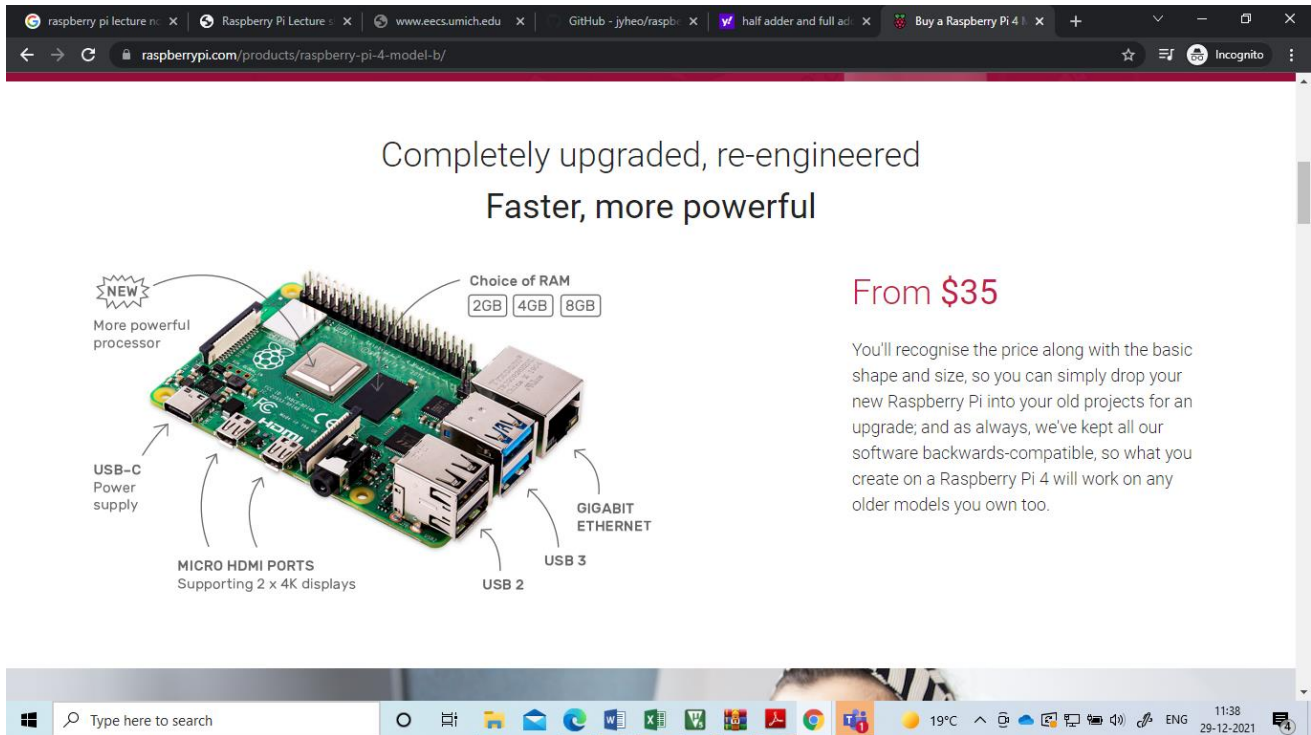
1. Open www.raspberrypi.org



2. Go to product downloads and select hardware.



3. Select Raspberry Pi 4



Completely upgraded, re-engineered
Faster, more powerful

From \$35

You'll recognise the price along with the basic shape and size, so you can simply drop your new Raspberry Pi into your old projects for an upgrade; and as always, we've kept all our software backwards-compatible, so what you create on a Raspberry Pi 4 will work on any older models you own too.

4. Once it is with you then install its OS.



Raspberry Pi OS

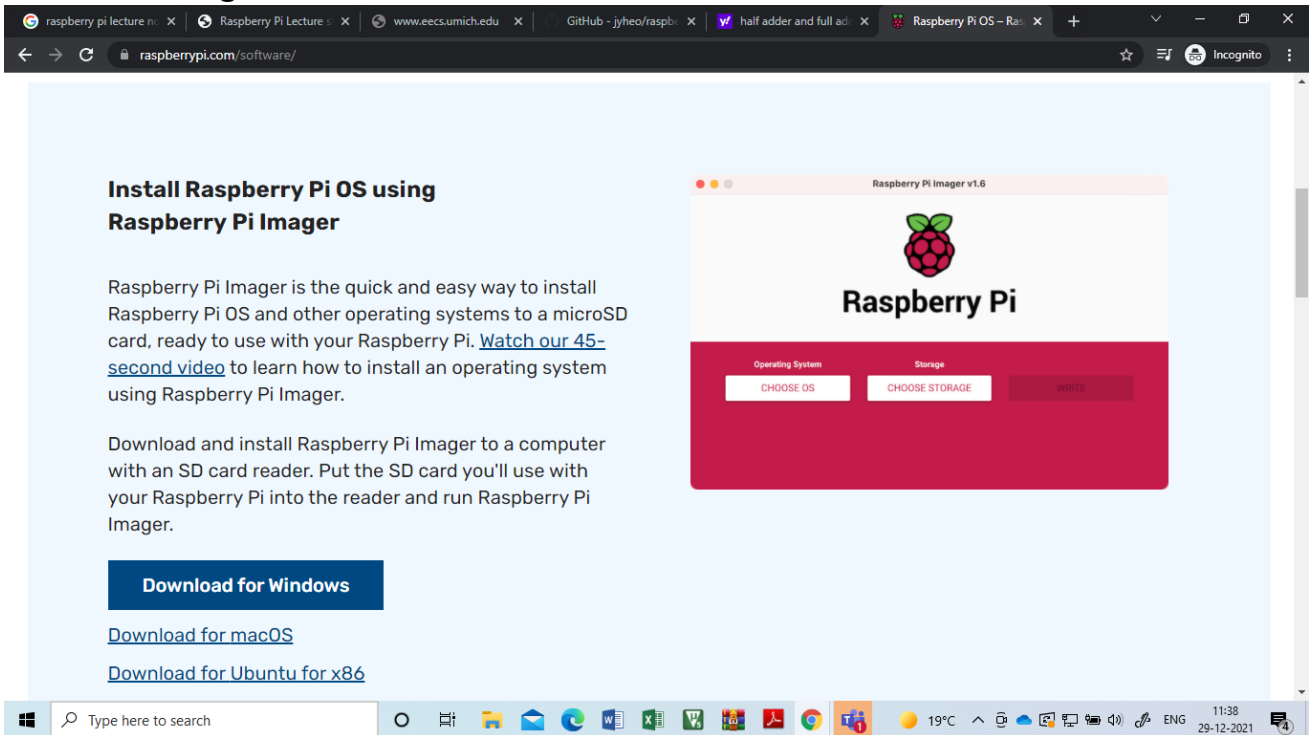
Your Raspberry Pi needs an operating system to work. This is it. Raspberry Pi OS (previously called Raspbian) is our official supported operating system.

Install Raspberry Pi OS using Raspberry Pi Imager

Raspberry Pi Imager is the quick and easy way to install Raspberry Pi OS and other operating systems to a microSD

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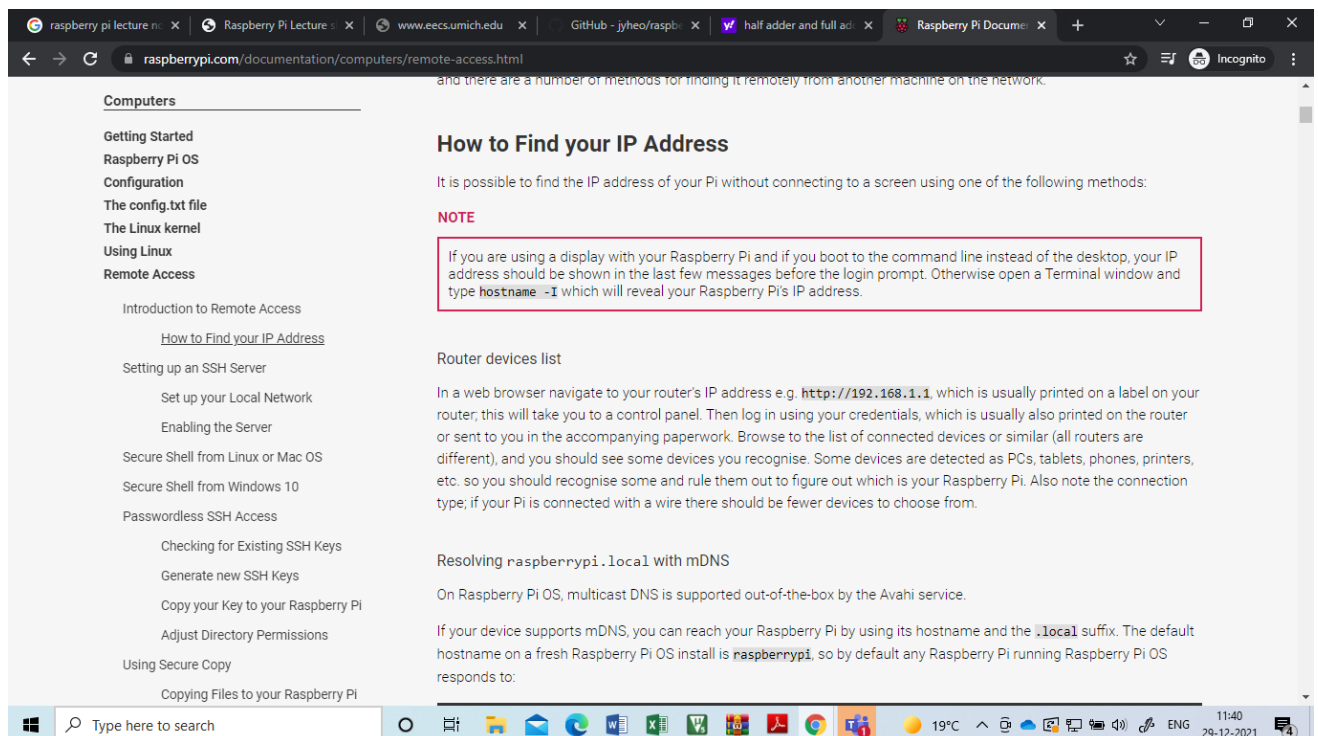
5. Download Raspberry Pi OS for windows or linux for operating system in which you are working with.



The screenshot shows the Raspberry Pi website's software page. The main heading is "Install Raspberry Pi OS using Raspberry Pi Imager". Below this, there is a paragraph explaining that the Raspberry Pi Imager is a quick and easy way to install the OS and other operating systems to a microSD card. A link to a video is provided. Another paragraph instructs users to download and install the Raspberry Pi Imager to a computer with an SD card reader. Below the text, there are three buttons: "Download for Windows", "Download for macOS", and "Download for Ubuntu for x86". To the right of the text is a screenshot of the Raspberry Pi Imager v1.6 application interface, which shows the Raspberry Pi logo and the text "Raspberry Pi". Below the logo are three buttons: "CHOOSE OS", "CHOOSE STORAGE", and "WRITE". The browser's address bar shows "raspberrypi.com/software/". The Windows taskbar at the bottom shows the search bar and various application icons.

6. Install OS on memory chip and connect keyboard , mouse and monitor..

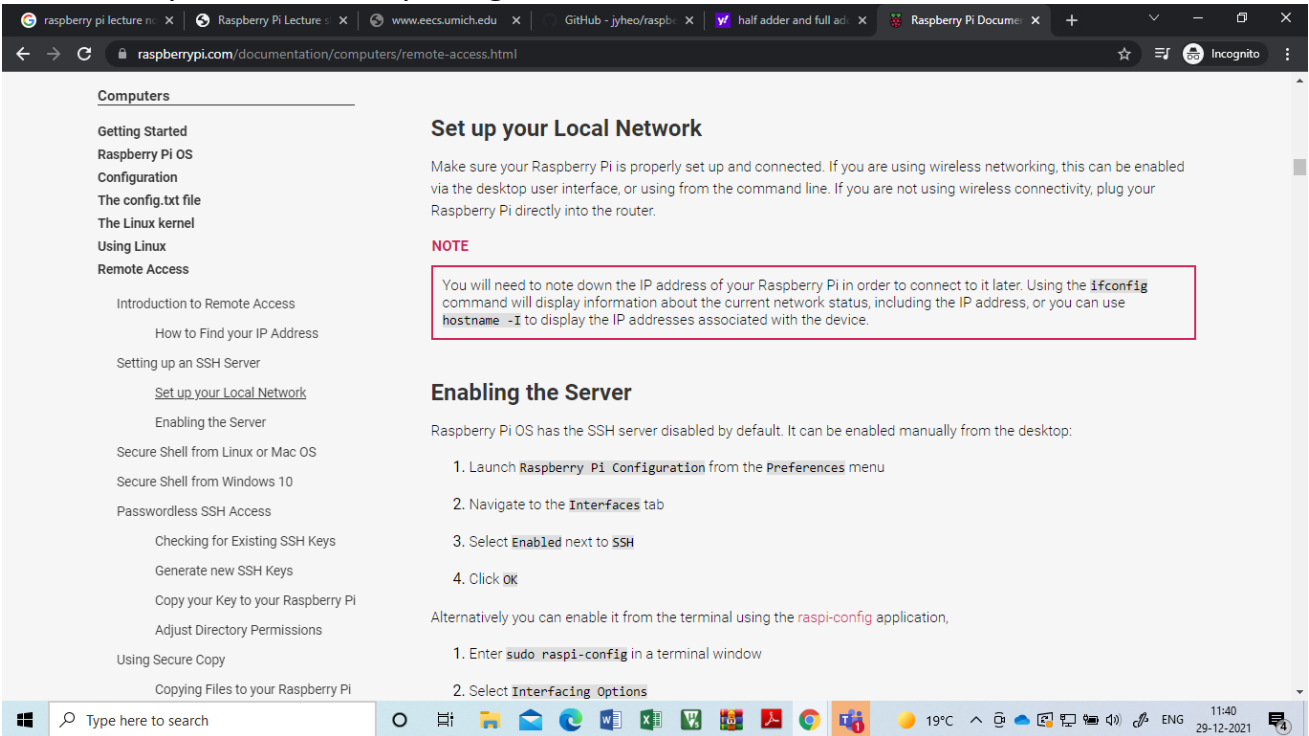
7. Find out IP address of PI.



The screenshot shows the Raspberry Pi documentation page for "How to Find your IP Address". The page is titled "Computers" and has a sidebar with a list of topics: "Getting Started", "Raspberry Pi OS", "Configuration", "The config.txt file", "The Linux kernel", "Using Linux", and "Remote Access". The "Remote Access" section is expanded, showing a list of topics: "Introduction to Remote Access", "How to Find your IP Address", "Setting up an SSH Server", "Set up your Local Network", "Enabling the Server", "Secure Shell from Linux or Mac OS", "Secure Shell from Windows 10", "Passwordless SSH Access", "Checking for Existing SSH Keys", "Generate new SSH Keys", "Copy your Key to your Raspberry Pi", "Adjust Directory Permissions", "Using Secure Copy", and "Copying Files to your Raspberry Pi". The main content area is titled "How to Find your IP Address" and contains a paragraph explaining that it is possible to find the IP address of your Pi without connecting to a screen. A "NOTE" box contains the text: "If you are using a display with your Raspberry Pi and if you boot to the command line instead of the desktop, your IP address should be shown in the last few messages before the login prompt. Otherwise open a Terminal window and type `hostname -I` which will reveal your Raspberry Pi's IP address." Below the note is a section titled "Router devices list" which explains how to find the IP address of your router. The bottom of the page shows the Windows taskbar with the search bar and various application icons.

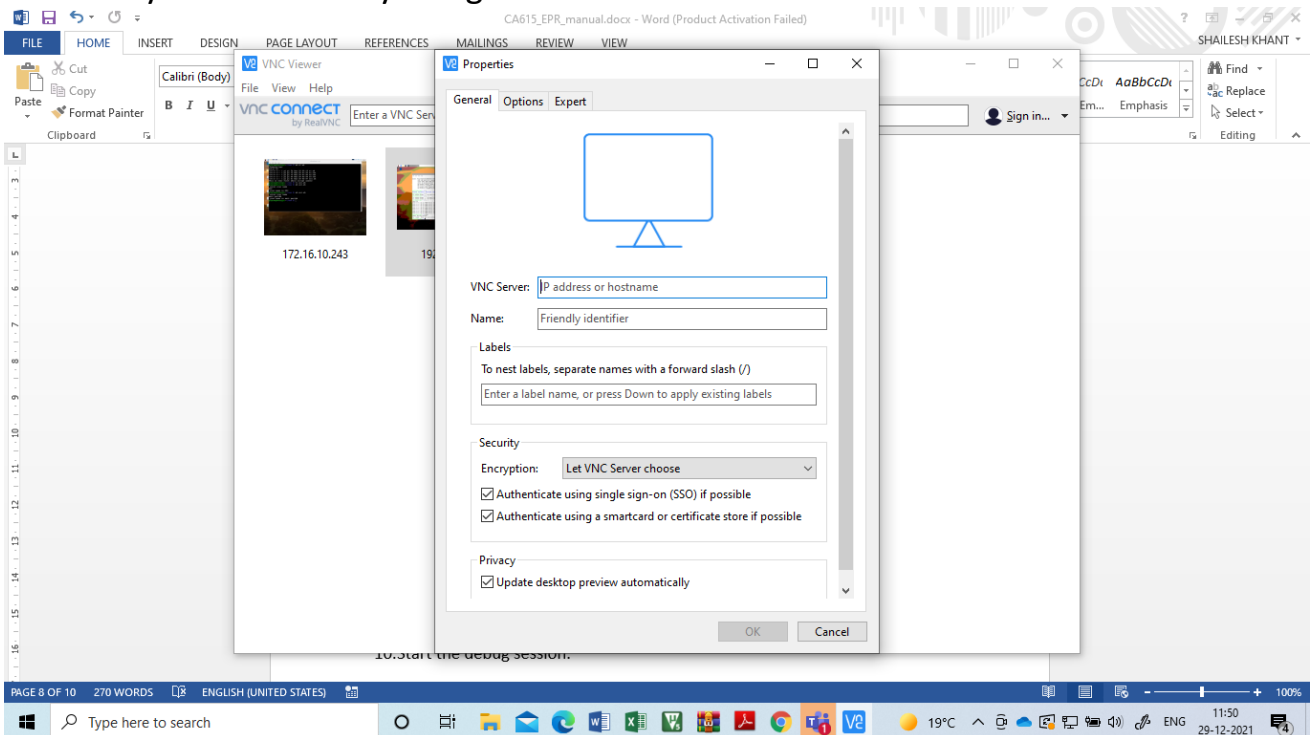
Tool Based Manual || BSCIT || SEM VI || CA615: Embedded Programming using Raspberry PI

8. Access your PI remotely using SSH.



The screenshot shows the Raspberry Pi documentation website in a web browser. The left sidebar lists various topics under 'Computers', including 'Getting Started', 'Raspberry Pi OS', 'Configuration', 'The config.txt file', 'The Linux kernel', 'Using Linux', and 'Remote Access'. The main content area is titled 'Set up your Local Network' and includes a 'NOTE' box stating: 'You will need to note down the IP address of your Raspberry Pi in order to connect to it later. Using the `ifconfig` command will display information about the current network status, including the IP address, or you can use `hostname -I` to display the IP addresses associated with the device.' Below this, the 'Enabling the Server' section explains that the SSH server is disabled by default and provides a list of steps to enable it: 1. Launch Raspberry Pi Configuration from the Preferences menu, 2. Navigate to the Interfaces tab, 3. Select Enabled next to SSH, 4. Click OK. It also mentions an alternative method using the `raspi-config` application in a terminal window.

9. Access your PI remotely using VNC Viewer.



The screenshot shows a Windows desktop with a Microsoft Word document open in the background. In the foreground, the VNC Viewer application is running, displaying a remote desktop session. The VNC Viewer's 'Properties' dialog box is open, showing the 'General' tab. The 'VNC Server' field is set to 'IP address or hostname', and the 'Name' field is set to 'Friendly identifier'. The 'Security' section shows 'Encryption' set to 'Let VNC Server choose', with checkboxes for 'Authenticate using single sign-on (SSO) if possible' and 'Authenticate using a smartcard or certificate store if possible' both checked. The 'Privacy' section has 'Update desktop preview automatically' checked. The 'Labels' section is also visible, showing a list of labels and a field to enter a new label.

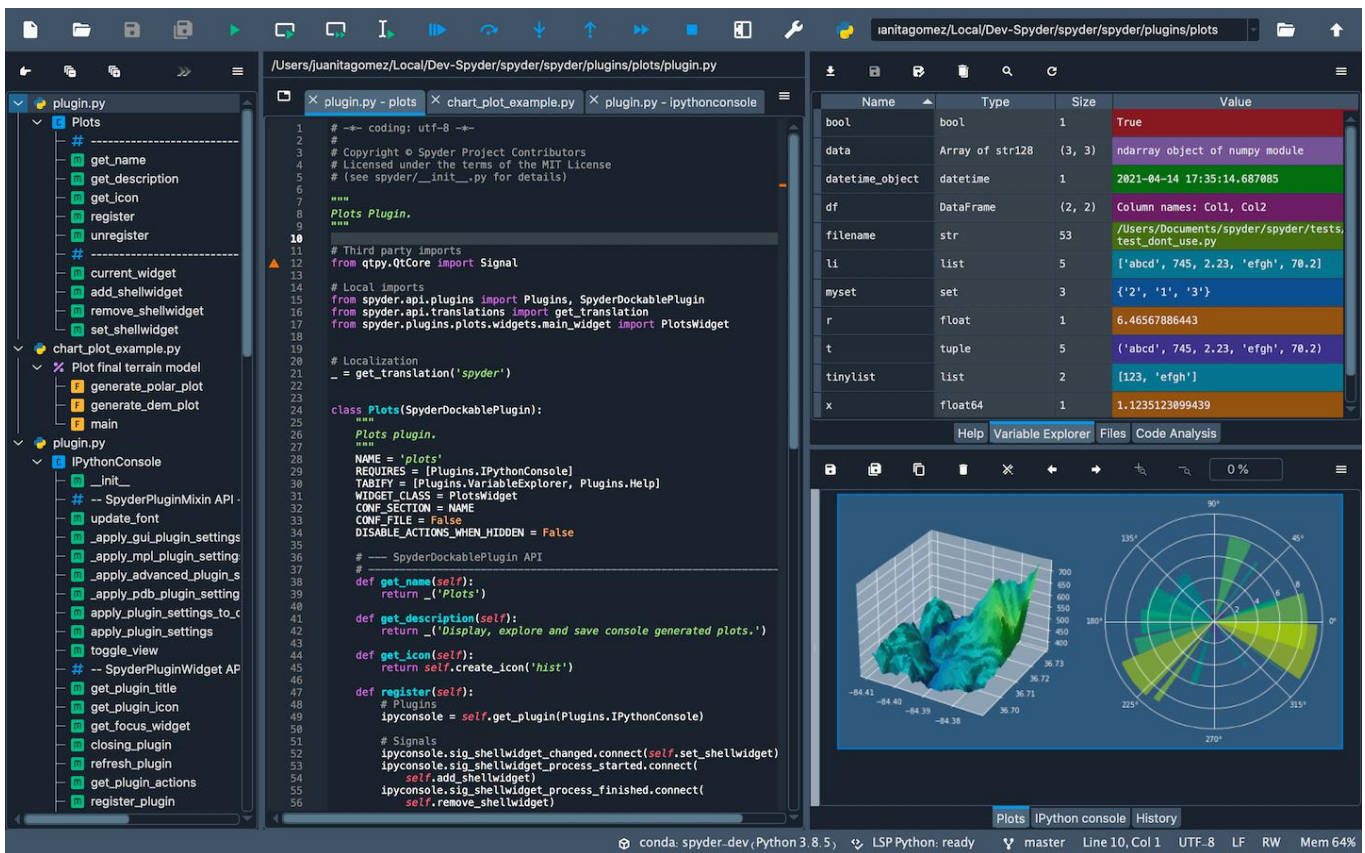
10. Start working with raspberry PI projects.

11. Use python for programming hardware using raspberry PI.

12. Install the packages as needed by the projects.

Tool 3:

Welcome to Spyder (Python Interpreter)



The screenshot shows the Spyder Python IDE interface. The central code editor displays the 'plugin.py' file, which is a Spyder plugin for the 'plots' package. The code includes imports for Qt, QtCore, and QtOpenGL, and defines a 'Plots' class that inherits from 'SpyderDockablePlugin'. The class implements methods for getting the name, description, icon, and registering the plugin. The right sidebar shows the 'Variable Explorer' with a table of variables and their values. The bottom panel displays a 3D surface plot and a polar plot.

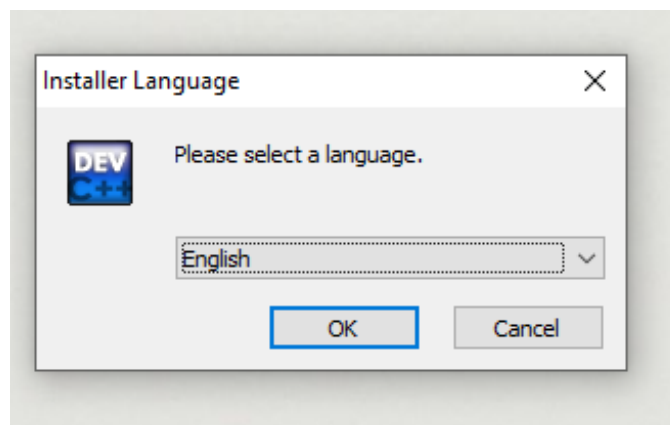
Name	Type	Size	Value
bool	bool	1	True
data	Array of str128	(3, 3)	ndarray object of numpy module
datetime_object	datetime	1	2021-04-14 17:35:14.687085
df	DataFrame	(2, 2)	Column names: Col1, Col2
filename	str	53	/Users/Documents/spyder/spyder/tests/test_dont_use.py
li	list	5	['abcd', 745, 2.23, 'efgh', 70.2]
myset	set	3	{'2', '1', '3'}
r	float	1	6.46567886443
t	tuple	5	('abcd', 745, 2.23, 'efgh', 70.2)
tinylist	list	2	[123, 'efgh']
x	float64	1	1.1235123899439

Spyder is a powerful scientific environment written in Python, for Python, and designed by and for scientists, engineers and data analysts. It features a unique combination of the advanced editing, analysis, debugging, and profiling functionality of a comprehensive development tool with the data exploration, interactive execution, deep inspection, and beautiful visualization capabilities of a scientific package.

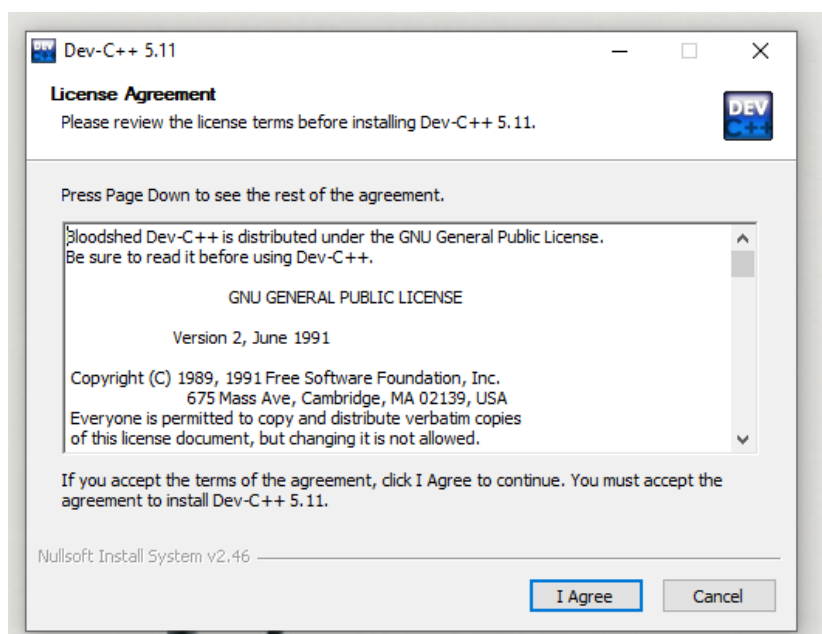
Installation and Getting Started under Windows OS

Follow the steps mentioned in following figure to install Dev c++ on Windows Operating System.

1. Go to <https://sourceforge.net/projects/orwelldvcpp/>
2. Click on Download and wait for five seconds.
3. Double click on Dev-Cpp 5.11 TDM-GCC 4.9.2 Setup file

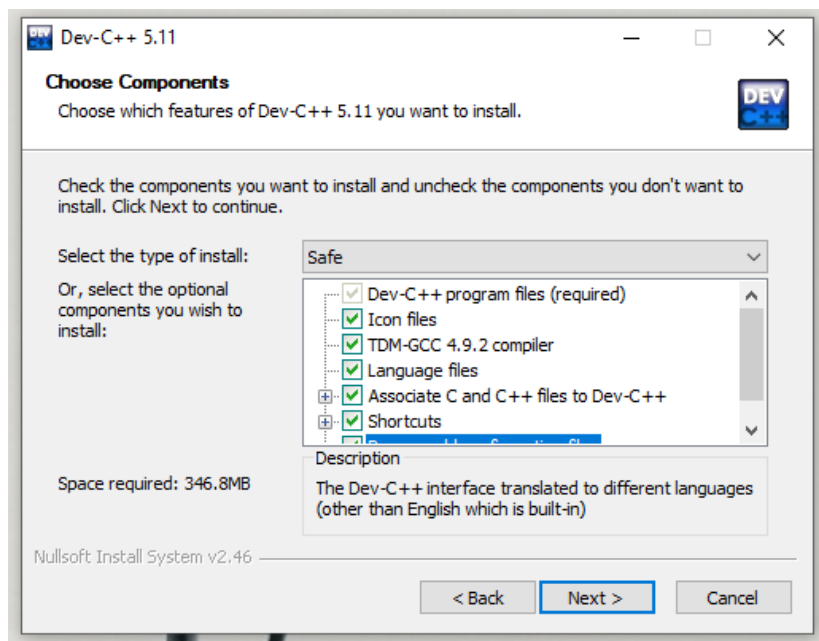


4. Click On Ok.

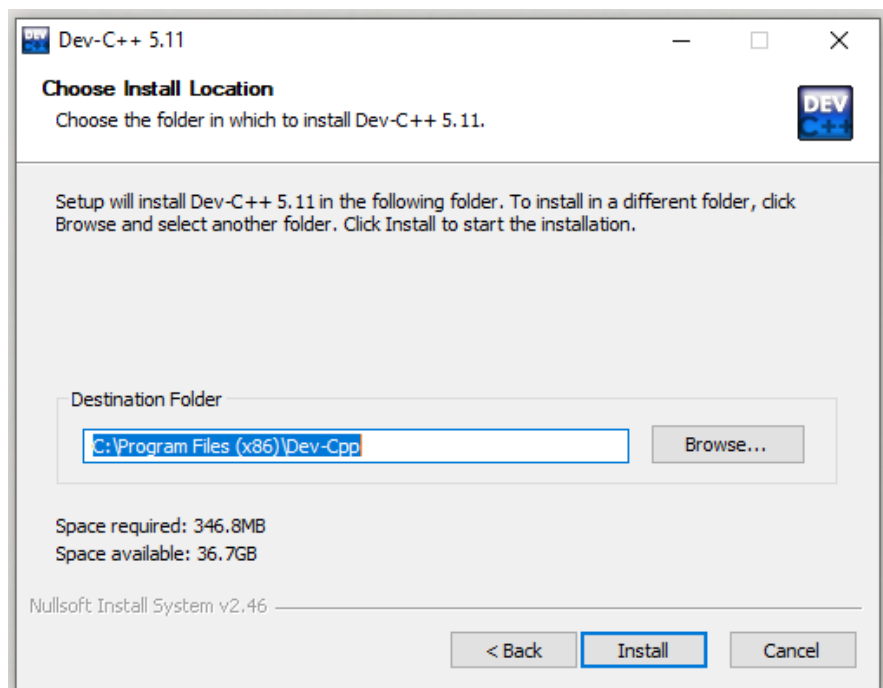


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5. Click On I Agree.

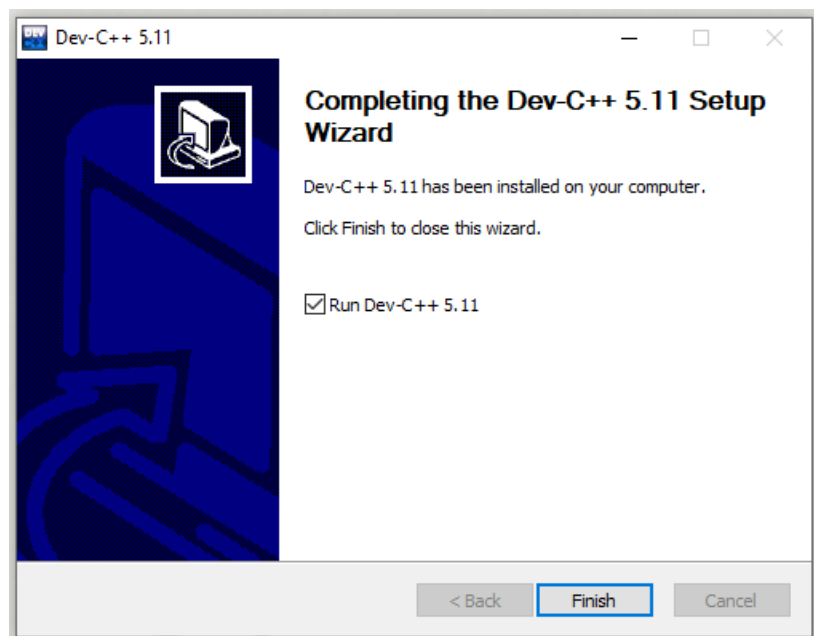


6. Click on Next.

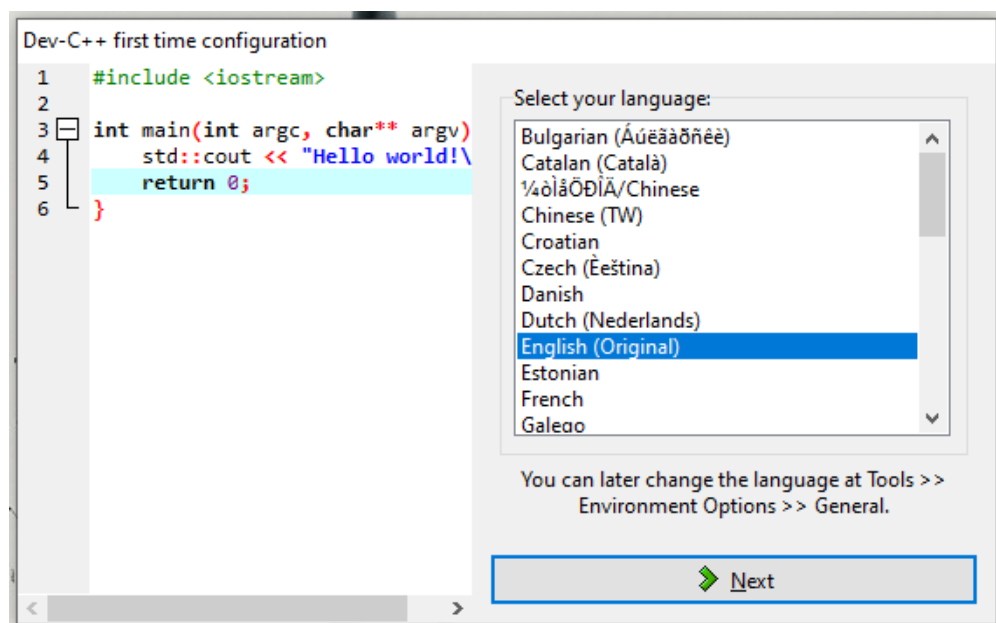


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7. Click on Install.

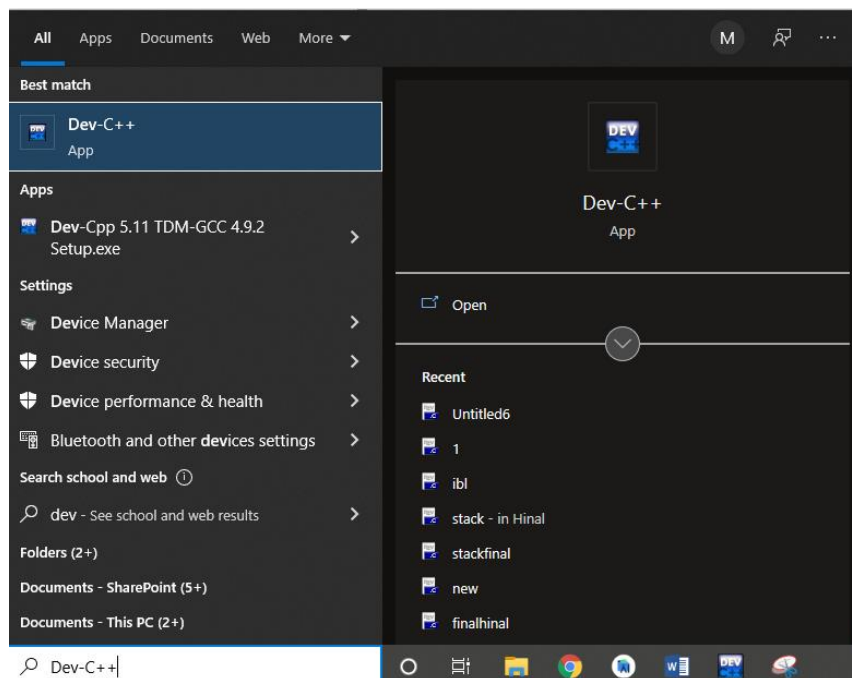


8. Click on finish.

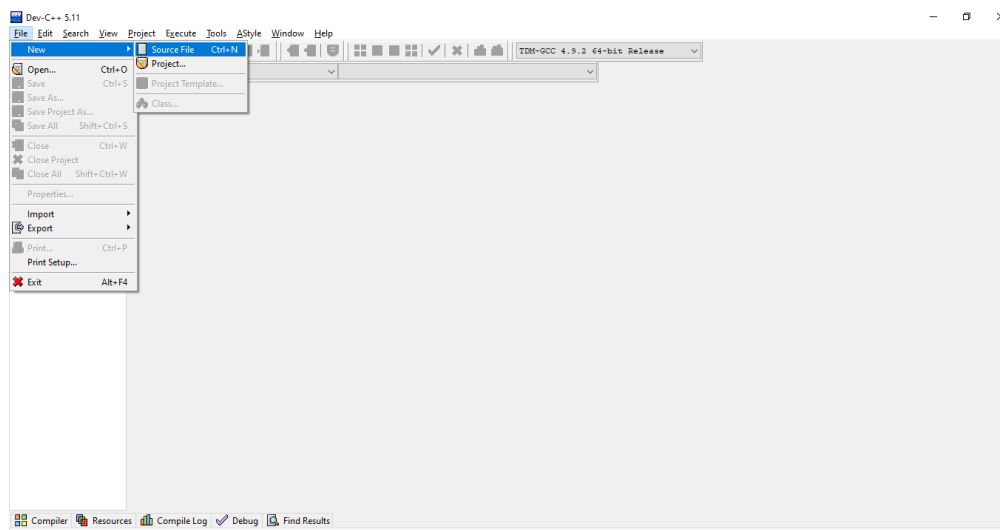


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9. Click on Next, Next Okay.

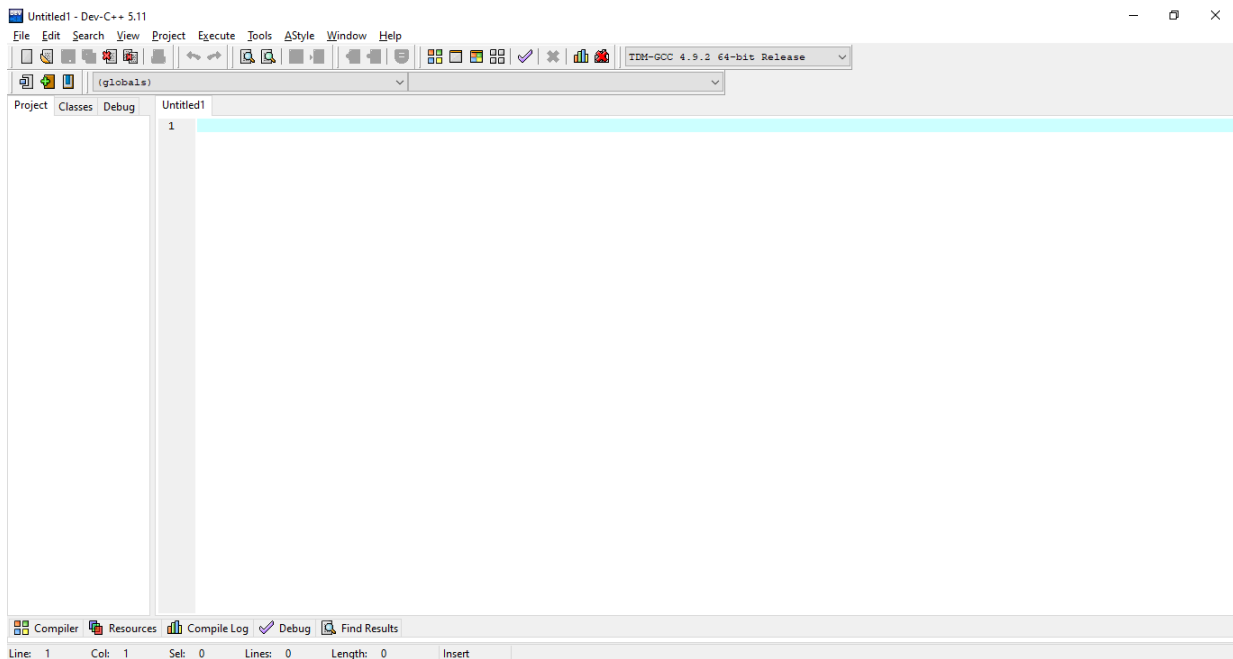


10. Open Dev C++.

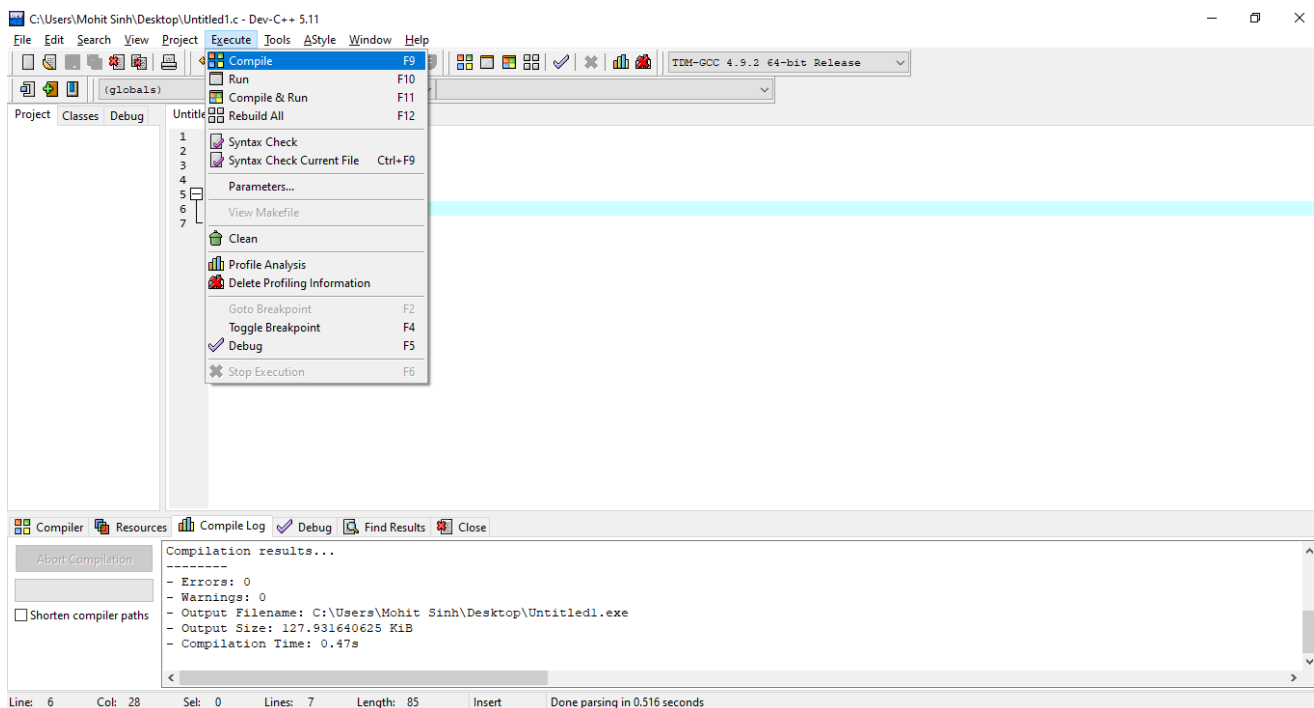


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11. IDE of Dev C++.

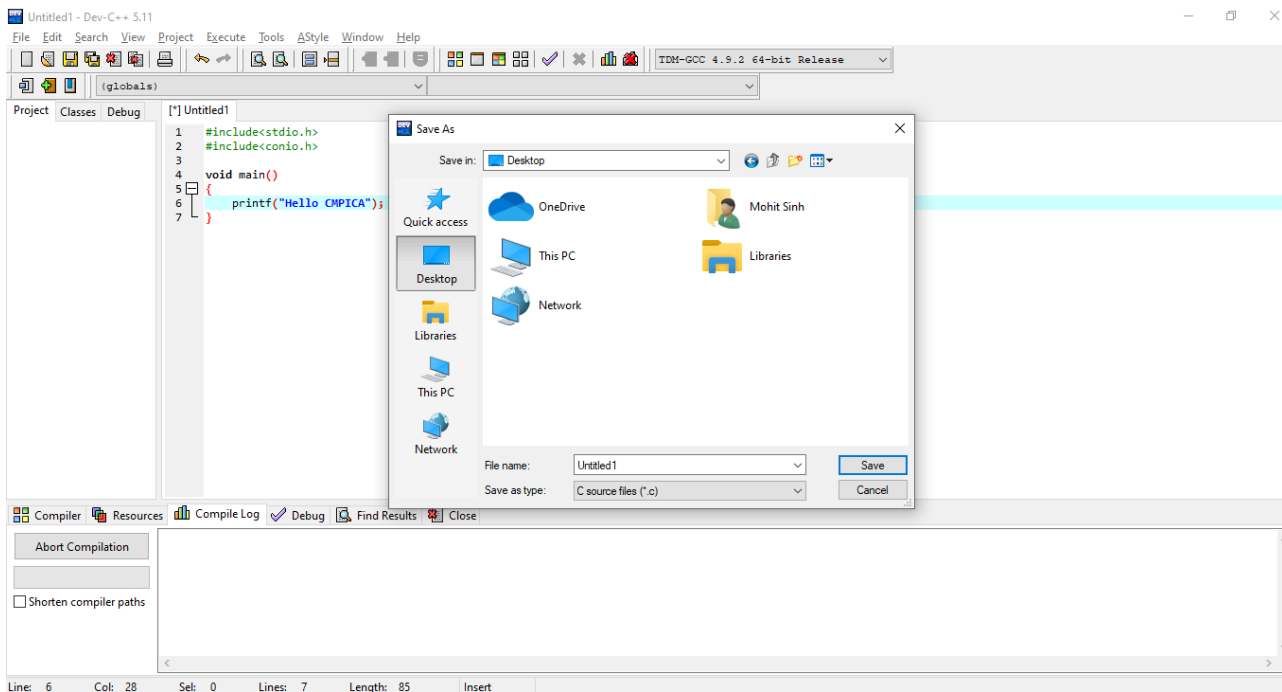


12. New file for c programming.

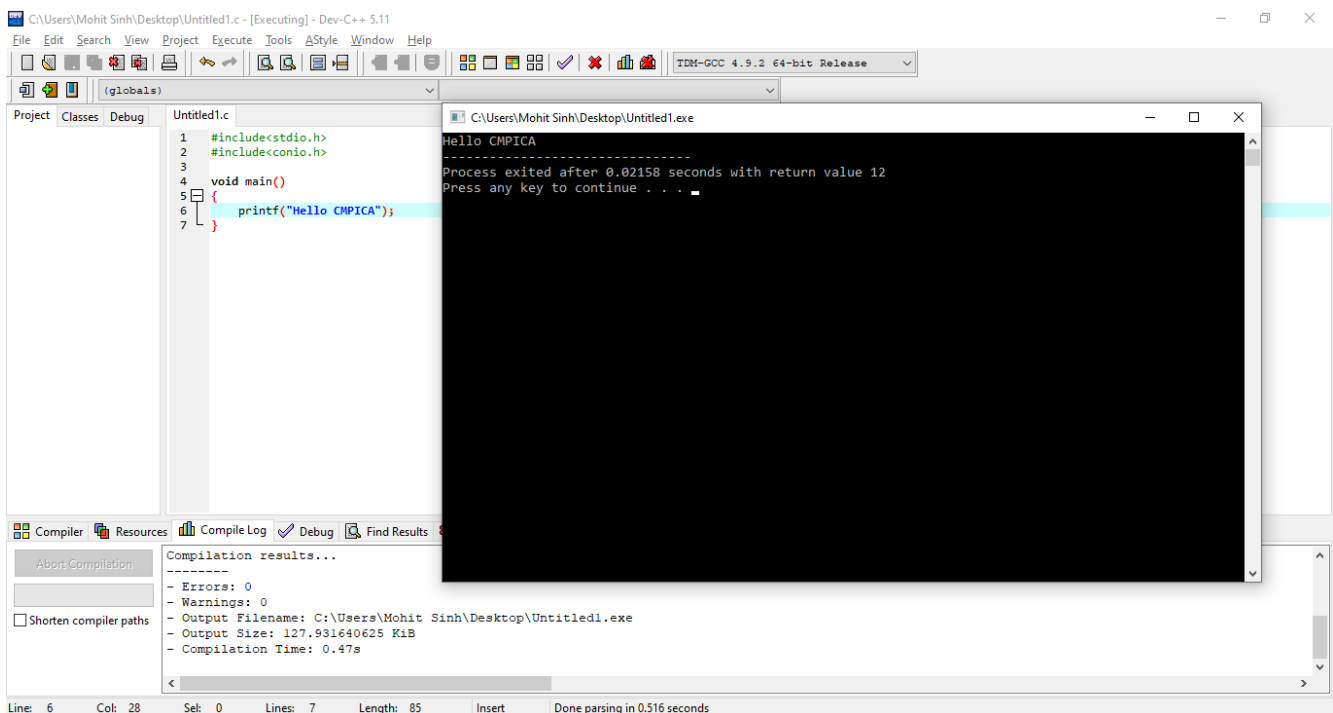


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13. From the execute tab of menu bar you have to select compile option.



14. To run the program, save the file with appropriate name and extension.



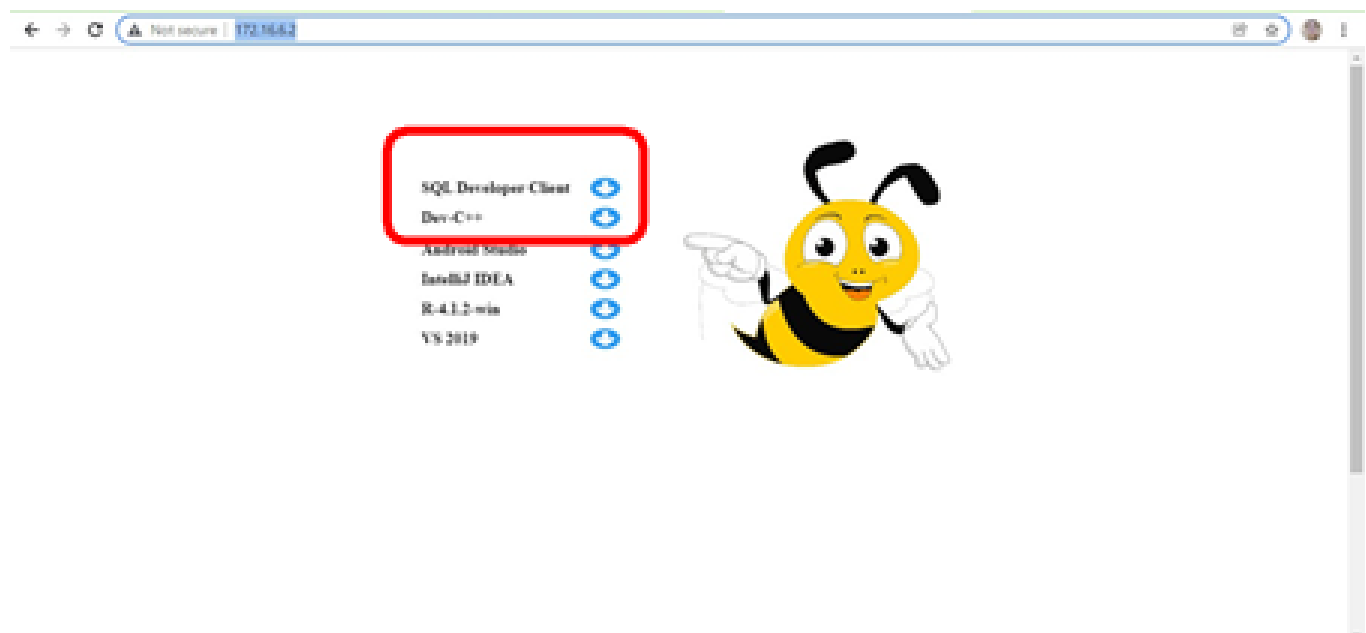
The black screen window is the console window that display the output of the program.

Objectives

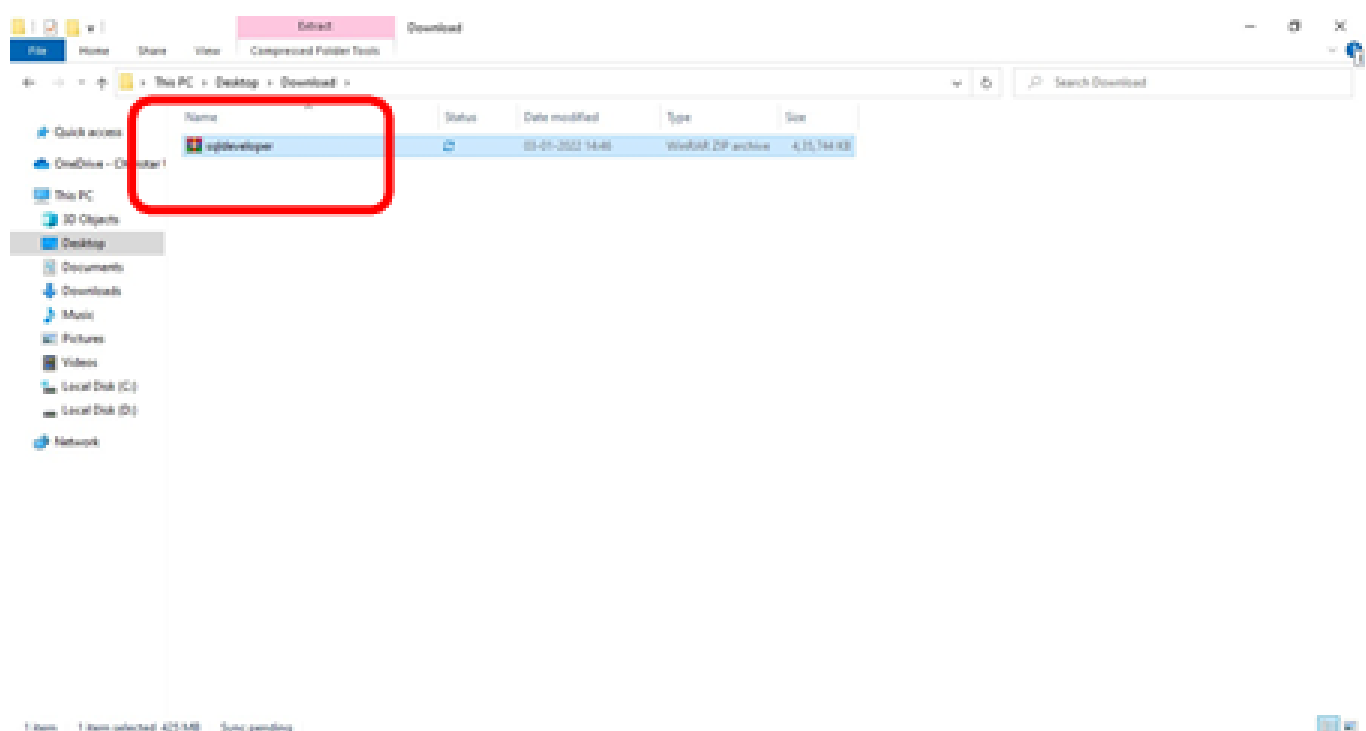
- To make students acquaint with installation of Database Client Software.

Step: 1 Go to <http://172.16.6.2/>

Download SQL Developer Client



This will download the zip with name sqldeveloper. Unzip it.





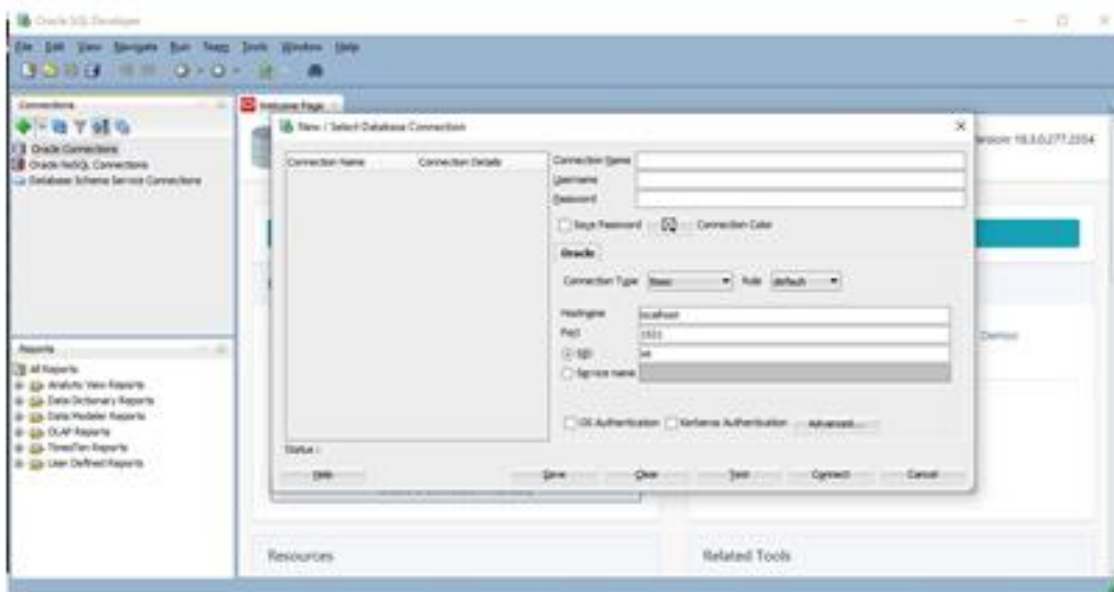
CHARUSAT

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY



Tool Based Manual || BCA || SEM II || CA130 : Database Fundamentals

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Step: 4 Give Proper Connection Name for example CMPICA

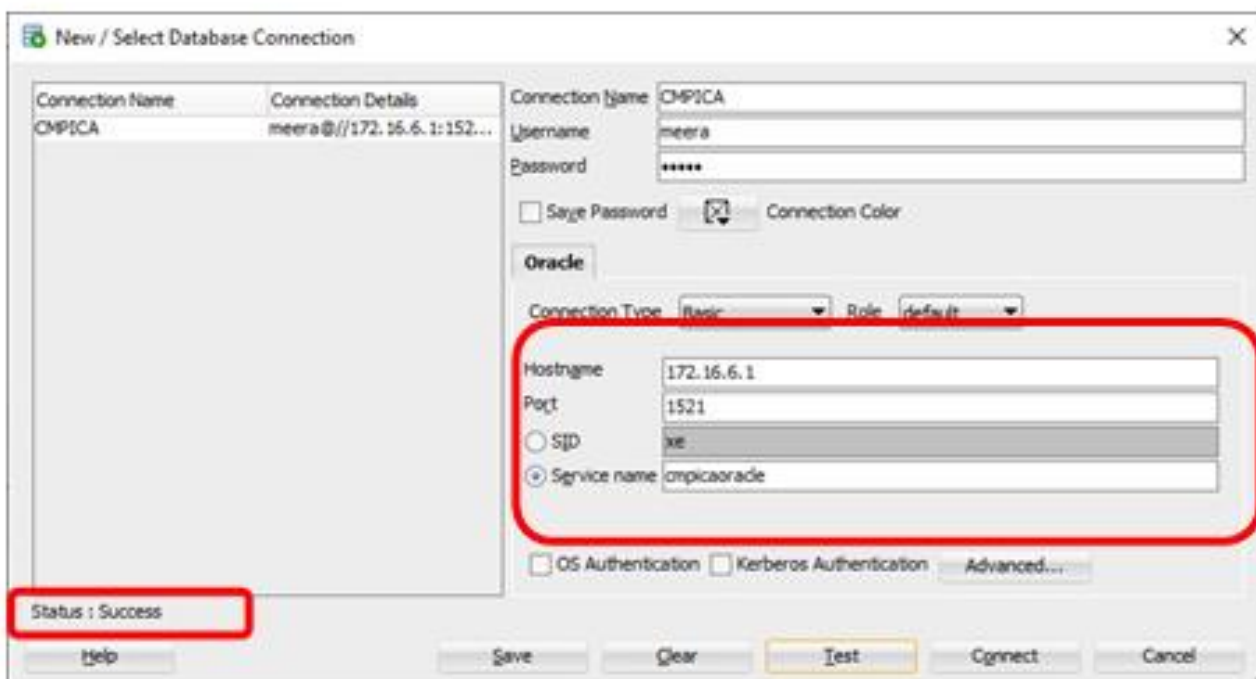
Enter Username and Password provided to you in respective fields.

In Hostname write **172.16.6.1**

Click service name radio button and enter **cmpicaoracle**

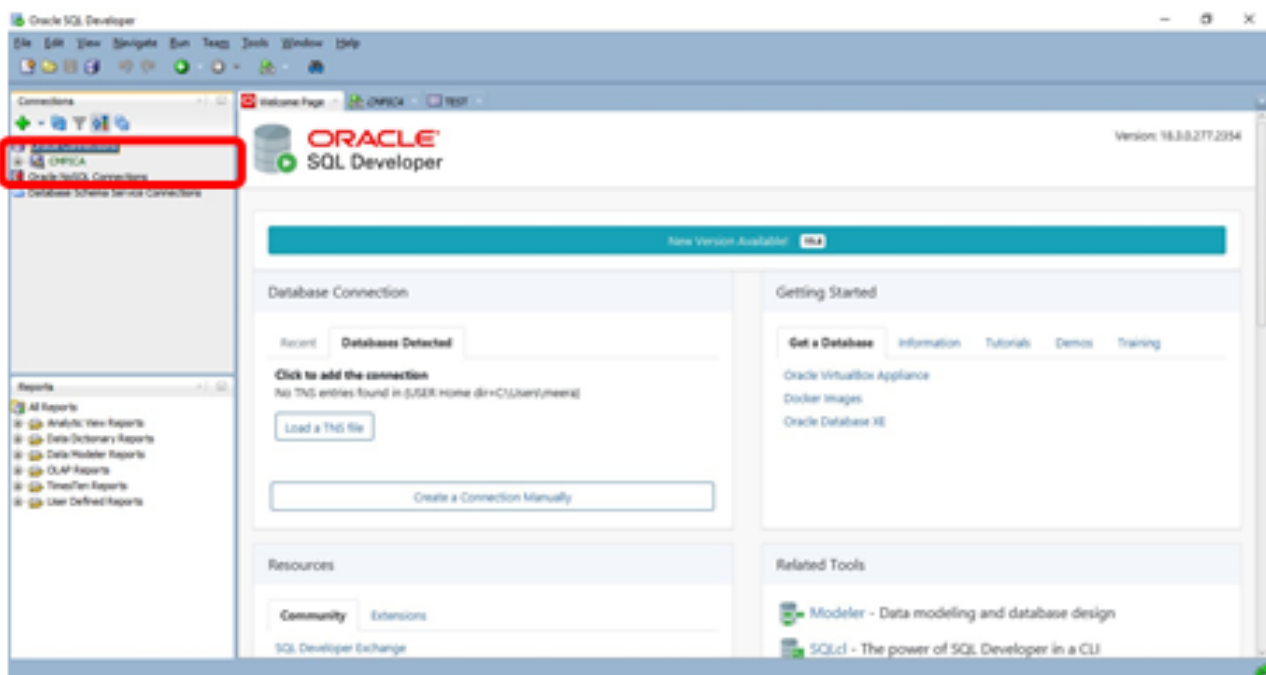
KEEP REST OF THE CONFIGURATION AS IT IS. And Click on Test button. The **SUCCESS** should get displayed.

Now click on Connect Button.

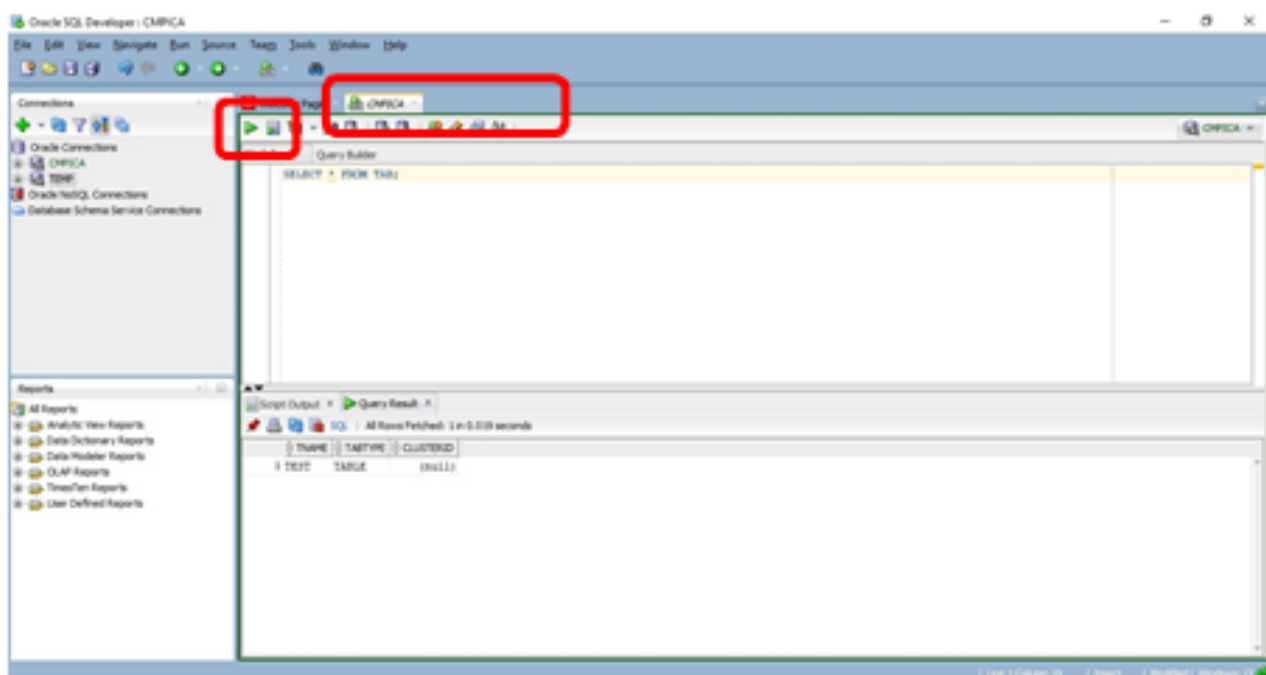


Your newly created connection should get displayed on the left panel.

Tool Based Manual || BCA || SEM II || CA130 : Database Fundamentals



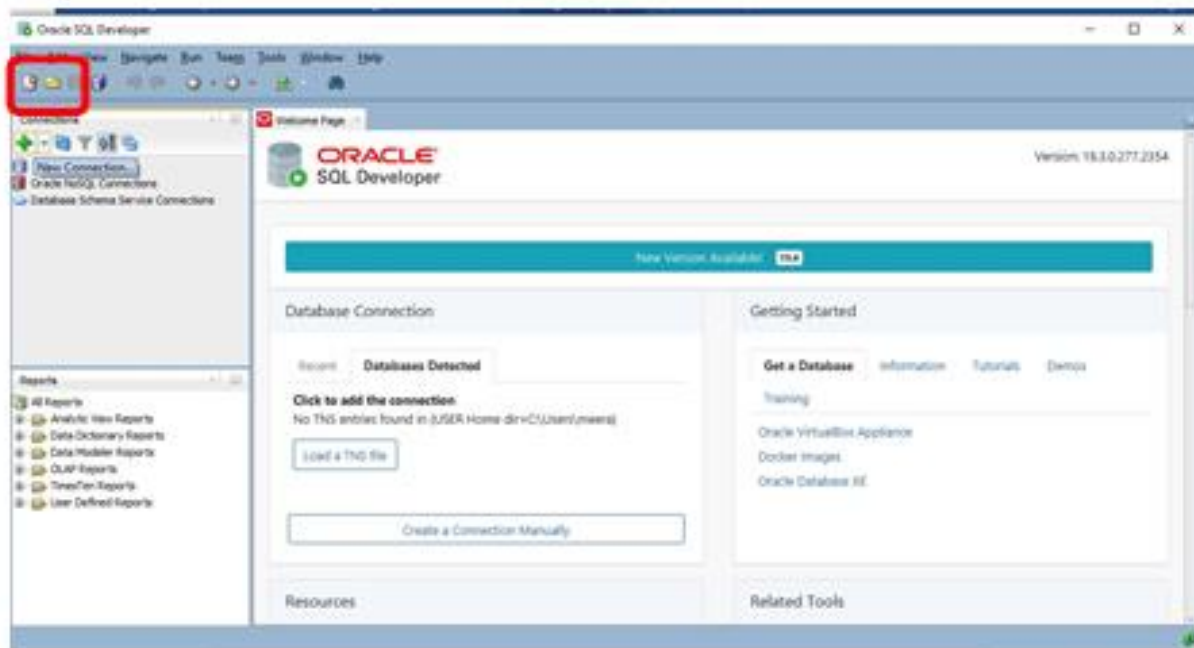
Step 5 On the right side panel observe the Tab with CMPICA (Connection Name). Write your query and Run your query (execute) using  icon **OR** press Ctrl + Enter.



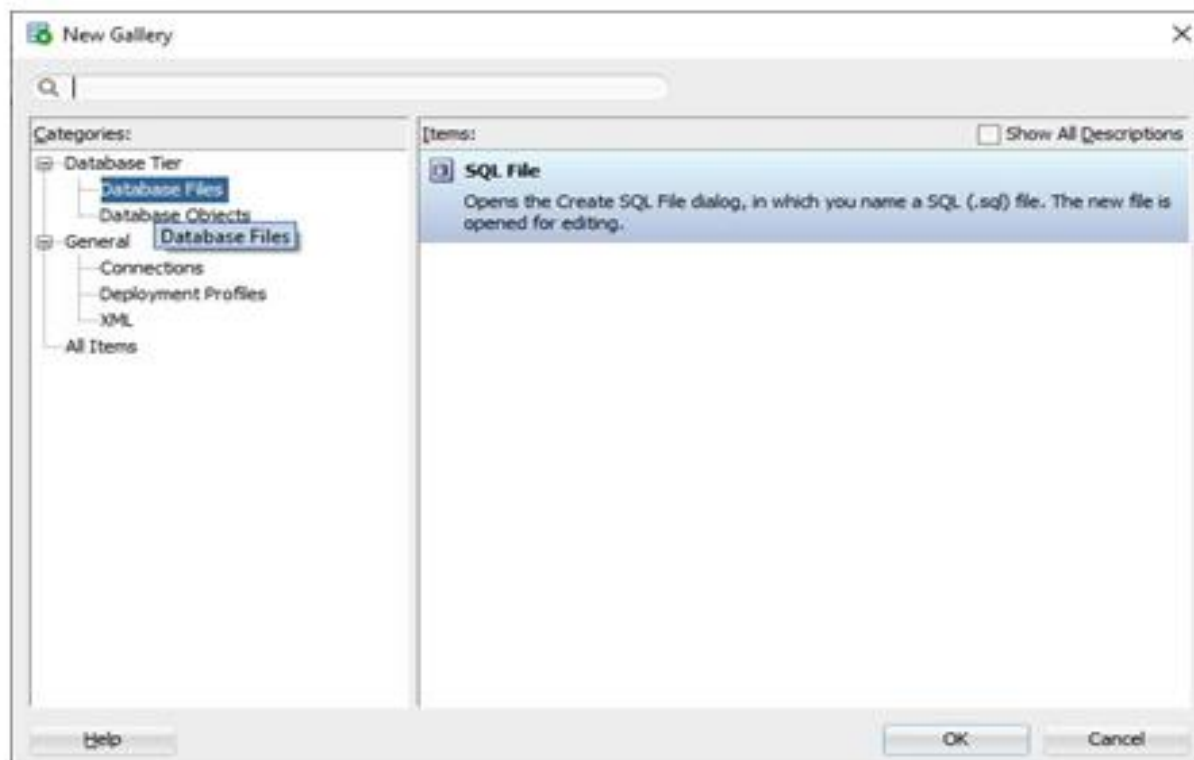
Step 6 IF you want to create a new SQL file then

In the right panel click on File -> New symbol,  or press Ctrl + N and Then select database -> SQL File and click OK. Please refer below screenshots.

Tool Based Manual || BCA || SEM II || CA130 : Database Fundamentals

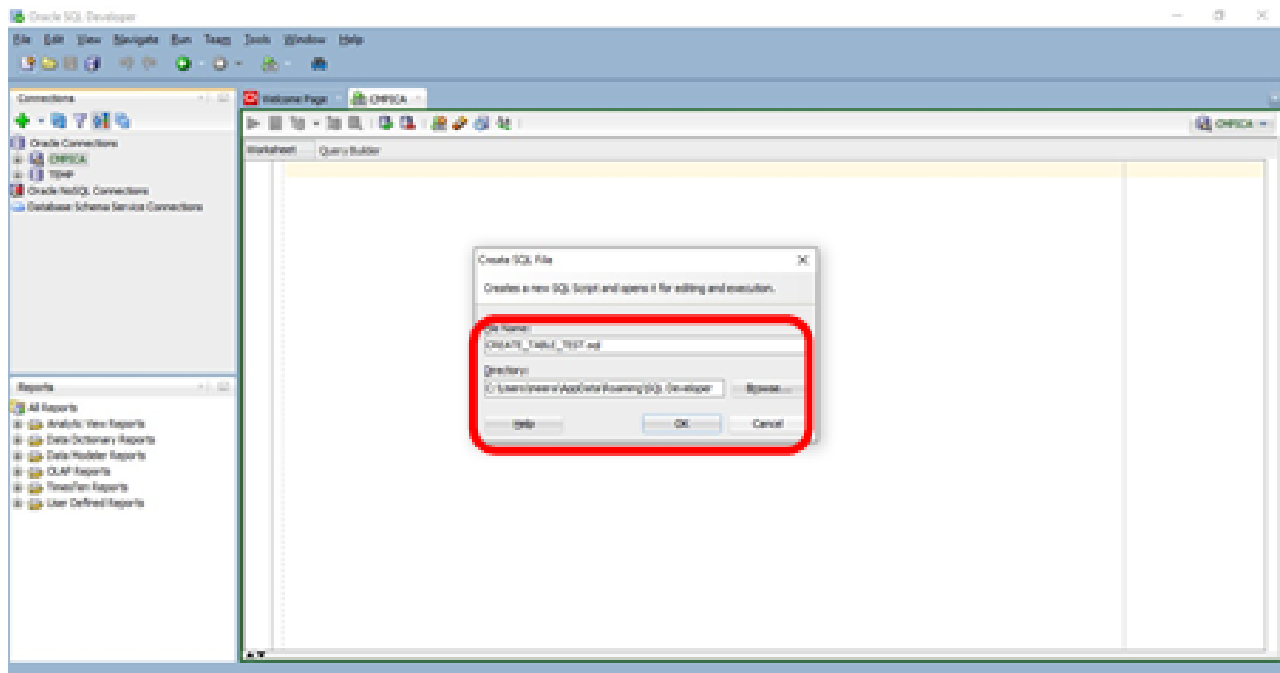


The following screen will appear.



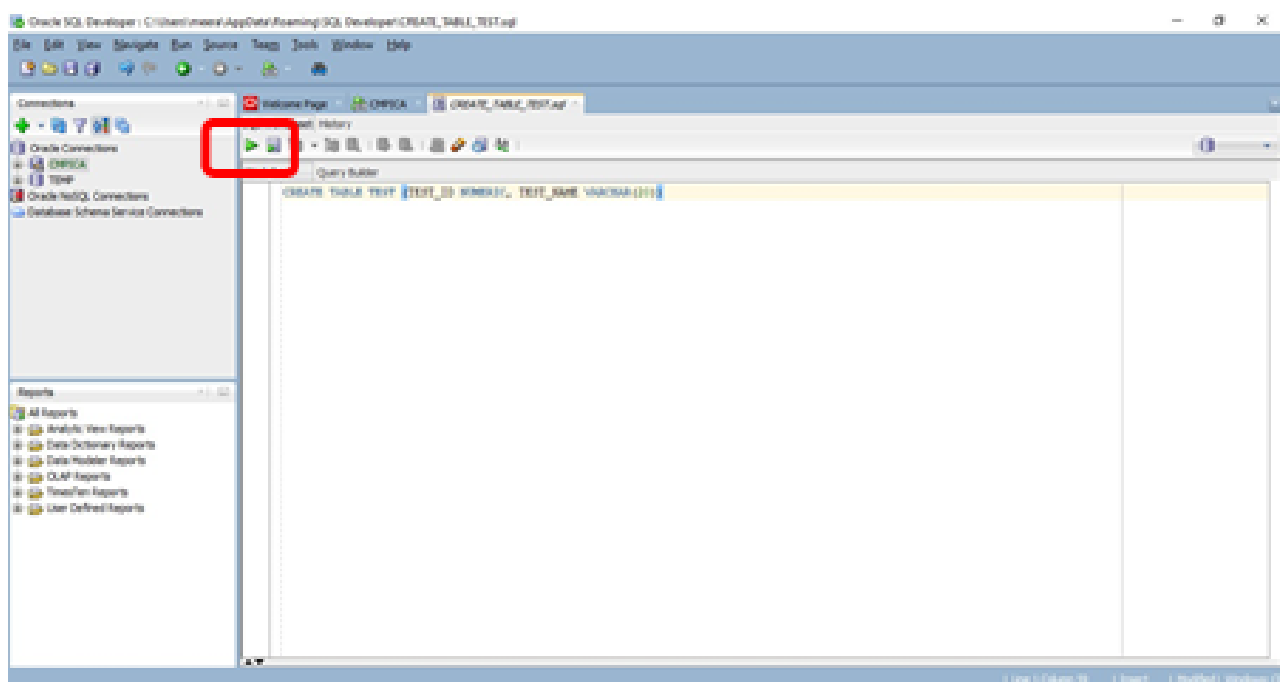
Give proper file name and save to the proper location. [IT IS ADVISED TO SAVE ALL THE SQL IN ONE FOLDER]

Tool Based Manual || BCA || SEM II || CA130 : Database Fundamentals



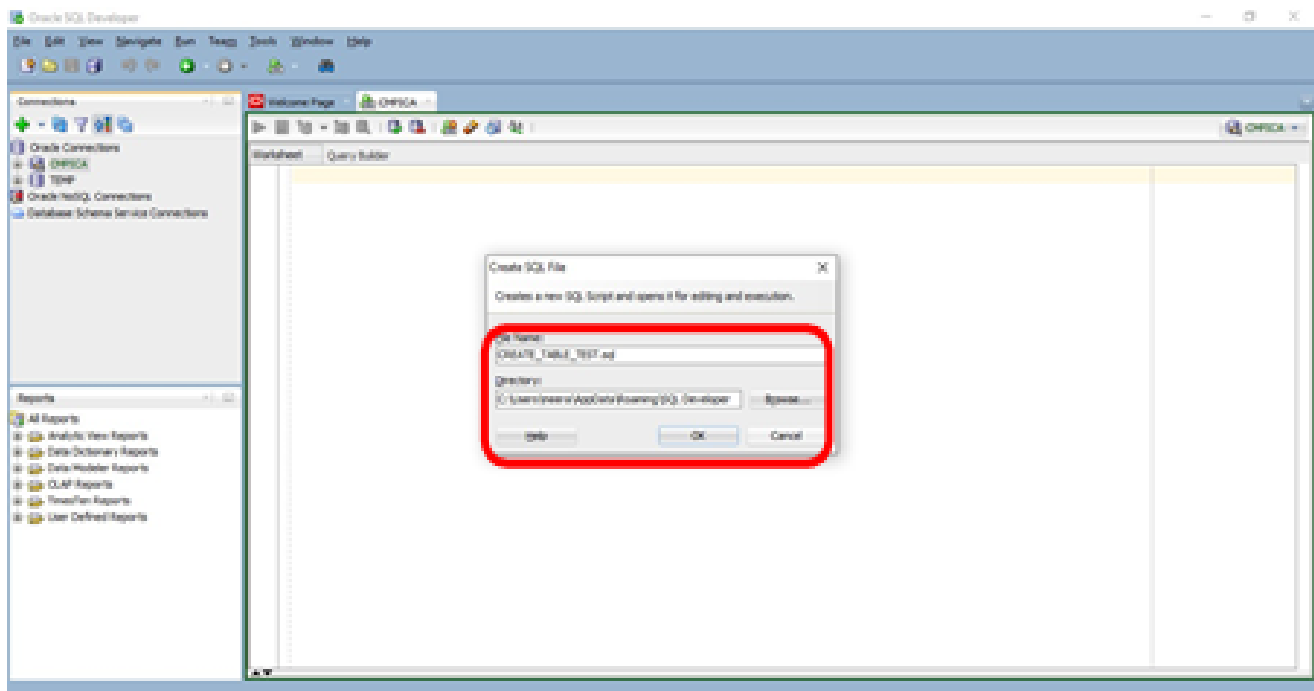
Click OK.

Step 7 Write your query and Run your query (execute) using  icon **OR** press Ctrl + Enter. Select the appropriate Connection in the dialogue box and press OK.



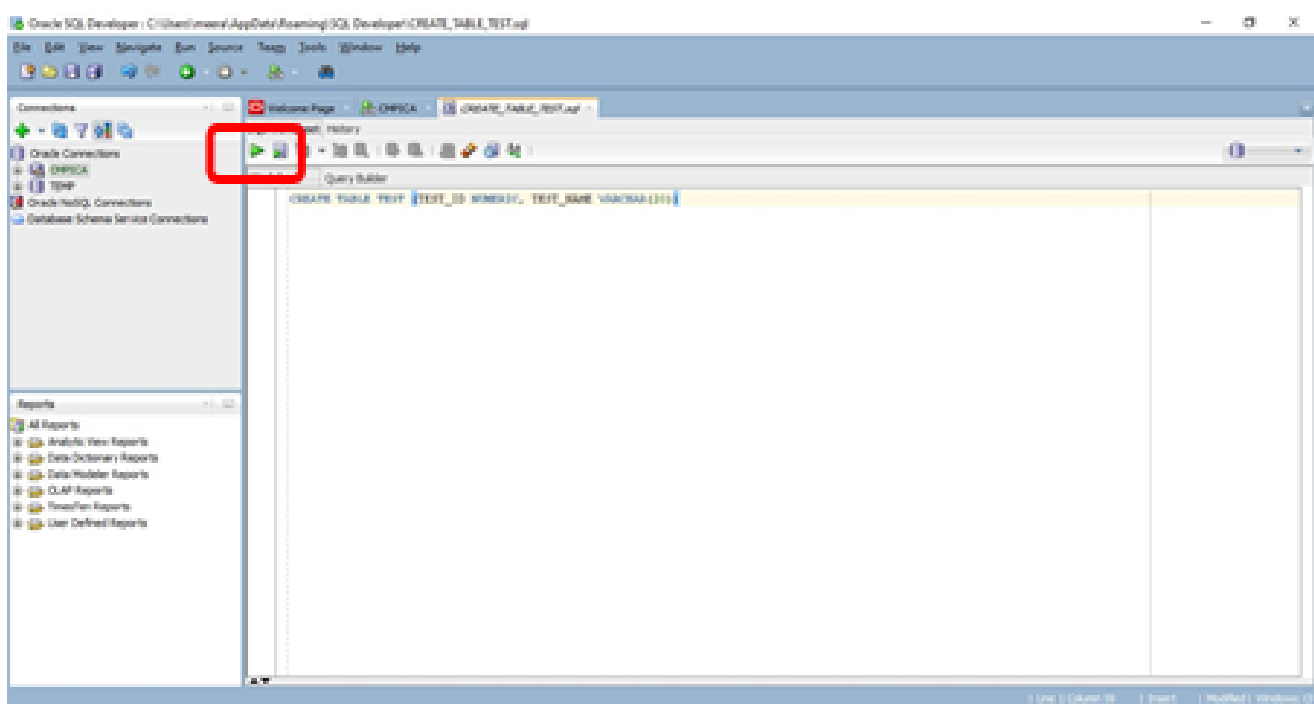
Select the appropriate connection and press OK.

Tool Based Manual || BCA || SEM II || CA130 : Database Fundamentals



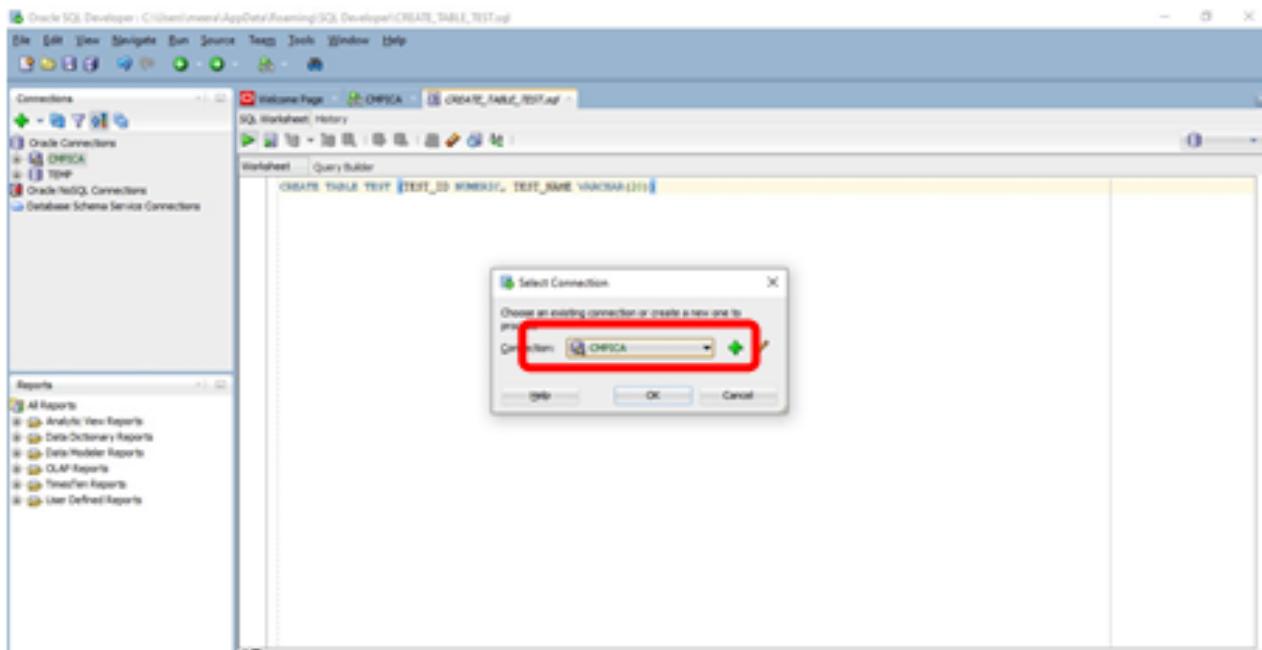
Click OK.

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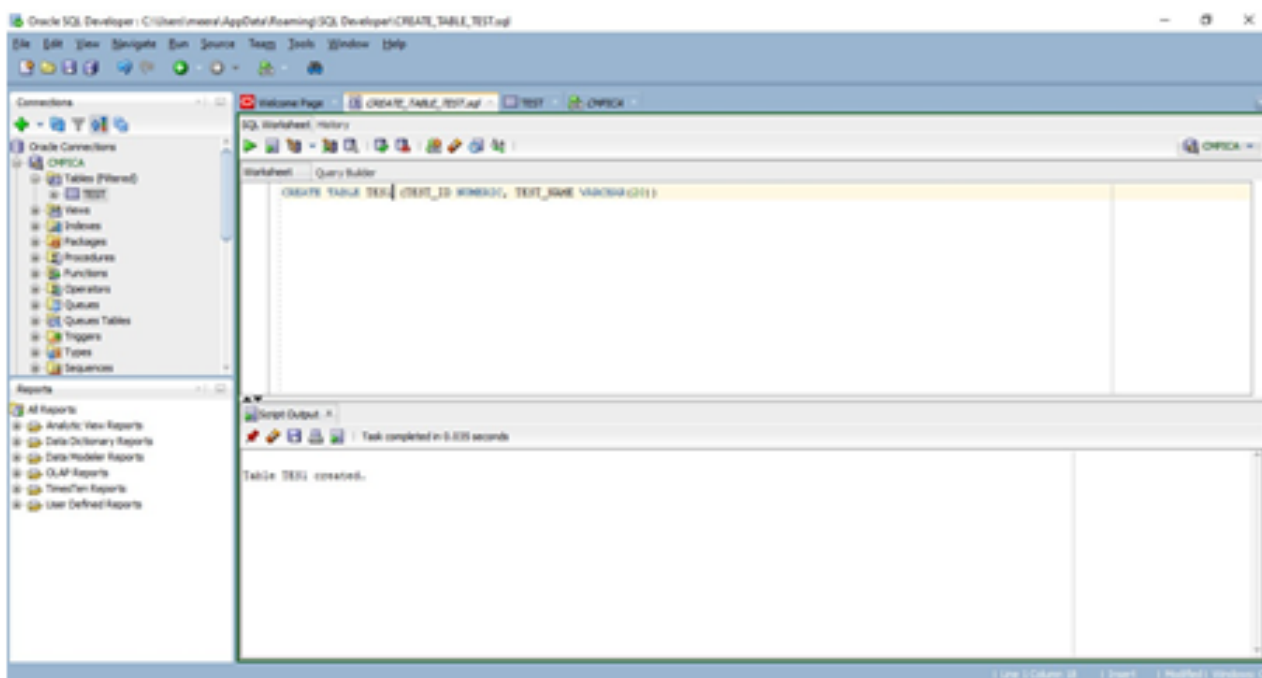


Select the appropriate connection and press OK.

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Observe execution result in bottom panel.



Congratulations!!! Your Oracle SQL Developer Setup is completed successfully.

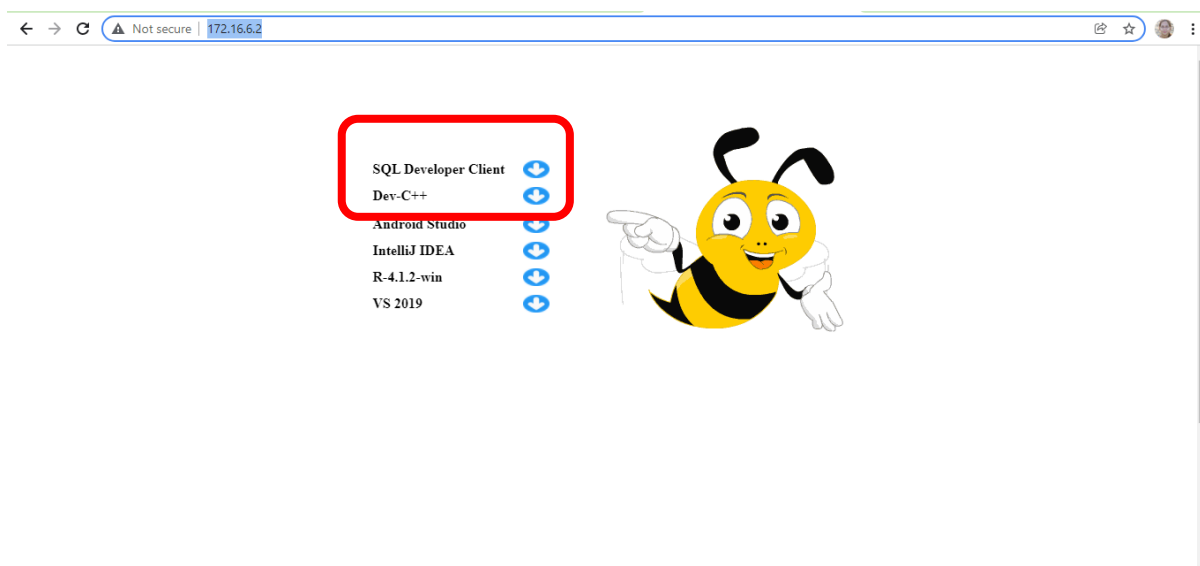
Tool Based Manual || BSc-IT || SEM IV || CA507 : Advanced Database System

Objectives

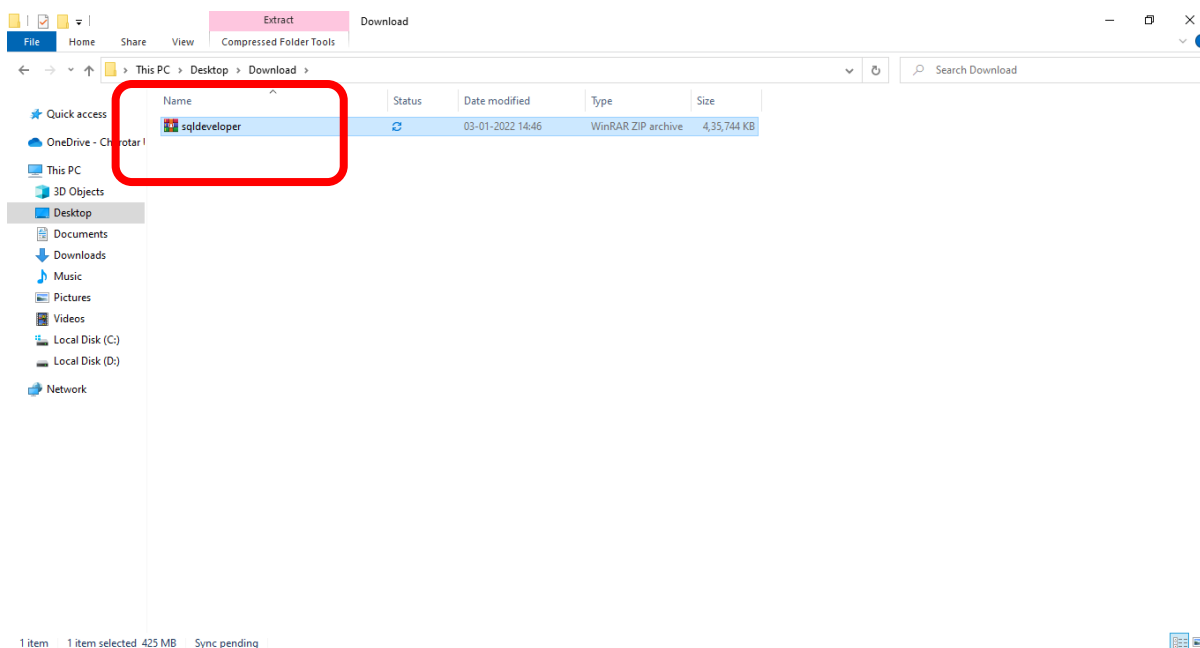
- To make students acquaint with installation of Database Client Software.

Step: 1 Go to <http://172.16.6.2/>

Download SQL Developer Client

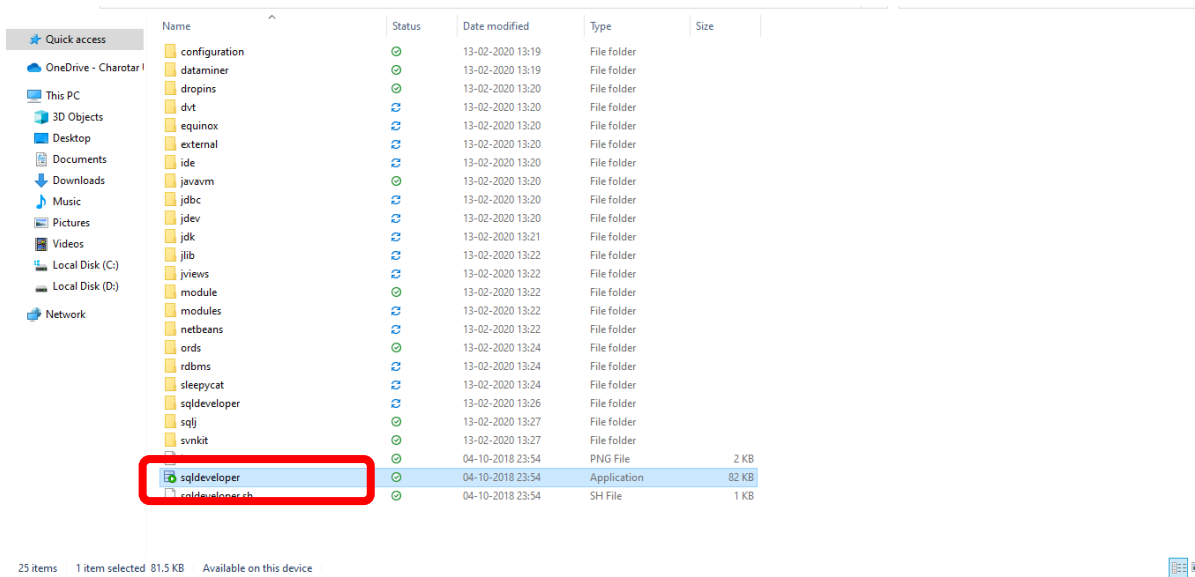


This will download the zip with name **sqldeveloper**. Unzip it.

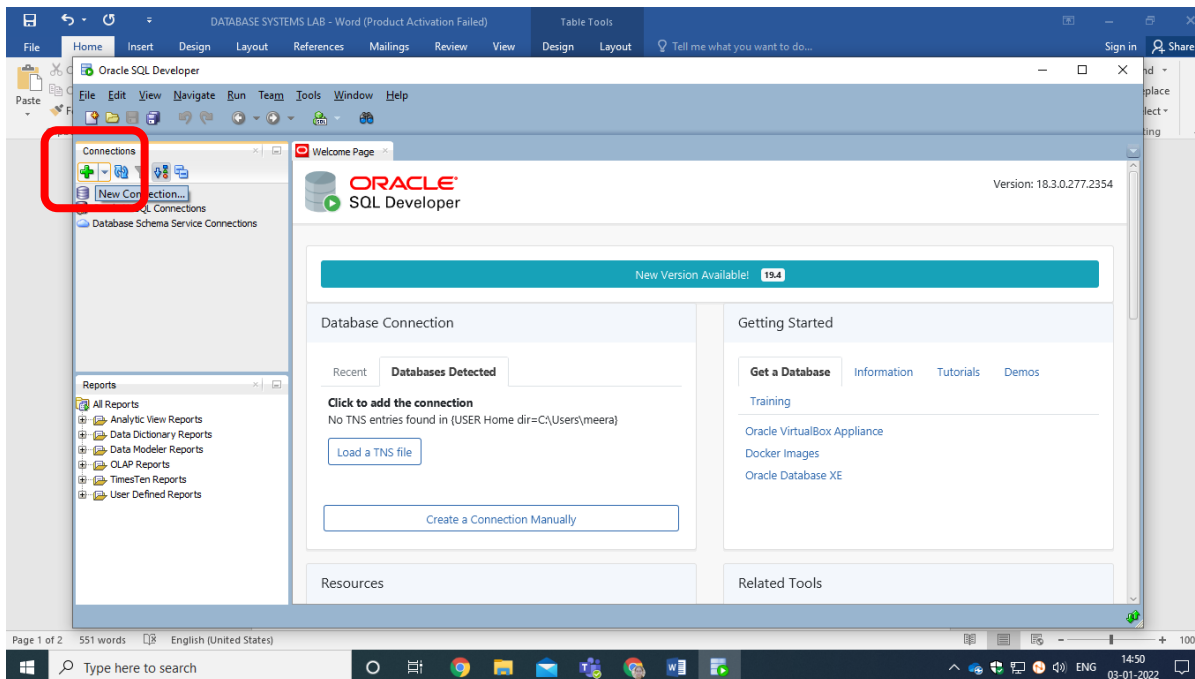


Step: 2 Double click on **sqldeveloper** Application

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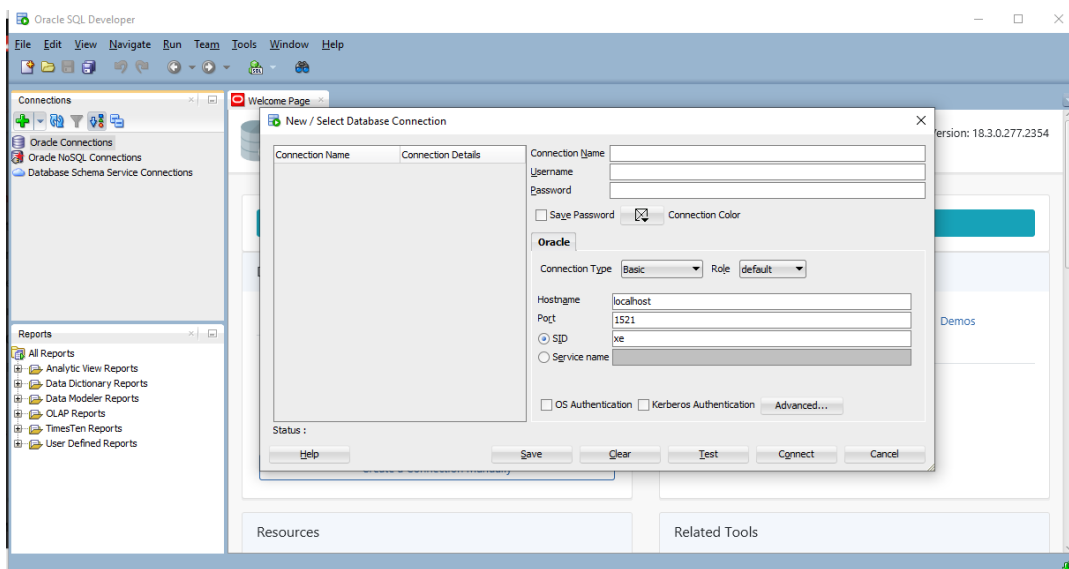


It will open the Oracle SQL Developer.



Step: 3 Click on  Sign, the following screen will appear.

Tool Based Manual || BSc-IT || SEM IV || CA507 : Advanced Database System



Step: 4 Give Proper Connection Name for example CMPICA

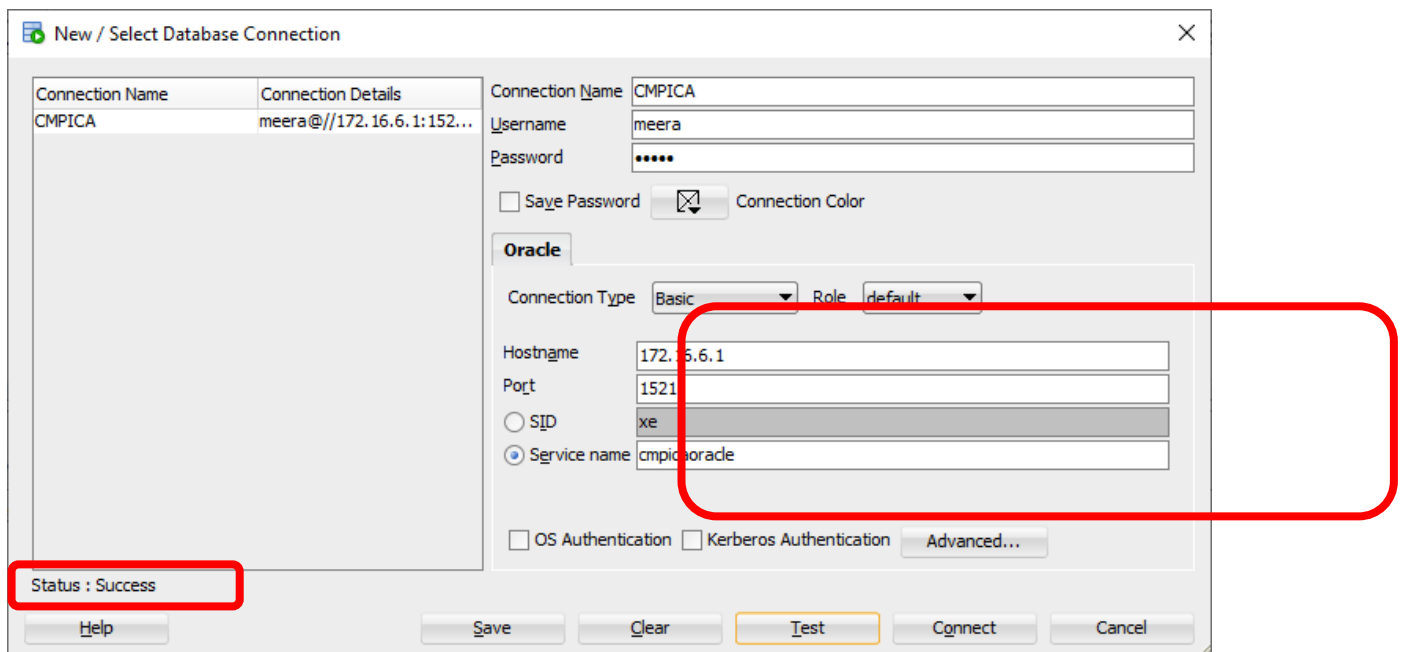
Enter Username and Password provided to you in respective fields.

In Hostname write **172.16.6.1**

Click service name radio button and enter **cmpicaoracle**

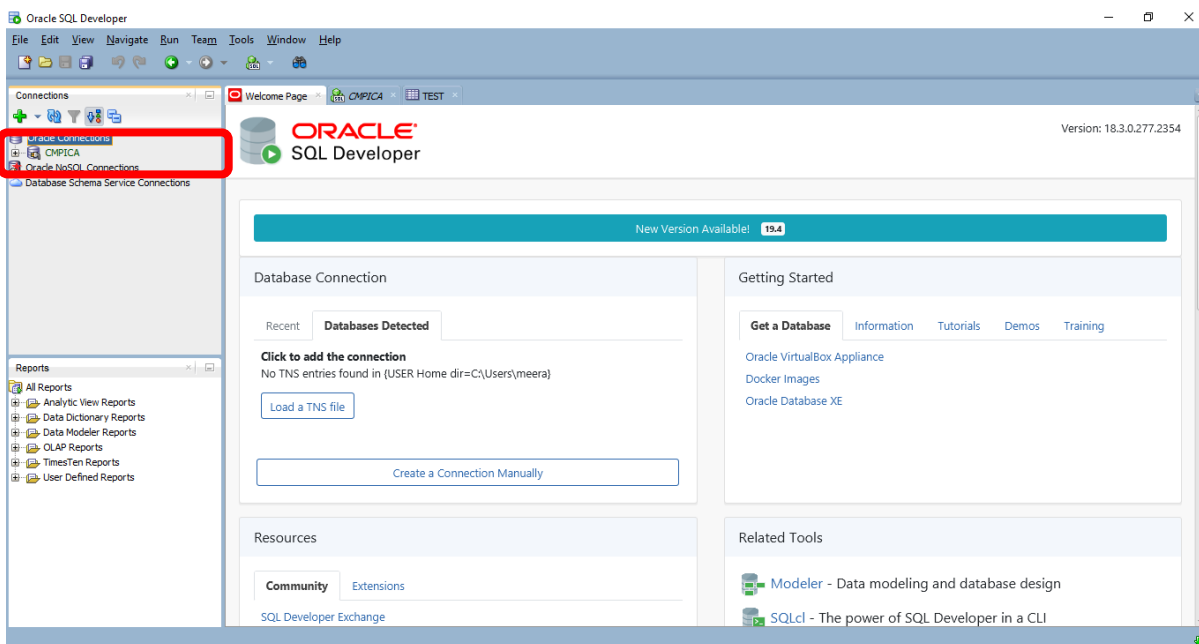
KEEP REST OF THE CONFIGURATION AS IT IS. And Click on Test button. The **success** should get displayed.

Now click on Connect Button.

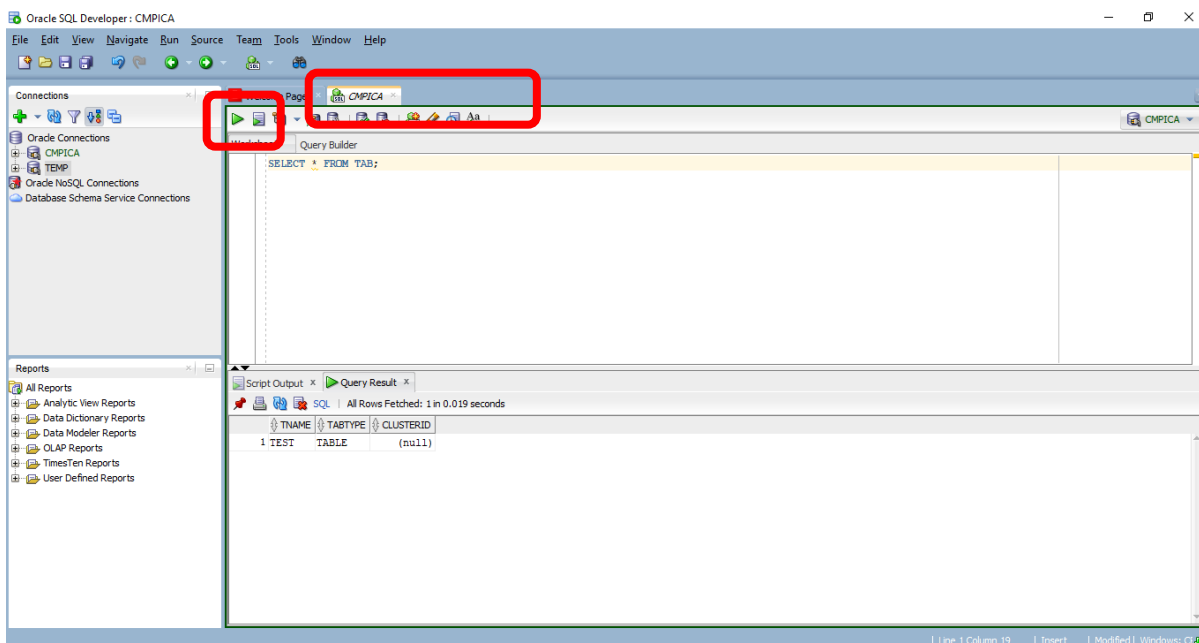


Your newly created connection should get displayed on the left panel.

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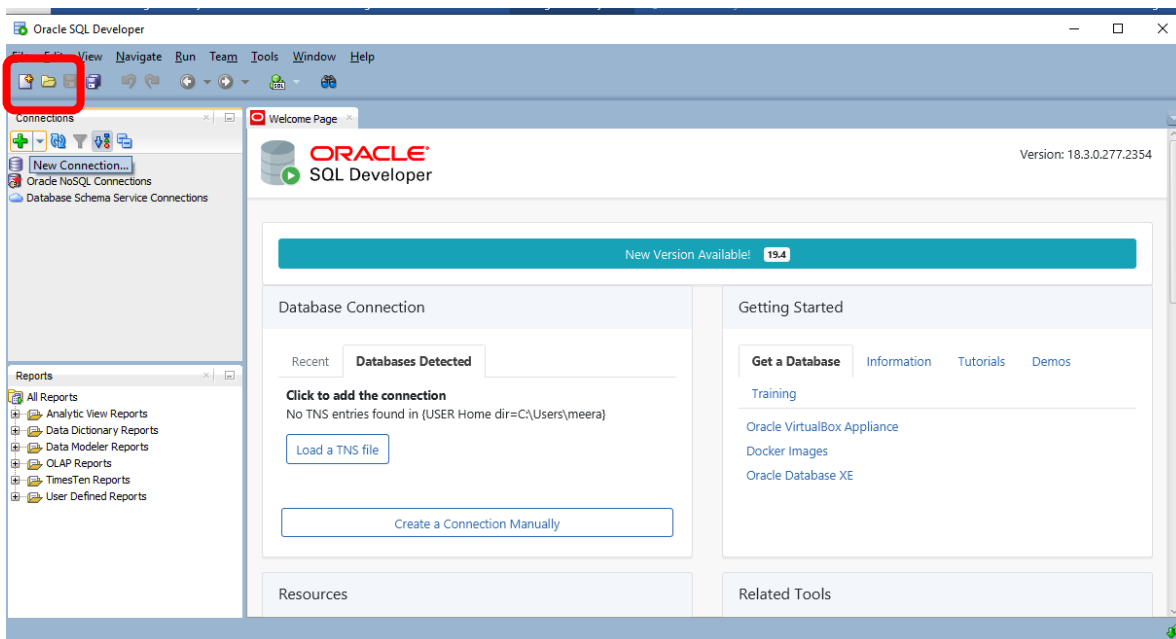
Step 5 On the right side panel observe the Tab with CMPICA (Connection Name). Write your query and Run your query (execute) using  icon **or** press Ctrl + Enter.



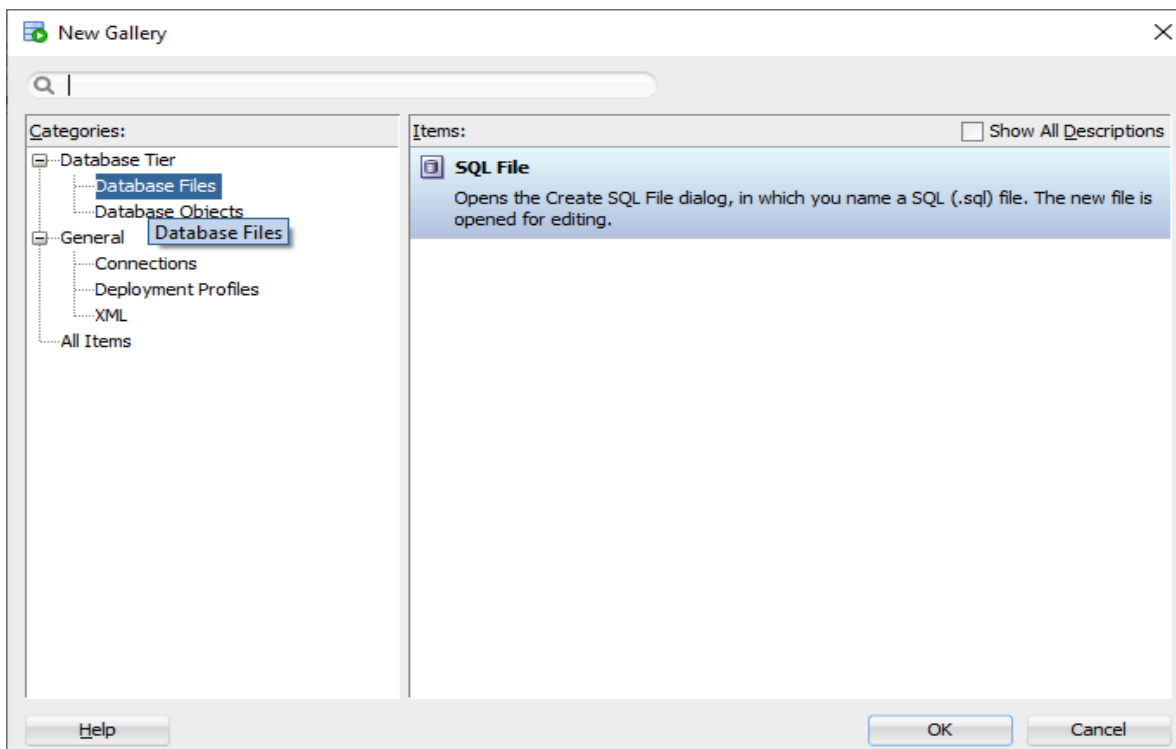
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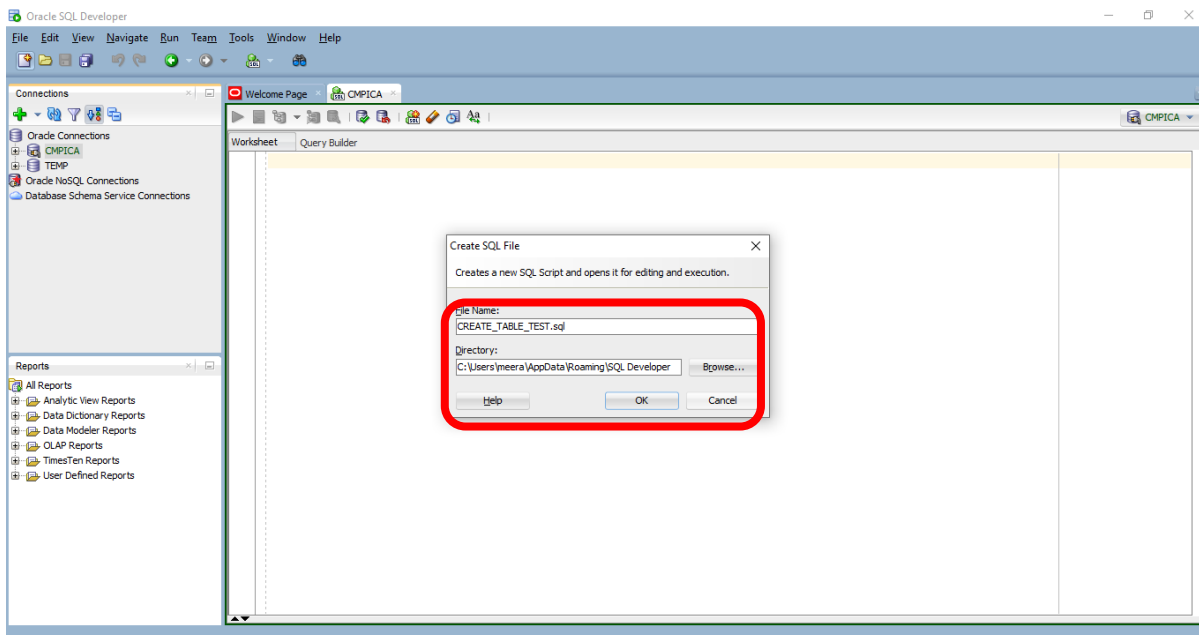


The following screen will appear.



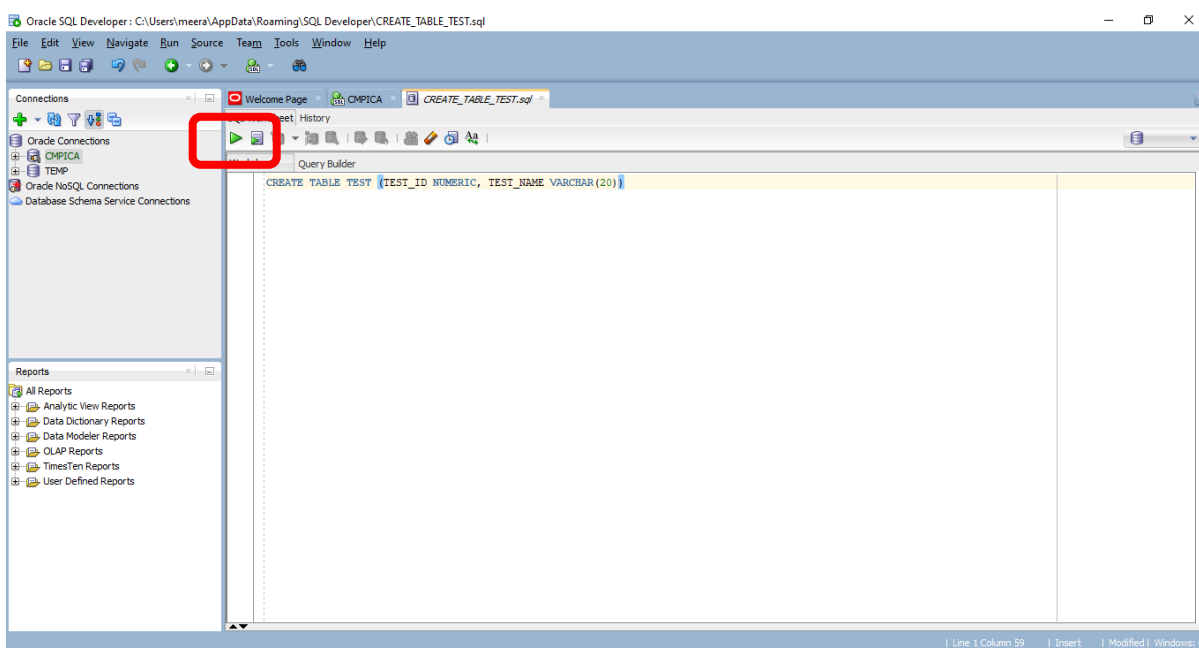
Give proper file name and save to the proper location. **[IT IS ADVISED TO SAVE ALL THE SQL IN ONE FOLDER]**

Tool Based Manual || BSc-IT || SEM IV || CA507 : Advanced Database System



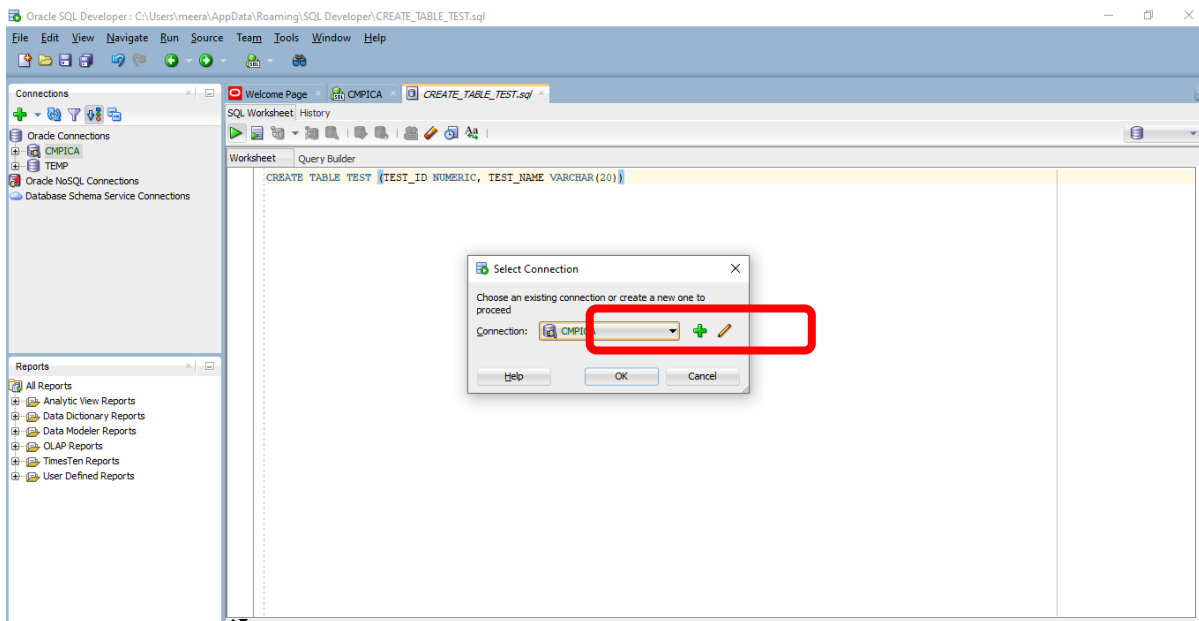
Click OK.

Step 7 Write your query and Run your query (execute) using  icon **or** press Ctrl + Enter. Select the appropriate Connection in the dialogue box and press OK.

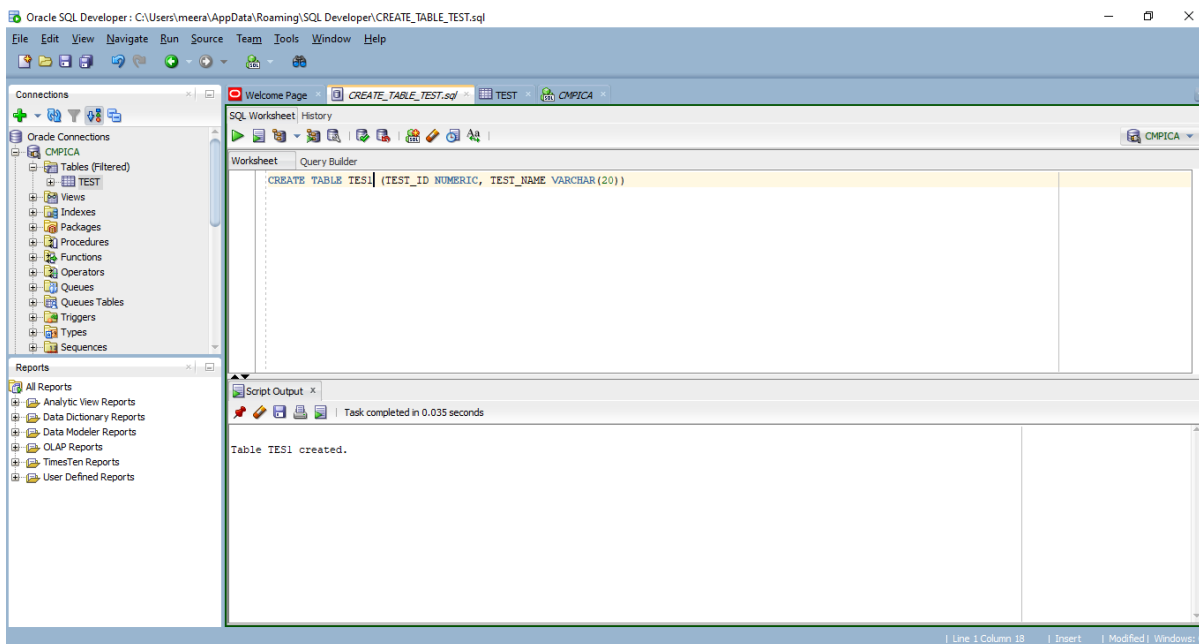


Select the appropriate connection and press OK.

Tool Based Manual || BSc-IT || SEM IV || CA507 : Advanced Database System



Observe execution result in bottom panel.



Congratulations!!! Your Oracle SQL Developer Setup is completed successfully.

Happy Database Exploration!!!

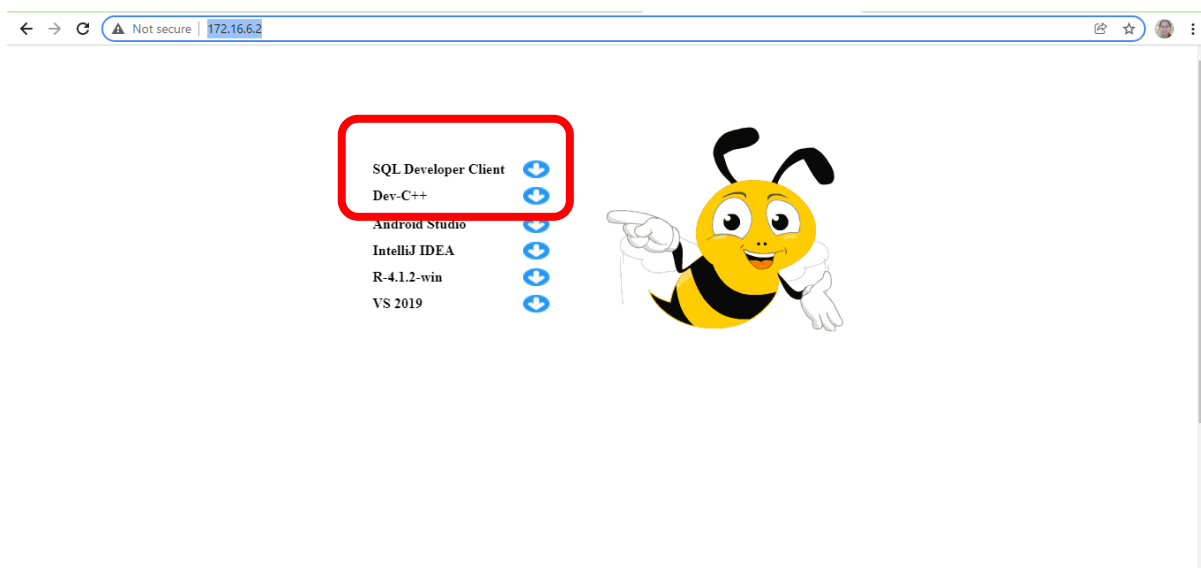
Tool Based Manual || BCA || SEM IV || CA222 : Advanced Database System

Objectives

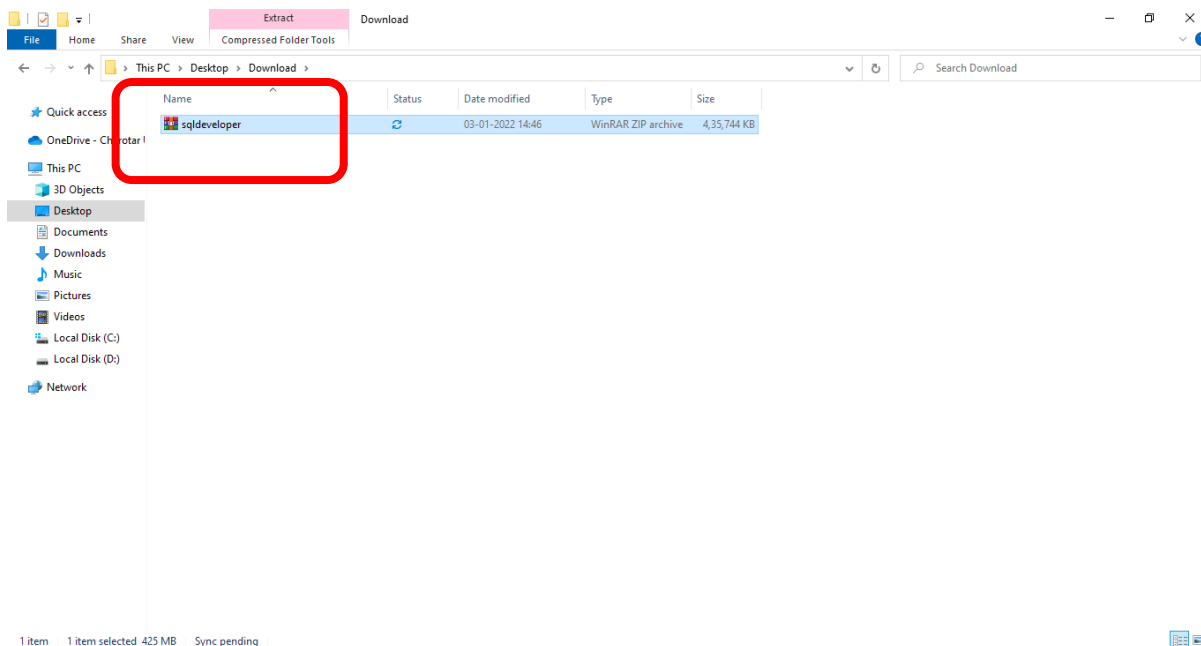
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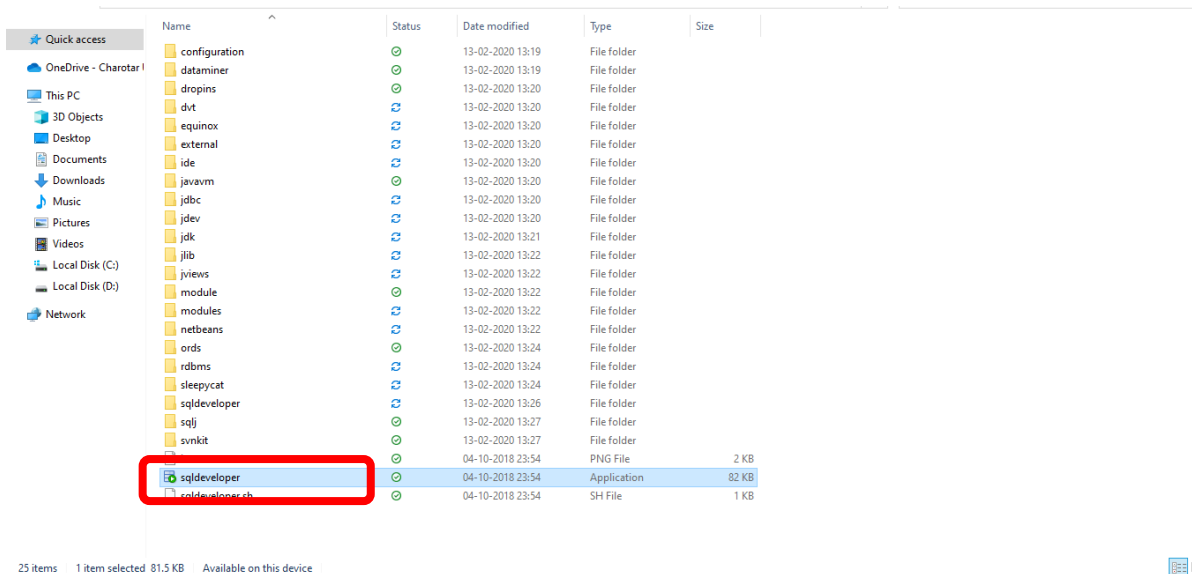


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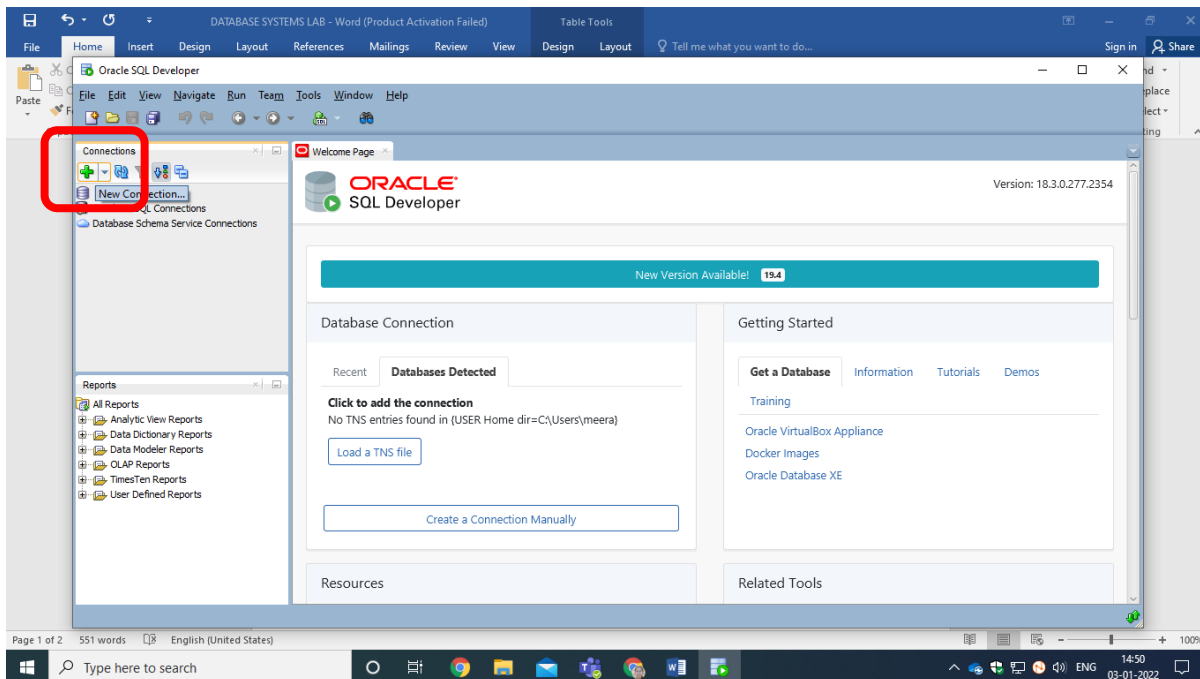


Step: 2 Double click on **sqldeveloper** Application

Tool Based Manual || BCA || SEM IV || CA222 : Advanced Database System

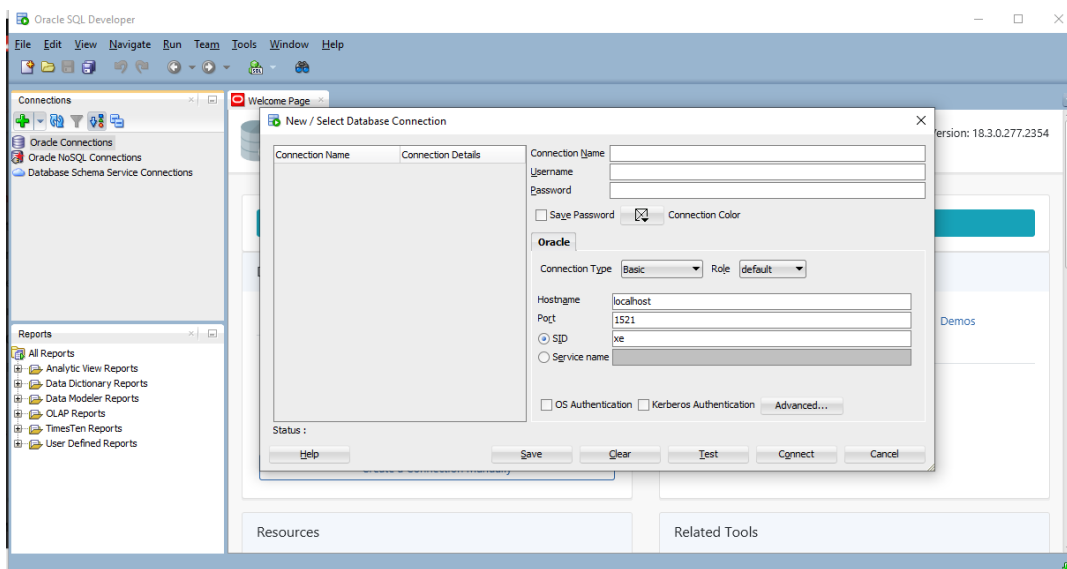


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Tool Based Manual || BCA || SEM IV || CA222 : Advanced Database System



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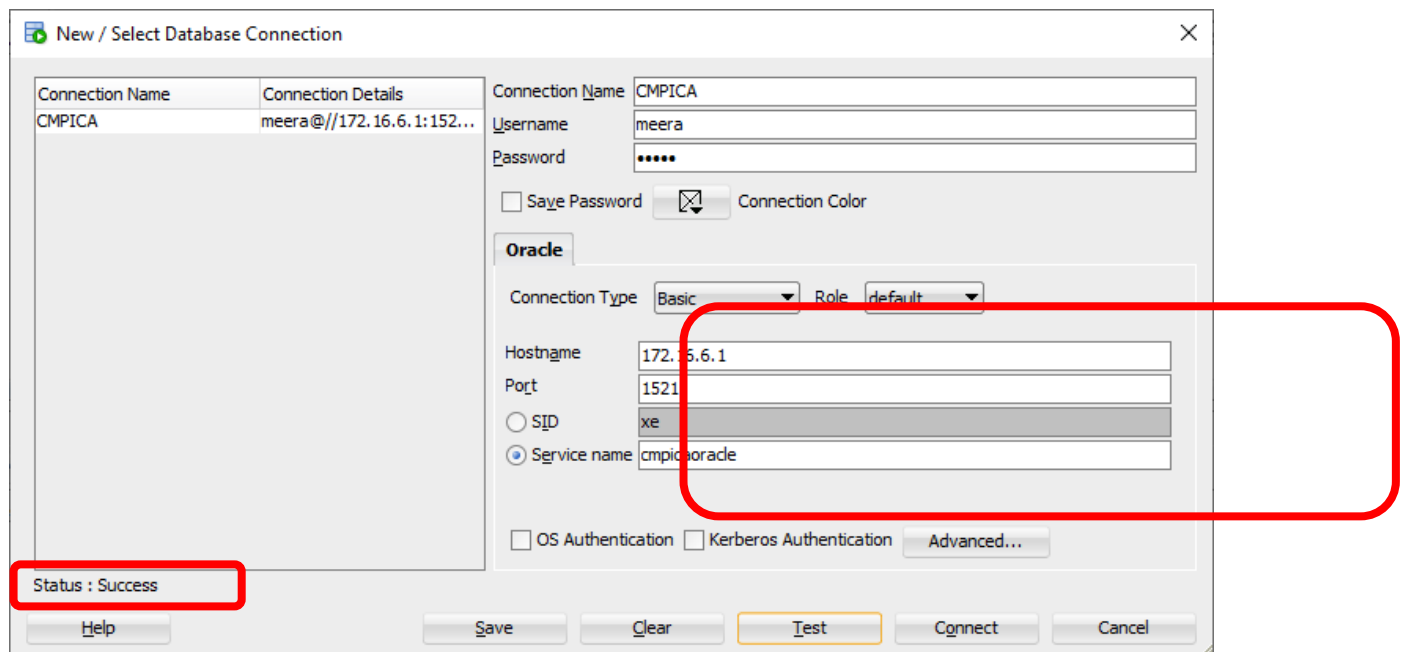
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Click service name radio button and enter **cmpicaoracle**

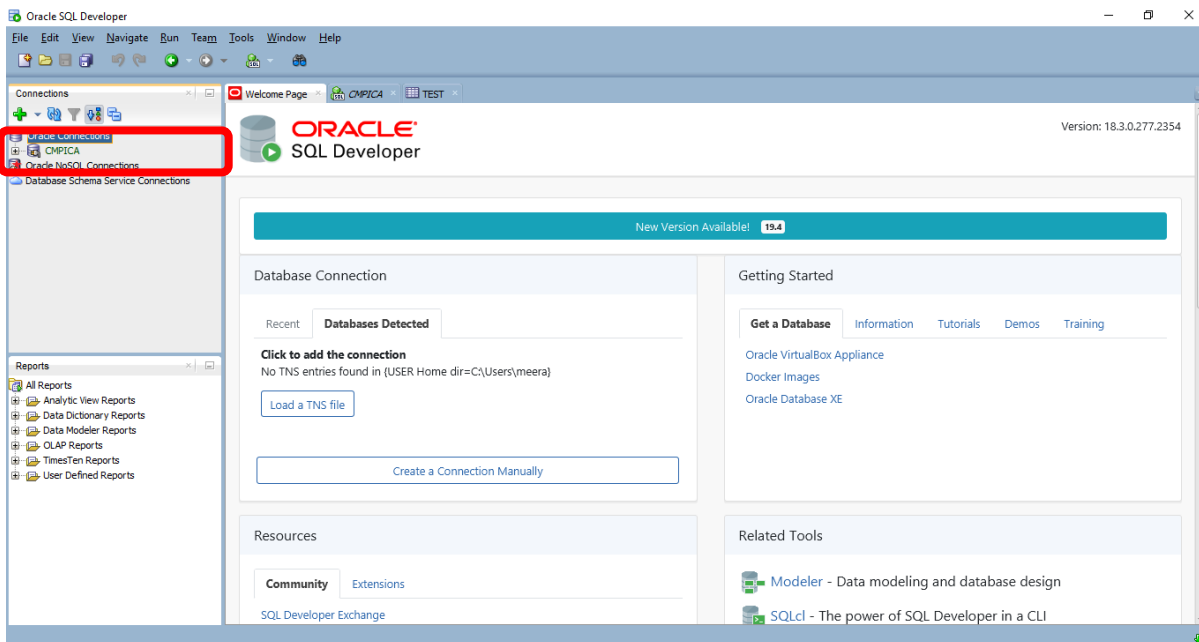
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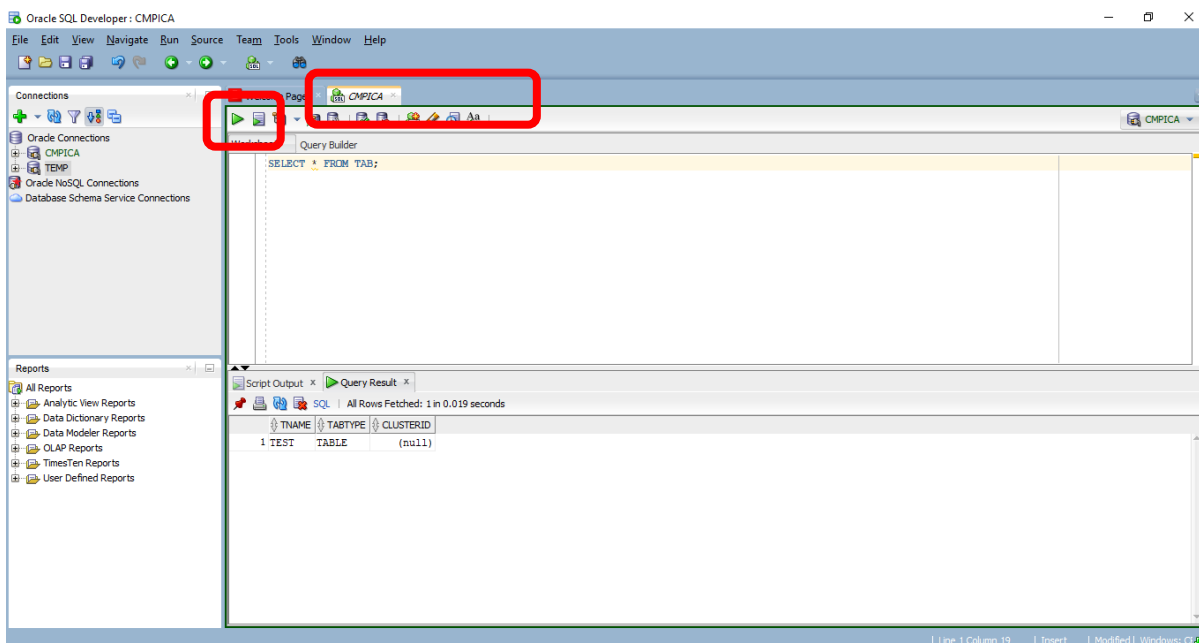


Your newly created connection should get displayed on the left panel.

Tool Based Manual || BCA || SEM IV || CA222 : Advanced Database System



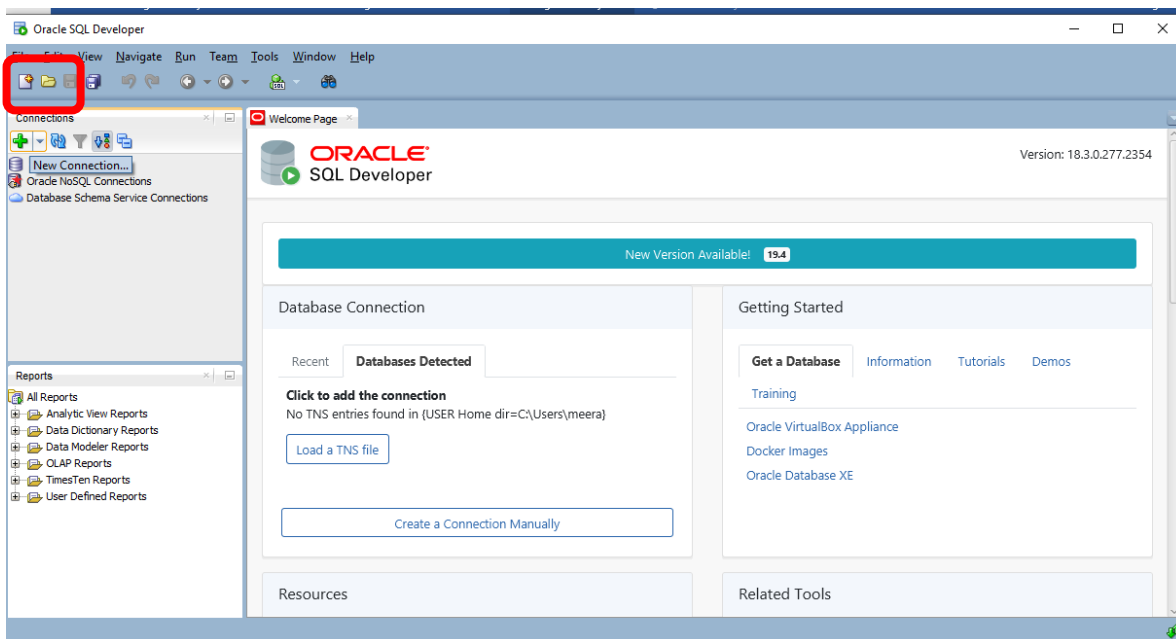
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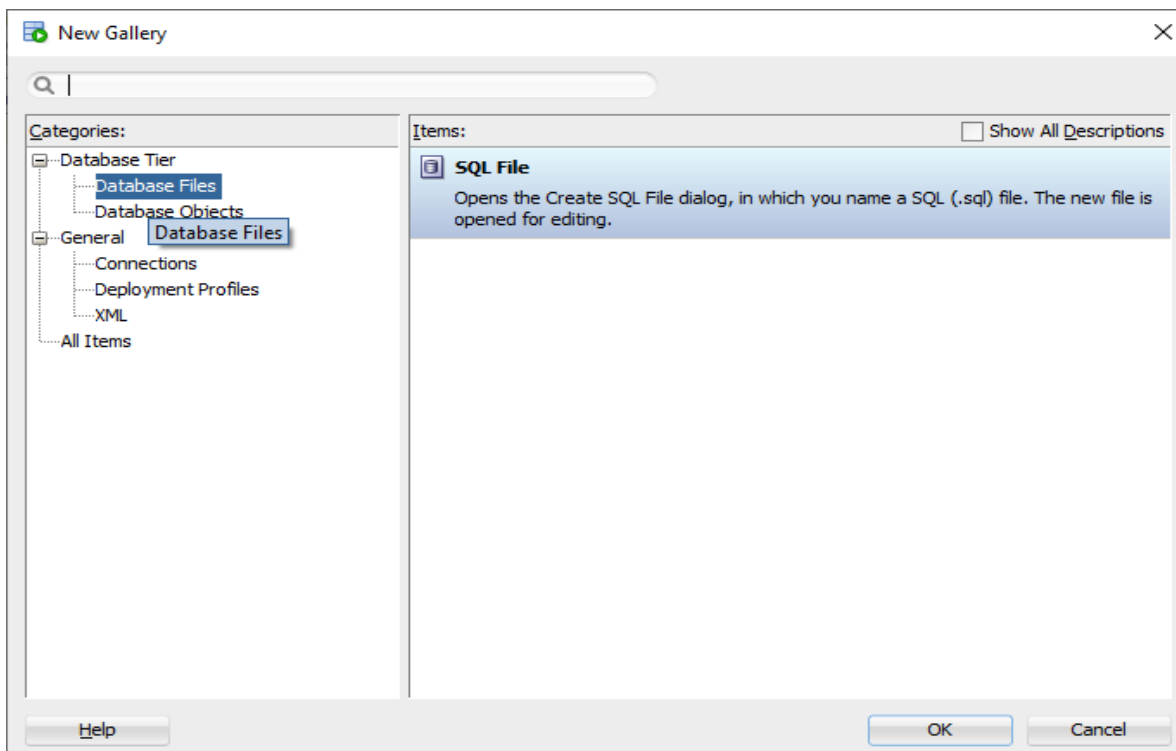
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Tool Based Manual || BCA || SEM IV || CA222 : Advanced Database System

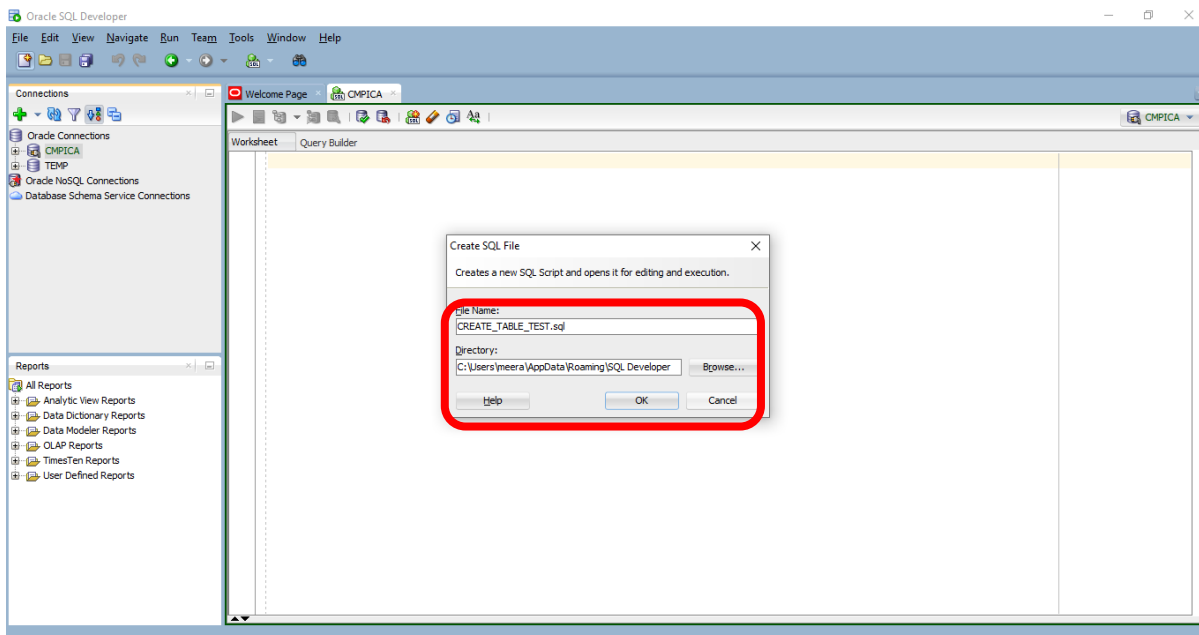


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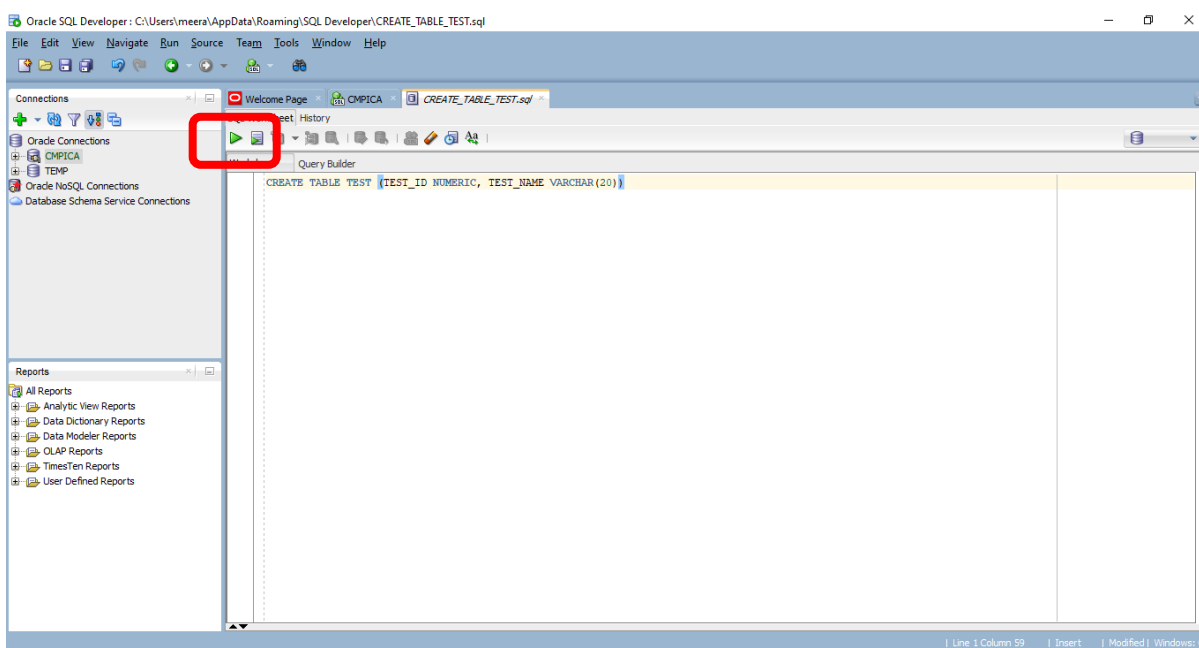
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Tool Based Manual || BCA || SEM IV || CA222 : Advanced Database System



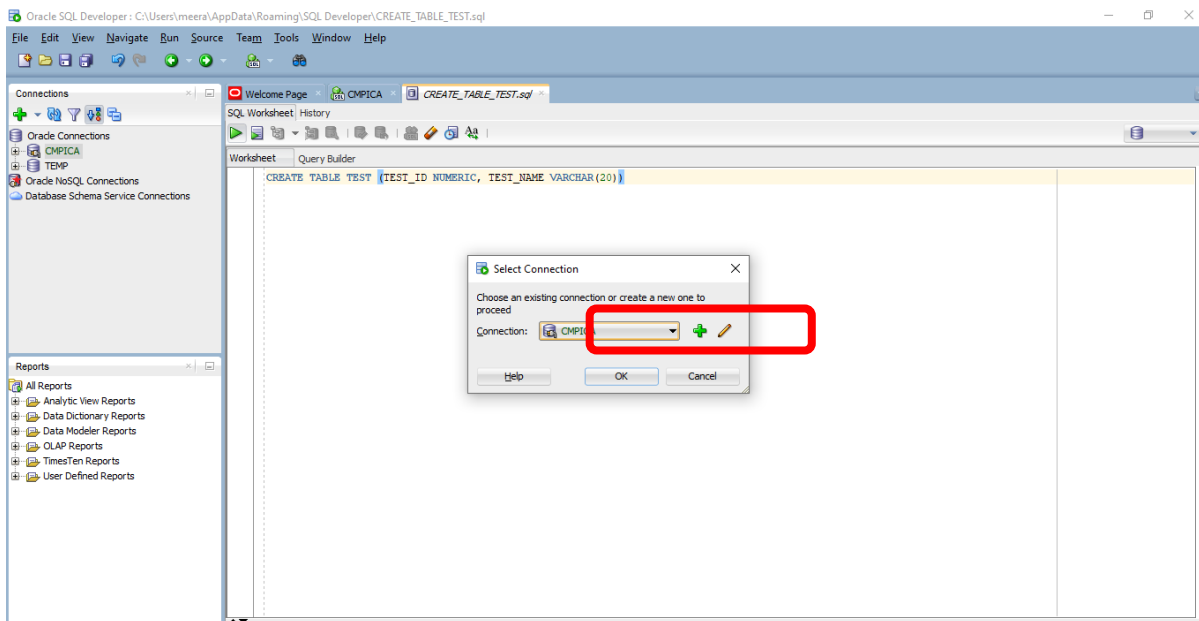
Click OK.

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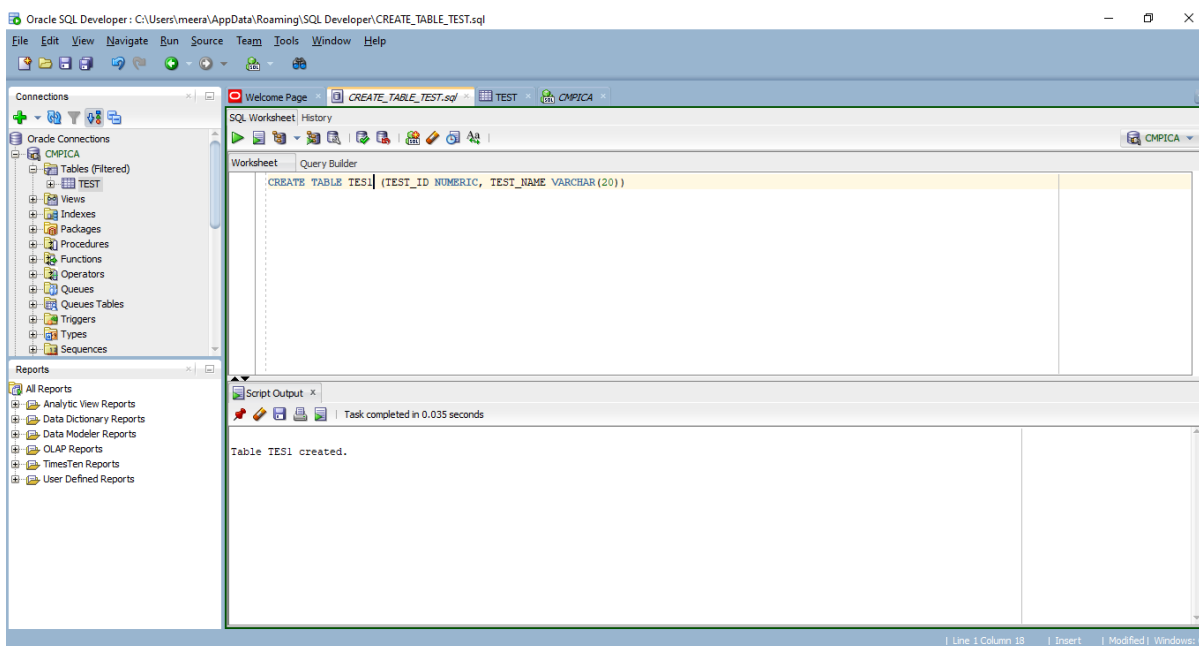


Select the appropriate connection and press OK.

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Observe execution result in bottom panel.



Congratulations!!! Your Oracle SQL Developer Setup is completed successfully.

Happy Database Exploration!!!